COMPLETE EXAM-PREPARATION TUTORIAL

The PMP® Exam Masterclass

From first principles to exam-ready judgement: every domain, every task, the formulas, the mnemonics, and a deep dive into RAM versus RACI.

PMP

People 33%
Process 41%
Business Environment 26%

Aligned to the July 2026 Exam Content Outline and the PMBOK® Guide, Eighth Edition

BEFORE YOU START

How to use this tutorial

MNEMONIC

Navy boxes with a gold edge give you a memory hook. They recur at intervals so recall is spaced, not crammed.

EXAM TIP

Blue boxes show how PMI turns the idea into a scenario question, and which answer pattern usually wins.

EXAM TRAP

Orange boxes flag distractors: answers that sound decisive but skip analysis, escalate too early, or break PMI ethics.

STUDY RHYTHM

One module per sitting. Re-read the speaker notes, then close the deck and redraw each diagram from memory.

LEARNING ROADMAP

Seven modules, basics to exam day

01	**The 2026 exam and the PMI mindset**	Format, blueprint, eligibility, how questions are built
02	**Foundations and PMBOK 8**	Projects, structures, principles, domains, life cycles
03	**People domain · 33%**	Vision, conflict, leadership, stakeholders, communication
04	**Process domain · 41%**	Scope, schedule, finance, EVM, quality, procurement, agile
05	**Business Environment · 26%**	Governance, compliance, change, risk, external forces
06	**Deep dive: RAM versus RACI**	Accountability, the common misunderstanding, the R/A distinction
07	**Exam readiness**	Formula sheet, mnemonic sheet, practice questions, study plan

LEARNING SCIENCE

The study engine behind this deck

1

Understand

Read the concept and the annotated figure. Ask “why would PMI want this?”

2

Encode

Lock it in with the mnemonic. Say it aloud; write it once by hand.

3

Retrieve

Close the deck. Redraw the diagram. Check gaps against the slide.

4

Apply

Answer scenario questions and explain why each wrong option is wrong.

MNEMONIC · “U-E-R-A: You Each Remember Always”

Understand → Encode → Retrieve → Apply. Revisit each module after 1 day, 3 days, 7 days and 21 days (spaced repetition).

MODULE ONE

Know the exam before you study it

01

The July 2026 Exam Content Outline is the blueprint. Every question maps to one of its 26 tasks.

EXAM FORMAT · JULY 2026 ONWARD

The numbers you must know cold

180

questions in total: 170 scored + 10 unscored pretest

240

minutes of exam time (tutorial and survey excluded)

2

ten-minute breaks; you cannot return to a closed section

26

tasks across 3 domains (8 + 10 + 8)

MNEMONIC · "180 · 240 · 2 × 10 · 3 tries"

≈ 80 seconds per question. Three attempts within the 1-year eligibility window, then a 1-year wait.

PACING CHECKPOINTS

Aim for Q60 by ~78 min and Q120 by ~156 min, leaving a buffer for case-study sets.

DOMAIN WEIGHTS · 2021 VERSUS 2026

Business Environment more than tripled

People · 8 tasks

42% (2021)

33% (2026)

Process · 10 tasks

50% (2021)

41% (2026)

Business Environment · 8 tasks

8% (2021)

26% (2026) · ×3.25

≈ Scored items per domain

People ≈ 56

Process ≈ 70

Business Environment ≈ 44

of 170 scored; exact counts vary by form

EXAM TIP

Old prep books under-teach governance, compliance, risk and change, which now sit in Business Environment.

DELIVERY APPROACHES ON THE EXAM

40% predictive · 60% agile and hybrid



Predictive

Scope fixed early, detailed upfront plan, baselines, formal change control, phase gates. Typical: construction, regulated builds.

Adaptive / agile

Scope emerges, short iterations, prioritised backlog, frequent feedback, self-organising team. Typical: software, AI products.

Hybrid

Mixes both: e.g. predictive hardware and agile software, or agile delivery inside predictive governance.

EXAM TRAP

Approaches are not isolated to any domain. Read every stem for clues (“sprint”, “backlog”, “baseline”, “CCB”) and answer in that context.

QUESTION FORMATS · PART 1 OF 2

Classic formats you will see most

Multiple choice · single response

Four options, one best answer. Often two are “right” in isolation; pick the one PMI would do **first** or **best**.

All modalities

Multiple response

“Choose two” or “choose three”. No partial credit is assumed: treat each option as true/false.

All modalities

Matching / drag-and-drop

Pair items across columns, e.g. contract type ↔ risk owner, or conflict mode ↔ scenario.

Computer-based test only

Enhanced matching

Drag boxes onto a chart or diagram, e.g. place stakeholders on a power/interest grid.

Computer-based test only

EXAM TIP

Read the last sentence of the stem first: it tells you whether PMI wants the first, next, best or most likely action.

QUESTION FORMATS · PART 2 OF 2

New and visual formats in 2026

NEW

Case or scenario sets

A detailed business or project story, sometimes with charts, followed by a series of questions on it.

NEW

Graphic-based

Interpret a burndown, control chart, S-curve, network diagram or matrix, then answer.

Point and click (hotspot)

Click the right area of an image, e.g. the point on a control chart that breaks the rule of seven.

Pull-down list

Choose the correct word or value from a drop-down inside a sentence or table.

MNEMONIC · "Can Graphs Point Places?"

Case · Graphic · Point-and-click · Pull-down: the four visual/interactive formats to practise on a simulator.

ELIGIBILITY · 2026 RULES

Education sets the experience you need

Education level	Experience leading projects
Secondary school (EQF 4 / ISCED 3–4)	60 months (5 years)
Associate / short-cycle tertiary (EQF 5 / ISCED 5)	48 months (4 years)
Bachelor’s or higher (EQF 6 / ISCED 6)	36 months (3 years)
Bachelor’s or postgraduate from a PMI GAC-accredited program	24 months (2 years)

KEY RULE

Experience must be non-overlapping, within the last 10 years (previously 8), in a professional setting.

DOES NOT COUNT

School projects, degree research, personal events, home improvement, routine operational work.

TRAINING · APPLICATION · AUDIT

35 hours, verified experience, one year to sit

35 h

Project management education

Aligned to the ECO. Books and practice exams alone do not count. 1 classroom hour = 1 training hour.

1 Dec

2026 live-class rule

Live instructor-led hours count only from an ATP, a China REP, or an accredited academic programme. Self-paced: any provider.

CAPM

Credit for prior learning

An active CAPM waives the 35-hour requirement. GAC coursework can also count.

Apply ✓ Possible audit ✓ Pay & schedule ✓ Sit within 1 year ✓ **60 PDUs / 3 yrs**

ANATOMY OF THE BLUEPRINT

Domain → Task → Enabler

DOMAIN
People

TASK
Help ensure knowledge transfer

ENABLERS
- Identify knowledge critical to the project
- Gather knowledge
- Foster an environment for knowledge transfer

Domain
High-level knowledge area essential to practice.

Task
The PM’s underlying responsibility. Every exam covers all tasks.

Enabler
Illustrative examples, not an exhaustive list.

DOMAIN I · PEOPLE · 33%

The eight People tasks

1 Develop a common vision	**5** Align stakeholder expectations
2 Manage conflicts	**6** Manage stakeholder expectations
3 Lead the project team	**7** Help ensure knowledge transfer
4 Engage stakeholders	**8** Plan and manage communication

MNEMONIC · "Very Calm Leaders Engage, Align, Manage, Know, Communicate"

Vision · Conflict · Lead · Engage · Align · Manage · Knowledge · Communicate, in task order 1 → 8.

DOMAIN II · PROCESS · 41%

The ten Process tasks

1	Integrated PM plan and plan delivery
2	Develop and manage project scope
3	Help ensure value-based delivery
4	Plan and manage resources
5	Plan and manage procurement
6	Plan and manage finance
7	Plan and optimise quality
8	Plan and manage schedule
9	Evaluate project status
10	Manage project closure

MNEMONIC · "Plans Shape Value; Resources Procure Finance; Quality Schedules Status Closure"

Plan · Scope · Value · Resources · Procurement · Finance · Quality · Schedule · Status · Closure.

DOMAIN III · BUSINESS ENVIRONMENT · 26%

The eight Business Environment tasks

1 Define and establish governance	**5** Plan and manage risk
2 Plan and manage compliance	**6** Continuous improvement
3 Manage and control changes	**7** Support organisational change
4 Remove impediments, manage issues	**8** Evaluate external environment changes

MNEMONIC · "Good Companies Change Issues Into Real Improvements, Organisationally & Externally"

Governance · Compliance · Change · Issues · Risk · Improvement · Organisational change · External.

WHAT CHANGED IN JULY 2026

Five shifts in emphasis

AI in scenarios

AI-assisted decisions, ethical and responsible use, data and security risks.

Sustainability

As a planning input, a quality cost, a compliance category and a risk.

Value and outcomes

Success = value worth the effort and expense, not only triple constraint.

Stakeholder engagement

Building trust and influence to reach objectives; aligning expectations.

Governance focus

Escalation paths, thresholds, OPAs, compliance and external change.

MNEMONIC

“A SVaGE”

AI · Sustainability · Value · Governance · Engagement: the 2026 lenses.

THE PMI MINDSET · PART 1

Think like PMI’s ideal project leader

1. **Assess before acting.** Gather facts, analyse impact, then respond.
2. **Be proactive.** Prevent problems with planning, risk work and communication.
3. **Talk directly first.** Private, face-to-face conversation before escalation.
4. **Follow the process.** Changes go through change control; no side deals.
5. **Serve and empower.** Coach, remove impediments, let the team decide how.
6. **Deliver value.** Prioritise by business value and stakeholder outcome.
7. **Act ethically.** Honesty, responsibility, respect, fairness: always.
8. **Tailor.** Fit the approach to the context; there is no one way.

MNEMONIC · “A Proactive Direct Process Serves Valuable Ethical Tailoring”

Assess · Proactive · Direct · Process · Serve · Value · Ethics · Tailor.

The answer-ranking ladder

Best · Analyse / assess the situation and impact

Then · Consult the plan, register or team

Then · Collaborate with those involved

Then · Use the agreed process (CR, risk response)

Later · Escalate to sponsor or steering group

Worst · Ignore, blame, force, or act alone

EXAM TIP

Escalation is correct only when the issue exceeds your authority or the agreed threshold.

EXAM TRAP

“Immediately tell the sponsor” and “fire the team member” are almost always wrong first moves.

RESULTS, RETAKES AND RENEWAL

Plan the whole certification cycle

STAGE 1

Eligibility year

Up to 3 attempts within 1 year of approval.

STAGE 2

Result report

Pass/fail plus a performance rating for each domain.

STAGE 3

After 3 fails

Wait 1 year from the last attempt, then reapply.

STAGE 4

Renewal

60 PDUs every 3 years plus the renewal fee.

Domain ratings

Above Target · Target · Below Target · Needs Improvement. Use them to aim a retake study plan.

Eligibility expired?

If the year lapses without a pass, reapply; no waiting period applies in that case.

MODULE TWO

Foundations and the PMBOK® Guide, 8th Edition

02

Projects, organisations, six principles, seven performance domains, five focus areas and the delivery-approach spectrum.

THE STARTING POINT

What exactly is a project?

A temporary initiative in a unique context undertaken to create value.

PMBOK® Guide definition as quoted in the PMP ECO 2026

Temporary

Has a defined beginning and end. The *outcome* may last for decades; the project does not.

Unique context

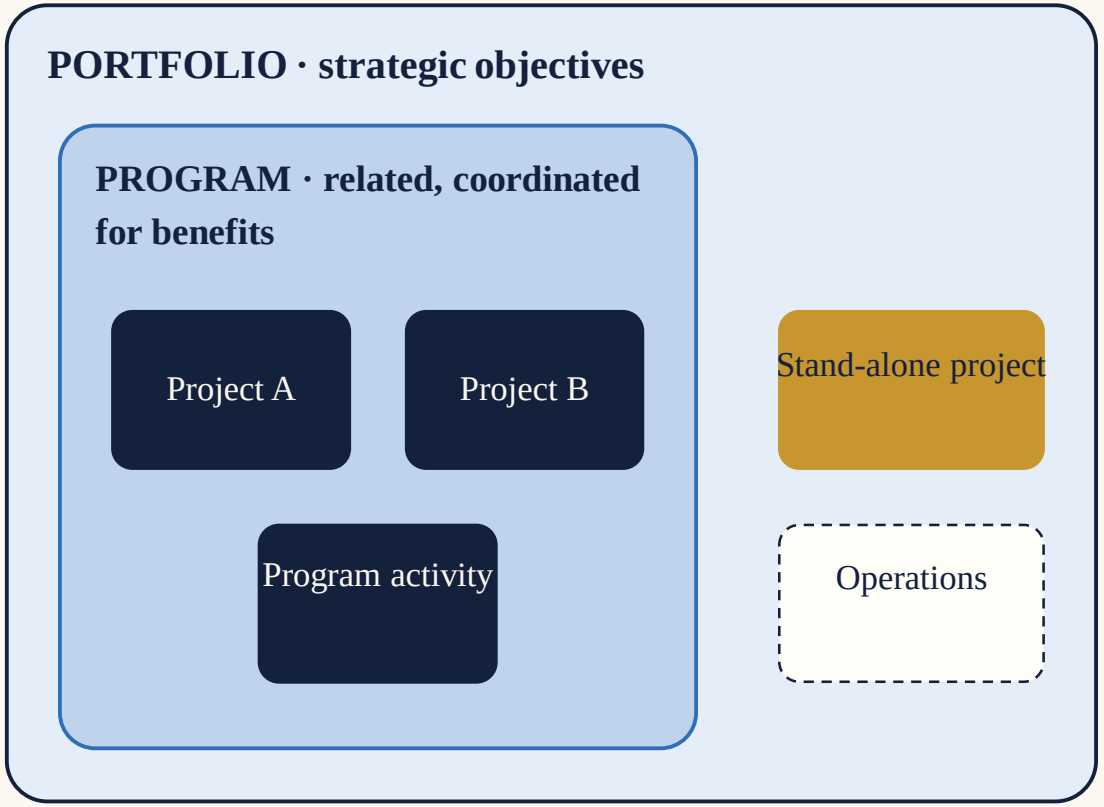
Distinct goals, environment, approach or stakeholders, even if the product is repeated.

Creates value

Deliverables lead to outcomes, outcomes lead to benefits and value for the organisation.

ORGANISATIONAL CONTEXT

Portfolio ⊃ Program ⊃ Project



Project: delivers outputs and outcomes; managed for success.

Program: realises benefits not available if managed separately.

Portfolio: selects and balances work to meet strategy; components need not be related.

Mnemonic: "Projects do things right; programs do the right things together; portfolios choose the right things."

PROJECTS VERSUS OPERATIONS

Change the business, or run the business

Dimension	Project	Operations
Duration	Temporary, defined end	Ongoing, repetitive
Purpose	Create change and new value	Sustain the organisation
Output	Unique deliverable or result	Standardised product or service
Team	Formed for the work, then released	Stable, functional teams
Risk profile	Higher uncertainty	Lower, well understood
Link point	Hands over at transition	Receives and sustains benefits

EXAM TIP · Process Task 10

“Validate readiness for transition (e.g., to operations)” is an ECO enabler: plan handover, training, support and acceptance criteria early.

PROJECT MANAGEMENT OFFICE

Three PMO types on a control dial



Low control

High control

Supportive

Consultative: templates, best practice, training, lessons learned. Acts as a repository.

Controlling

Requires compliance with frameworks, templates and governance; may audit projects.

Directive

Directly manages projects; project managers are assigned from and report to the PMO.

MNEMONIC · "S-C-D: Suggests, Checks, Drives"

Supportive suggests. Controlling checks. Directive drives the projects itself.

ORGANISATIONAL STRUCTURES

Who really holds the authority?

Structure	PM authority	PM role	Budget control	Team reports to
Functional	Little or none	Part-time coordinator	Functional manager	Functional manager
Weak matrix	Low	Expeditor / coordinator	Functional manager	Functional manager
Balanced matrix	Low to moderate	Full-time PM	Shared	Both managers
Strong matrix	Moderate to high	Full-time PM	Project manager	Both, PM dominant
Projectised	High to almost total	Full-time PM	Project manager	Project manager

MNEMONIC · "Fix With Balance, Stay Productive"

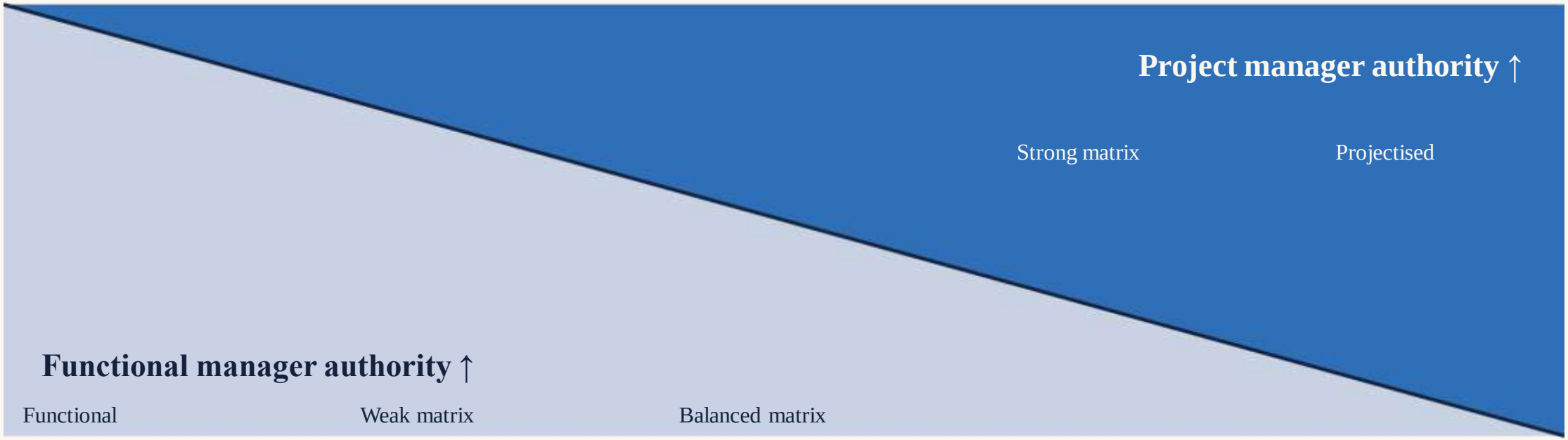
Functional → Weak → Balanced → Strong → Projectised: PM power rises left to right.

EXAM TRAP

In a matrix, team members have two bosses. Resource conflicts → negotiate with functional managers.

ANNOTATED FIGURE

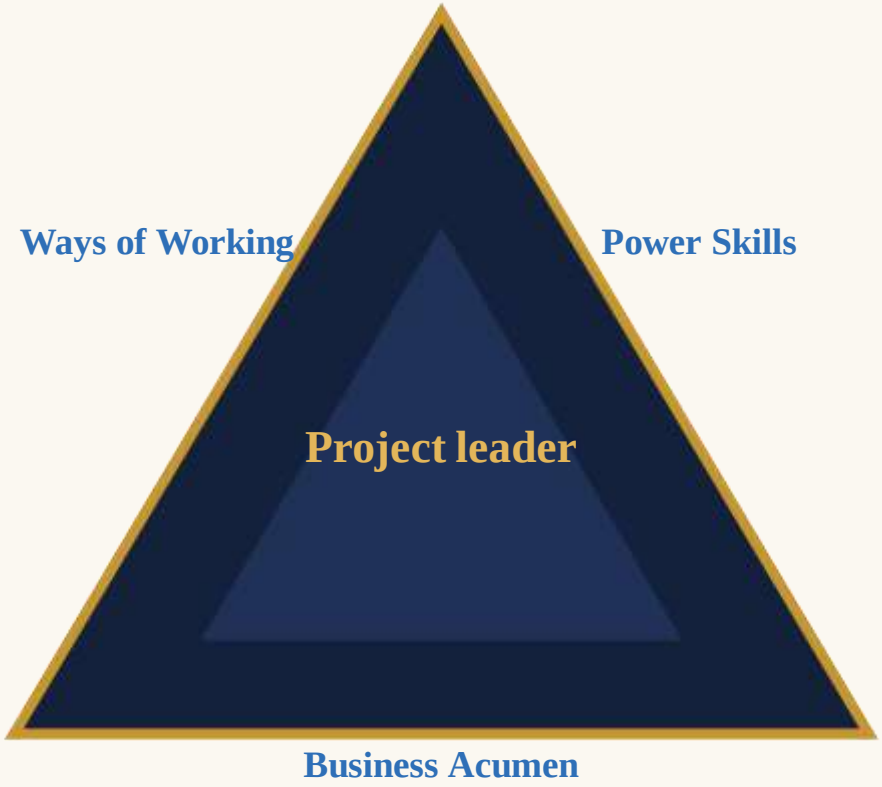
The authority seesaw



Read it: wherever you stand on the axis, the shaded height shows who decides. Low PM authority ⇒ win through influence, collaboration and negotiation.

THE PROJECT MANAGER'S COMPETENCIES

PMI Talent Triangle®



Ways of Working
Predictive, agile, hybrid, design thinking: the technical how.

Power Skills
Communication, collaborative leadership, empathy, problem solving.

Business Acumen
Strategy, markets, finance, regulation: why the project exists.

HOW THE STANDARD EVOLVED

PMBOK 6 → 7 → 8

6th Edition · 2017

10 knowledge areas
5 process groups
49 processes with ITTOs
Prescriptive, process-centric

7th Edition · 2021

12 principles
8 performance domains
No processes listed
Principle- and value-centric

8th Edition · 2025/26

6 principles
7 performance domains
5 focus areas
40 non-prescriptive processes

Takeaway: PMBOK 8 keeps 7's value mindset and brings back practical process guidance, with AI, PMOs and procurement expanded.

PMBOK® GUIDE 8TH EDITION · ARCHITECTURE

Why, what and how in one structure

6 Principles · the WHY	Holistic view · Value · Quality · Accountable leader · Sustainability · Empowered culture
7 Performance domains · the WHAT	Governance · Scope · Schedule · Finance · Stakeholders · Resources · Risk
5 Focus areas · the WHEN	Initiating · Planning · Executing · Monitoring & Controlling · Closing
40 Processes · the HOW	Non-prescriptive; inputs, tools & techniques, outputs as guidance; tailored to context.

THE SIX PRINCIPLES · OVERVIEW

Six principles that guide behaviour

H

Adopt a Holistic View

Systems thinking across the organisation

V

Focus on Value

Outcomes that matter, not just tasks done

Q

Embed Quality

In processes and deliverables, throughout

A

Be an Accountable Leader

Stewardship, ethics, responsible decisions

S

Integrate Sustainability

Environmental, economic, social impacts

E

Build an Empowered Culture

Trust, collaboration, psychological safety

MNEMONIC · "Happy Villagers Quietly Accept Sunny Evenings"

H · V · Q · A · S · E — Holistic, Value, Quality, Accountable, Sustainability, Empowered.

PRINCIPLE 1 OF 6

H

Adopt a Holistic View

What it means

Treat the project as a system of interacting parts inside a larger organisational system. A decision in one area (e.g. schedule) ripples into others (cost, risk, morale, operations).

How it shows up in scenarios

“Crashing will hit the date but overload the shared test lab used by another programme.” The holistic answer weighs the wider impact before acting.

EXAM TIP

Prefer answers that check dependencies, other projects and operations (ECO Process Task 8: “coordinate with other projects and operations”).

PRINCIPLE 2 OF 6

V

Focus on Value

What it means

Success is measured by outcomes and benefits that matter to stakeholders. Value can be financial, social, regulatory, strategic or experiential.

How it shows up in scenarios

“The feature set is on time, but customers aren’t using it.” Re-examine the business case and benefits; reprioritise the backlog by value.

MNEMONIC · “O-O-B-V”

Outputs → Outcomes → Benefits → Value. Deliverables are only the first link.

PRINCIPLE 3 OF 6

Q

Embed Quality

What it means

Build quality into processes and deliverables from the start. Prevention over inspection; quality is everyone’s job, owned by leadership.

How it shows up in scenarios

“Defects keep appearing in final testing.” Best answer: find the root cause and improve the process, not add more end-of-line inspectors.

EXAM TRAP

Gold plating (giving extras) is not quality. Quality = meeting requirements and fitness for use.

PRINCIPLE 4 OF 6

A

Be an Accountable Leader

What it means

Combine leadership with stewardship: own decisions and their consequences, act with integrity, care for entrusted resources, comply with law and ethics.

How it shows up in scenarios

“A status report would look better if a slipping milestone were left out.” Accountable leaders report truthfully and own the recovery plan.

LINK TO MODULE 6

Accountability is the “A” in RACI: one owner who answers for the outcome.

PRINCIPLE 5 OF 6

S

Integrate Sustainability

What it means

Consider environmental, economic and social impacts across the life cycle of the project and of its product, not just during delivery.

Where the ECO puts it

Critical information requirements (Process 1) · cost of quality (Process 7) · compliance categories (BE 2) · sustainability risks (BE 5).

MNEMONIC · “3 Ps”

People · Planet · Prosperity: the triple bottom line every option should be checked against.

PRINCIPLE 6 OF 6

E

Build an Empowered Culture

What it means

Create trust, collaboration and psychological safety so people raise problems early, decide locally and own their work.

How it shows up in scenarios

“The team waits for the PM to assign every task.” Coach them toward self-organisation; agree working agreements; delegate decisions.

EXAM TIP

Empowered ≠ unstructured. Clear roles (RACI) and ground rules are what make empowerment safe.

PMBOK 8 · PERFORMANCE DOMAINS

Seven domains of project performance

Governance
Oversight, decisions, compliance, policy alignment

Scope
Boundaries and quality expectations

Schedule
Timelines, sequencing, coordination

Finance
Budgets, cost performance, investment

Stakeholders
Engagement, shared understanding, value

Resources
People, materials, tools, facilities

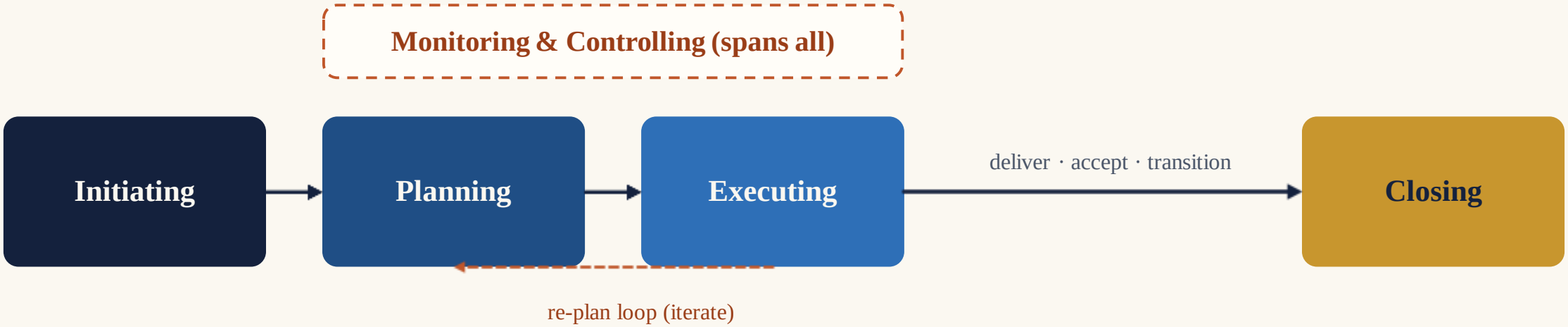
Risk
Threats and opportunities, proactively

Run as one integrated system

MNEMONIC · "Good Scouts Sail Far, Steering Rough Rivers"

Governance · Scope · Schedule · Finance · Stakeholders · Resources · Risk.

Five focus areas: the life of any project



MNEMONIC · "I Pet Every Mild Cat"

Initiating · Planning · Executing · Monitoring & Controlling · Closing.

EXAM TRAP

Focus areas are not phases. They repeat inside every phase or iteration.

ANNOTATED FIGURE · VALUE DELIVERY CHAIN

From strategy to realised value



Feedback loop: information on realised value flows back to strategy, so leaders can continue, change or stop investments. Benefits often appear *after* the project closes, so ownership transfers to operations or a business owner.

Exam tip: if a project is on track but the business case no longer holds, raise it through governance; the sponsor may change or cancel the project.

DEVELOPMENT APPROACHES

The delivery spectrum

Predictive

Scope, time, cost set early. Single delivery at the end. Change is controlled.

Iterative

Repeated cycles refine the whole product through feedback (prototypes).

Incremental

Delivers usable pieces in successive increments, adding functionality.

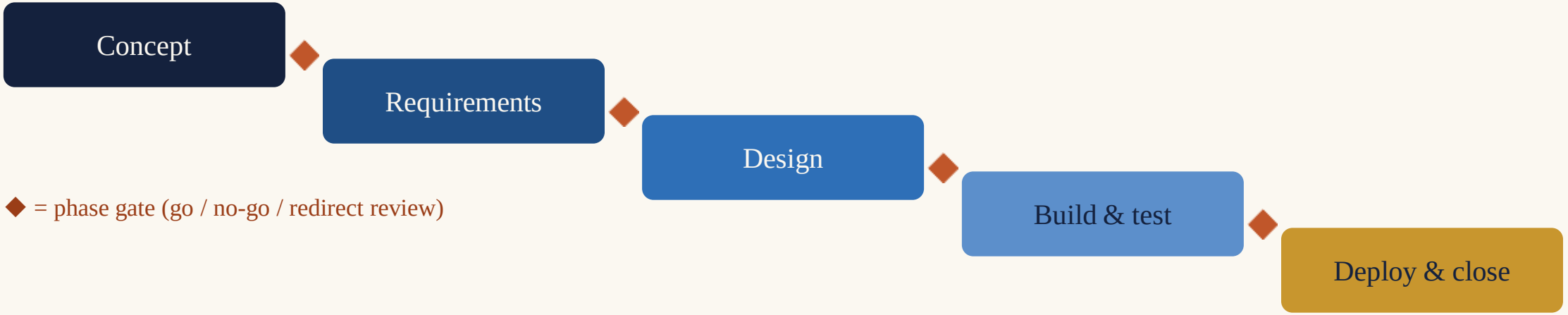
Agile

Both iterative and incremental; frequent, small, value-ordered deliveries.

Characteristic	Predictive end	Adaptive end
Requirements	Fixed, known upfront	Dynamic, emerge with feedback
Delivery cadence	Single delivery	Frequent small deliveries
Goal	Manage cost and control	Customer value via feedback

PREDICTIVE (WATERFALL) LIFE CYCLE

Phases, gates and baselines



Best when
Requirements stable, high regulation, costly change (bridges, plants).

Key artifacts
Scope, schedule and cost baselines; WBS; Gantt; change log.

Phase relationships
Sequential, or overlapping (fast tracking) with higher risk.

ADAPTIVE (AGILE) LIFE CYCLE

Fixed time and cost, flexible scope



Time · Cost estimated

Predictive

Scope estimated

Agile

The "inverted triangle": agile flexes what is built, not when or how much.

Timeboxes: 1–4 week iterations/sprints.

Backlog: ordered by value; top items refined.

Feedback: reviews and retrospectives each cycle.

Change: welcomed; enters via the backlog.

HYBRID APPROACHES

Four common hybrid patterns

1 · Agile then predictive



Prototype with sprints, then roll out predictively.

2 · Predictive then agile



Fixed infrastructure first, then iterative features.

3 · Combined in parallel



Hardware predictive, software agile, integrated at milestones.

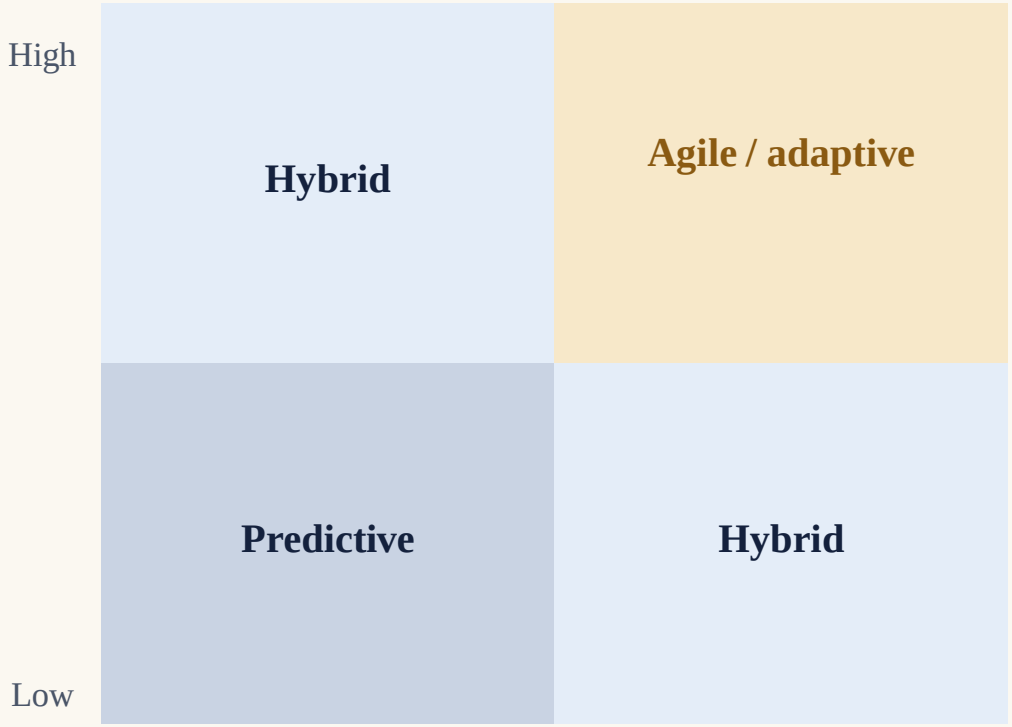
4 · Mostly predictive + agile parts



An uncertain component is run iteratively inside a plan.

Legend: navy = predictive phase · gold = agile iteration

Choosing the approach: a decision grid



Suitability filter questions

Culture: is there buy-in and trust in the team?

Team: size, co-location, experience, access to customer.

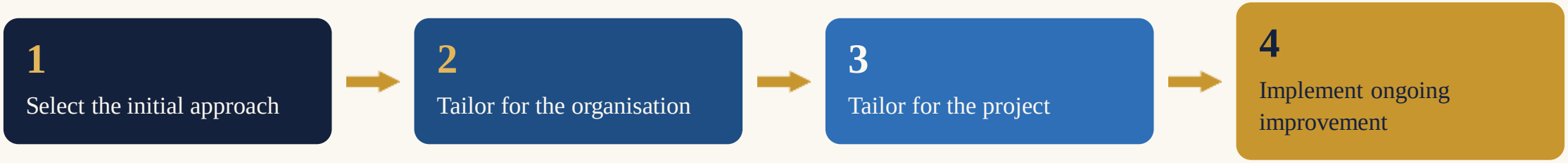
Project: rate of change, criticality, incremental delivery possible?

Constraints: regulation, contracts, safety cases.

Mnemonic "CTPC": Culture, Team, Project, Constraints.

TAILORING

Fit the method to the context



What gets tailored

Life cycle, processes, engagement, tools, methods and artifacts (ECO Process Task 9: “identify and tailor needed artifacts”).

Why tailor

More commitment from the team, focus on customer value, efficient use of resources, less waste.

Exam trap: tailoring never means skipping governance, ethics or legal compliance; it adjusts how, not whether.

INFLUENCES ON EVERY PROJECT

EEFs constrain; OPAs equip

Enterprise Environmental Factors

Conditions **not under the team's control**

Internal: culture, structure, infrastructure, IT software, staff capability, geographic spread
External: market, regulations, legal and social conditions, financial climate, physical environment

Organisational Process Assets

Plans, processes, policies, knowledge **the organisation owns**

Processes, policies, procedures: templates, change procedures, governance
Knowledge repositories: lessons learned, historical data, issue and defect logs, configuration libraries

MNEMONIC · "E = Environment you Endure; O = Organisation's Own"
BE Task 6: update OPAs with lessons learned. BE Task 1: build governance using OPAs.

MODULE THREE · DOMAIN I · 33%

03

People: lead, align, engage

Vision, conflict, leadership, power, team development, motivation, stakeholders, knowledge transfer and communication.

MODULE MAP · PEOPLE

Eight tasks around one leader



PEOPLE TASK 1 · DEVELOP A COMMON VISION

A shared vision is the team’s compass

ENABLER
Help ensure a shared vision with key stakeholders

ENABLER
Promote the shared vision

ENABLER
Keep the vision current

ENABLER
Root-cause any misunderstanding of the vision

Vision tools

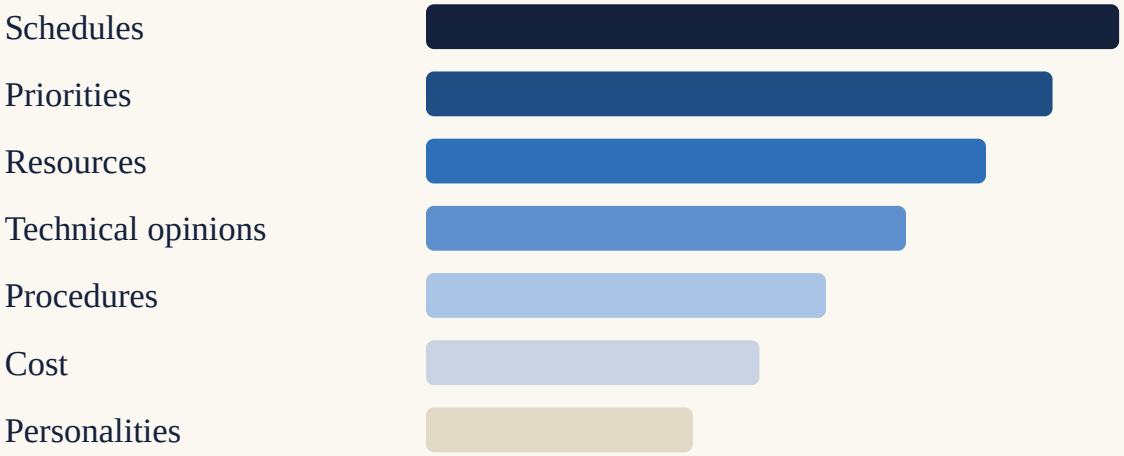
Project charter · product vision statement · elevator pitch · product box · vision board · team charter · roadmap.

Elevator pitch template

For [target], who [need], the [product] is a [category] that [benefit]. Unlike [alternative], ours [differentiator].

Exam tip: when two groups disagree about direction, the best first step is to revisit the agreed vision or charter together, then root-cause the gap.

Where project conflict comes from



Illustrative ranking from classic PM research (Thamhain & Wilemon): personality is the least frequent root cause.

MNEMONIC · "Seven Pirates Rarely Take Proper Care of People"

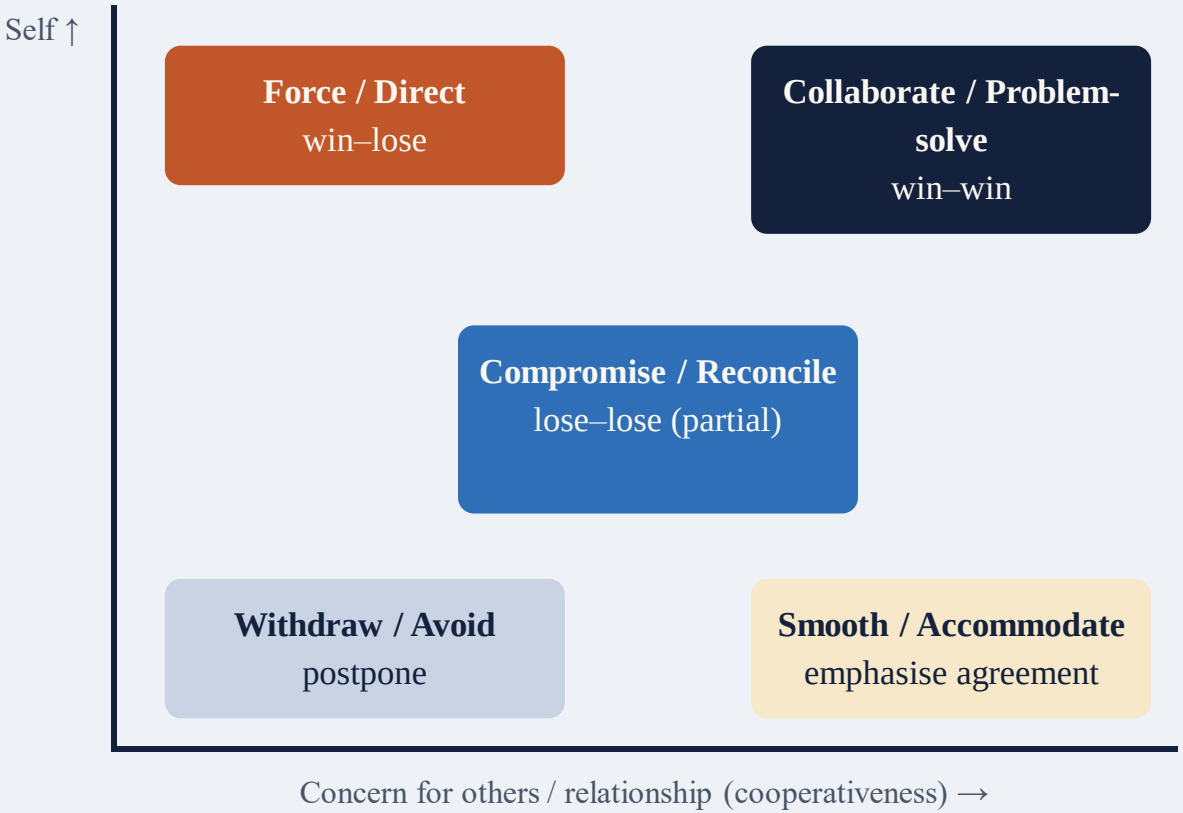
Schedules · Priorities · Resources · Technical · Procedures · Cost · Personalities.

ECO ENABLERS

Identify sources · analyse context · implement agreed strategy · communicate principles · set ground rules · rectify violations.

ANNOTATED FIGURE · CONFLICT-RESOLUTION TECHNIQUES

Five modes on two axes



MNEMONIC

“Five Wise Sheep Can Cooperate”

Force · Withdraw · Smooth · Compromise · Collaborate

Best default: collaborate. It addresses root cause and builds commitment.

Force only for emergencies, safety or ethics, when there is no time.

APPLYING THE MODES

Match the scenario to the technique

Scenario	Best technique
Two senior engineers disagree on architecture; time allows deep discussion	Collaborate · win–win
A safety hazard is reported; the crew wants to keep working	Force / direct · act now
Heated argument in a meeting; emotions are running high	Withdraw briefly, then resolve
Minor disagreement on report formatting; relationship matters	Smooth · stress agreement
Two equally powerful managers both want the same tester this week	Compromise · split time
Team member repeatedly breaks ground rules	Private conversation first

Exam trap: “Escalate to HR” or “ask the functional manager to discipline” is rarely first. Talk privately, then use agreed processes.

Agree the rules before you need them

SAMPLE TEAM CHARTER

Values: respect, candour, quality first
Working agreements: core hours 10:00–15:00; cameras on for ceremonies
Decision process: consensus by fist-of-five; PO decides on scope ties
Communication: chat for quick, ticket for work, call for conflict
Conflict process: talk directly within 24 h, then facilitator
Meeting rules: start on time, agenda first, actions logged

WHO WRITES IT

The team, facilitated by the PM. Ownership drives adherence.

When violated

Remind the person privately, refer to the charter, then involve the team. Everyone polices the rules, not only the PM.

Mnemonic "VWDCC": Values, Working agreements, Decisions, Communication, Conflict.

Leadership styles you must recognise

Servant
Serves first; removes impediments; grows people. Default for agile.

Transformational
Inspires with vision and ideals; drives change and innovation.

Transactional
Rewards and corrections for goals met or missed; management by exception.

Laissez-faire
Hands-off; team decides. Works with expert, self-motivated teams.

Charismatic
Energy and conviction inspire; risk of dependency on the leader.

Interactional
A blend of transactional, transformational and charismatic.

MNEMONIC · "STT-LCI"

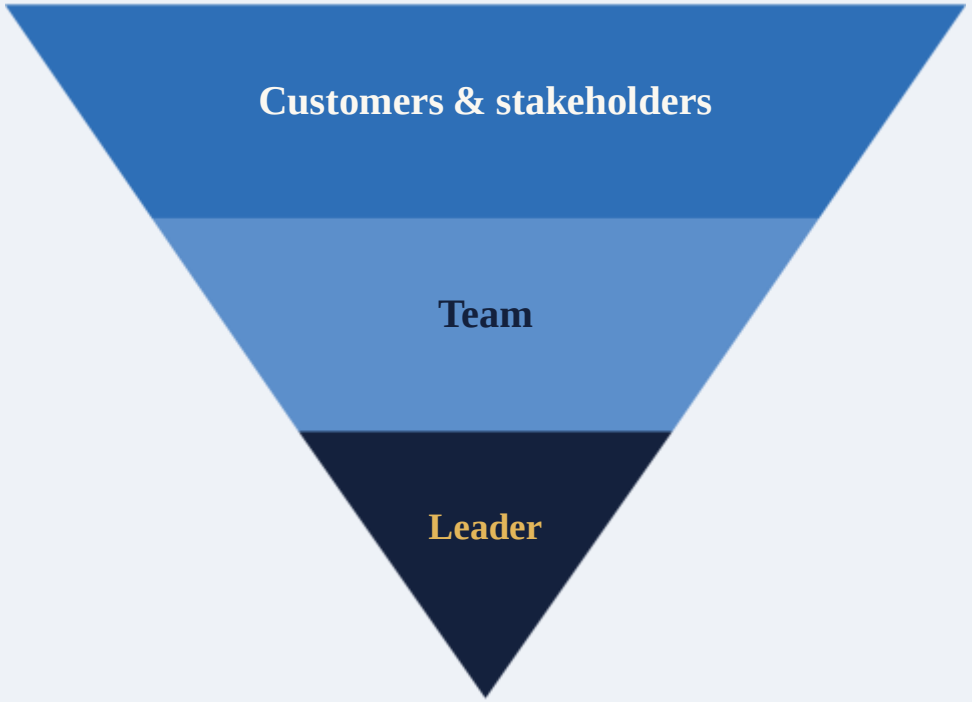
"Serve, Then Transform; Let Charisma Interact."

ECO ENABLER

"Determine an appropriate leadership style" — situation decides.

ANNOTATED FIGURE · SERVANT LEADERSHIP

Turn the org chart upside down



Shield the team from interruptions and diversions.

Remove impediments the team cannot remove itself.

Communicate and re-communicate the vision.

Provide resources, coaching and encouragement.

Mnemonic "SRCP": Shield, Remove, Communicate, Provide.

SITUATIONAL LEADERSHIP (HERSEY–BLANCHARD)

Match the style to team maturity

S1 · for D1

Directing

Low competence, high commitment. High task, low support.

S2 · for D2

Coaching

Some competence, low commitment. High task, high support.

S3 · for D3

Supporting

Competent, variable commitment. Low task, high support.

S4 · for D4

Delegating

High competence and commitment. Low task, low support.

MNEMONIC · “Don’t Cry, Support Delegates”

Directing → Coaching → Supporting → Delegating as the team matures.

LINK TO TUCKMAN

Forming ≈ direct · Storming ≈ coach · Norming ≈ support · Performing ≈ delegate.

SOURCES OF POWER (FRENCH & RAVEN, EXTENDED)

Influence without formal authority

Earned (personal) · best

Expert · knowledge and skill

Referent · respect, admiration, trust

Relational / informational · networks, data

Positional (formal)

Legitimate · the role itself

Reward · recognition, bonuses

Coercive / penalty · punishment · worst

MNEMONIC · "Really Excellent Leaders Can Reward"

Referent · Expert · Legitimate · Coercive · Reward (the classic five).

EXAM RANKING

Expert and referent = most effective; coercive = least. Reward is better than penalty.

TEAM DECISION-MAKING TECHNIQUES

How groups decide, fast and fairly

Fist of five

0–2 | 3 | 4–5

Fingers show support. Anyone at 0–2 explains concerns before proceeding.

Roman voting

Thumbs up, sideways, down. Quick read of consensus.

Delphi

Anonymous expert rounds with a facilitator; avoids bias and groupthink.

Voting rules

Unanimity · majority (>50%) · plurality (largest block) · autocratic.

Multicriteria analysis

Weighted scoring matrix to compare options objectively.

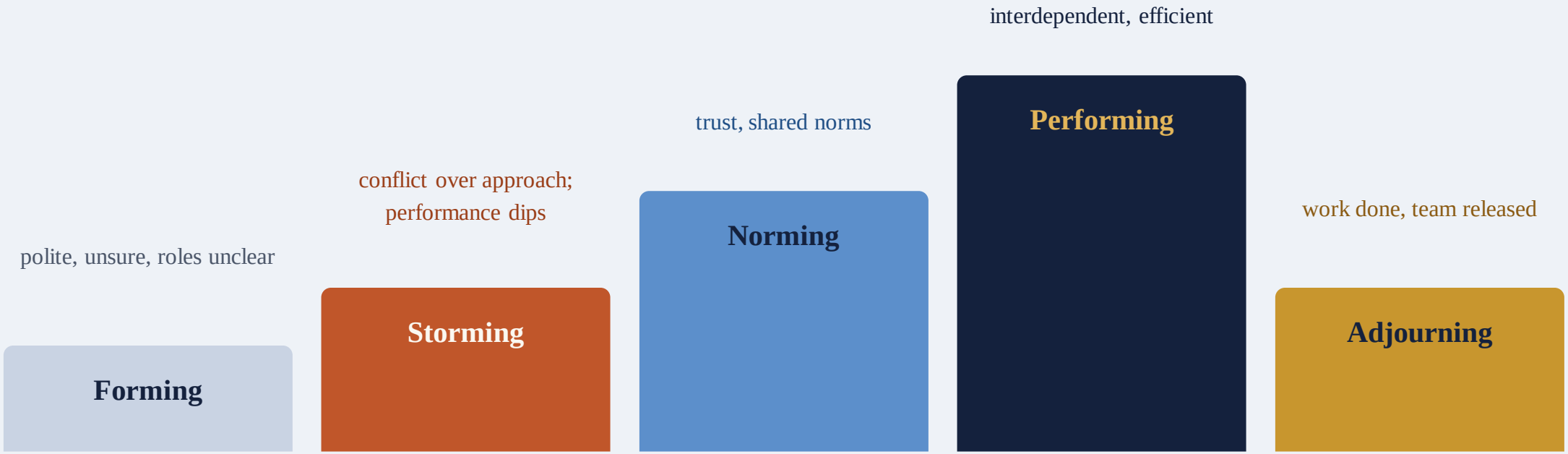
Nominal group

Brainstorm, then rank ideas by voting; balances voices.

Exam tip: in agile, prefer participatory techniques that build commitment; in the RACI, the single “A” makes the final call when consensus fails.

ANNOTATED FIGURE · TUCKMAN LADDER

How teams develop



MNEMONIC · "Fresh Students Need Practice Always"

Forming · Storming · Norming · Performing · Adjourning.

EXAM TRAP

Adding or losing a member can send a team back to forming. Stages can stall or repeat.

HIGH-PERFORMING TEAMS

Psychological safety and team health



Vertical axis: accountability / performance standards

Goal: high safety *and* high standards = learning zone.

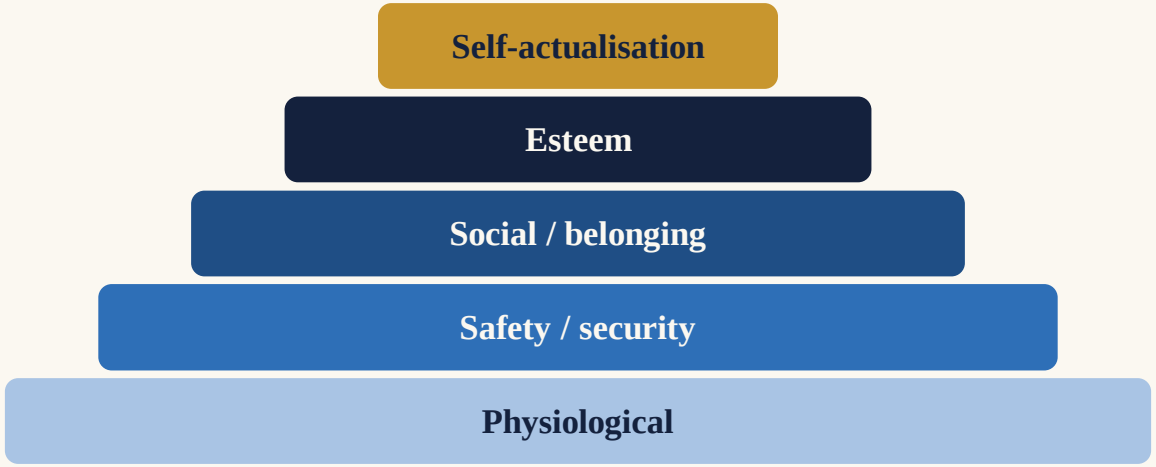
Signals: people admit mistakes, ask questions, challenge ideas.

Leader moves: model fallibility, thank bad news, blameless retros.

Exam tip: safety does not lower accountability: clear ownership (RACI) plus safety gives the learning zone.

MOTIVATION THEORY 1 · MASLOW

Hierarchy of needs



Rule: lower needs must be met before higher needs motivate.

On projects: job security fears block engagement; recognition feeds esteem; mastery feeds self-actualisation.

Mnemonic: “Please Stop Sending Emails, Sam” — Physiological, Safety, Social, Esteem, Self-actualisation (bottom up).

MOTIVATION THEORY 2 · HERZBERG

Two-factor theory

Hygiene factors

Prevent dissatisfaction · do **not** motivate

- Salary and benefits
- Working conditions
- Company policy
- Supervision
- Job security
- Relationships with peers

Motivators

Create real, lasting motivation

- Achievement
- Recognition
- The work itself
- Responsibility
- Advancement
- Personal growth

Exam tip: a pay rise removes complaints but does not create lasting motivation. Responsibility and recognition do. **Mnemonic:** “Hygiene Holds, Motivators Move.”

MOTIVATION THEORY 3 · MCGREGOR AND OUCHI

Theory X, Theory Y, Theory Z

X

People avoid work

Need close supervision, control and threats. Micromanagement. Leads to resentment.

Y

People seek work

Self-directed and creative when committed. Trust and delegate. Fits agile.

Z

Loyalty and wellbeing

Ouchi: long-term employment, collective decisions, care for staff and family.

Mnemonic: “X = eXternal control; Y = Yes, I want to; Z = Zen-like loyalty.” The exam favours Theory Y behaviour from the PM.

MOTIVATION THEORIES 4–6

McClelland, Vroom and Pink

McClelland · acquired needs

Achievement → challenging goals, feedback
Affiliation → team roles, harmony
Power → leadership, influence

Vroom · expectancy

Motivation = **Expectancy** (effort → performance) × **Instrumentality** (performance → reward) × **Valence** (value of reward). Any zero ⇒ zero.

Pink · Drive

For knowledge work, intrinsic motivators win:
Autonomy
Mastery
Purpose

MNEMONICS

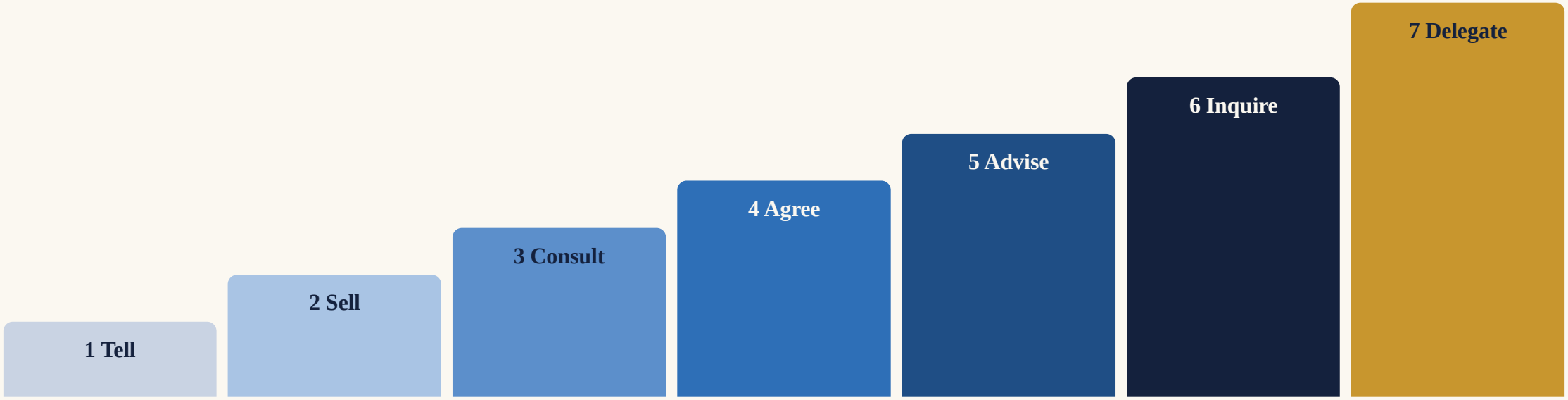
McClelland "AAP" · Vroom "E × I × V" · Pink "AMP" (Autonomy, Mastery, Purpose).

EXAM TIP

An unattainable bonus target kills motivation: expectancy is zero.

EMPOWERING THE TEAM

The delegation ladder



From Management 3.0 delegation poker: agree per decision area how much authority the team holds, then record it.

Mnemonic: “Tired Students Can Always Answer In December.”

DISTRIBUTED AND DIVERSE TEAMS

Leading across distance and difference

Challenges

Time zones · language · culture · isolation · lost tacit knowledge · tool overload · trust takes longer.

Remedies

Overlap hours · virtual co-location (always-on video, shared boards) · explicit working agreements · rotating meeting times · kick-off in person if possible.

Diversity, equity, inclusion

ECO enabler: “support the team’s varied experiences, skills, and perspectives.” Diverse teams decide better when every voice is heard.

Cultural awareness

Power distance, directness, attitudes to time and hierarchy differ. Ask, observe, adapt; never assume.

Exam tip: when a remote member is disengaged, first reach out one-to-one to understand why, then adjust communication or working agreements.

Assess, reward, develop

Assess

Team KPIs, 360 feedback, retrospectives, skills matrix, velocity stability.

Recognise

Timely, specific, fair. Prefer team-based over win–lose individual rewards.

Develop

Training, pairing, T-shaped skills, stretch assignments, mentoring.

T-shaped people

Deep in one specialism (the stem), broad across others (the bar). Generalising specialists reduce bottlenecks.

Exam trap

“Employee of the month” rewards in a collaborative agile team can damage teamwork. Reward the team.

PEOPLE TASK 4 · ENGAGE STAKEHOLDERS

The stakeholder engagement cycle

1
Identify
Early and continuously

→

2
Analyse
Power, interest, attitude

→

3
Plan
Tailor communication

→

4
Engage
Execute the plan

→

5
Monitor
Re-assess, adjust

Stakeholder: anyone who may affect, be affected by, or perceive themselves affected by the project, its decisions or outcomes.

Build trust and influence: ECO enabler. Consistency, transparency and delivering on promises create influence.

Mnemonic "I Always Plan Every Meeting": Identify · Analyse · Plan · Engage · Monitor.

ARTIFACT · STAKEHOLDER REGISTER

What a good register records

Name / role	Interest	Power	Interest level	Current	Desired	Strategy
[Sponsor]	ROI, timing	High	High	Supportive	Leading	Manage closely
[Ops manager]	Smooth handover	High	Low	Neutral	Supportive	Keep satisfied
[End users]	Ease of use	Low	High	Resistant	Supportive	Keep informed
[Regulator]	Compliance	High	Medium	Unaware	Neutral	Keep satisfied
[Facilities]	Space, access	Low	Low	Neutral	Neutral	Monitor

Typical fields: identification (name, role, contact), assessment (requirements, expectations, influence), classification (internal/external, category).

Sensitive: attitude ratings like “resistant” may need restricted access. Share judiciously.

ANNOTATED FIGURE · POWER / INTEREST GRID

Four strategies on one grid



MNEMONIC

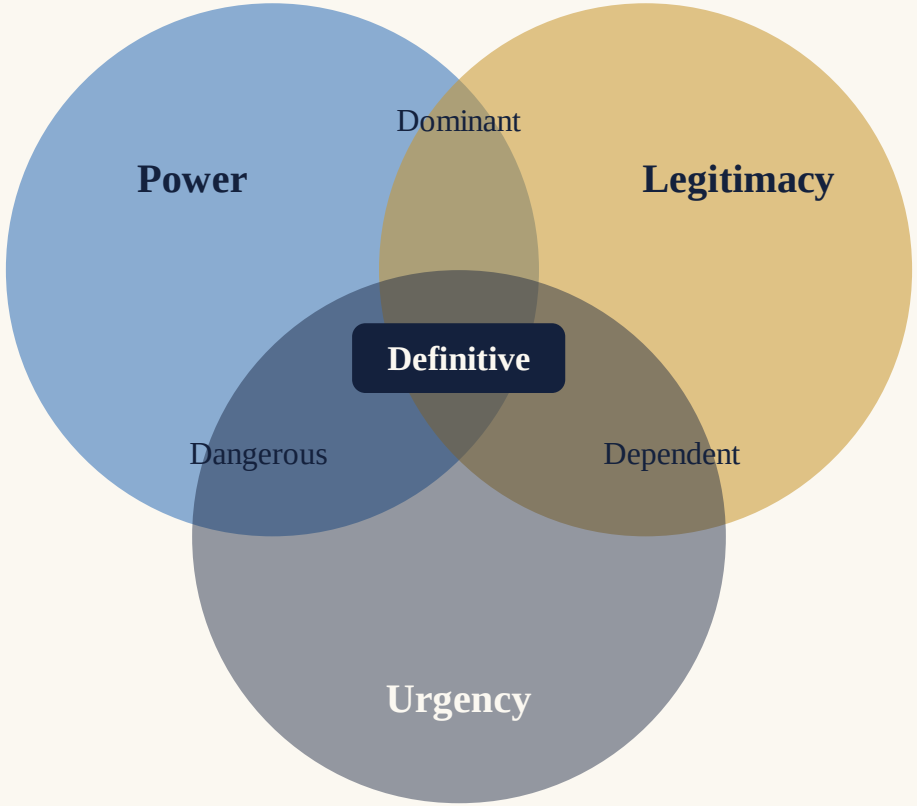
"High power? Satisfy. Both high? Manage. Only interest? Inform. Neither? Monitor."

Variants: power/influence, influence/impact, and the 3-D stakeholder cube.

Likely as an **enhanced-matching** item: drag each stakeholder into a quadrant.

ANNOTATED FIGURE · SALIENCE MODEL

Power, legitimacy and urgency



One attribute: dormant (P), discretionary (L), demanding (U) — latent.

Two: dominant (P+L), dangerous (P+U), dependent (L+U) — expectant.

All three: definitive — highest priority.

Best for large, complex communities where a 2×2 grid is too coarse.

STAKEHOLDER ENGAGEMENT ASSESSMENT MATRIX

Current (C) versus desired (D)

Stakeholder	Unaware	Resistant	Neutral	Supportive	Leading
[Sponsor]				C	D
[End users]		C		D	
[Regulator]	C		D		
[Ops manager]			C	D	
[Facilities]			C D		

MNEMONIC · "Under Rain, Nobody Stays Long"

Unaware · Resistant · Neutral · Supportive · Leading.

READ THE GAP

Distance C → D = engagement effort. Biggest gap here: end users.

STAKEHOLDER ENGAGEMENT PLAN

Turning analysis into action

Strategies

Per stakeholder or group: workshops, demos, one-to-ones, steering briefings.

Information needs

What, how often, in what format, from whom — the bridge to the communications plan.

Change handling

How new stakeholders are added; how shifting attitudes trigger re-planning.

Relationships

Interrelationships and coalitions among stakeholders.

Metrics

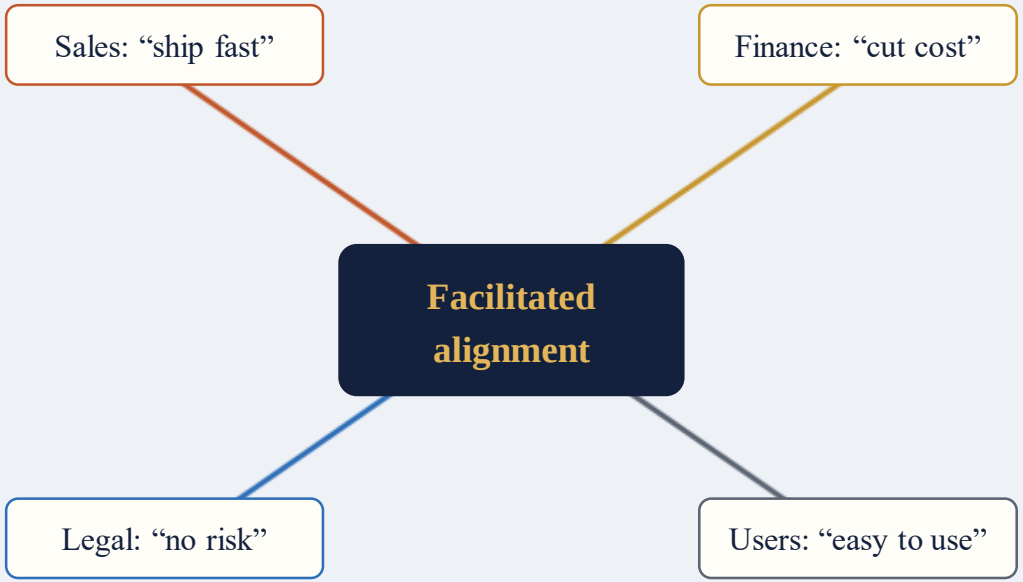
Satisfaction scores, attendance, sentiment, adoption.

Optimise alignment

ECO: align stakeholder needs, expectations and project objectives.

Exam trap: this plan may contain sensitive judgements; it is not always distributed to every stakeholder.

From conflicting asks to shared priorities



1 Categorise stakeholders (role, need, influence).

2 Identify expectations, explicit and implicit.

3 Facilitate joint discussion against objectives and value.

4 Mentor stakeholders new to the approach.

“CIFM”: Categorise, Identify, Facilitate, Mentor.

MENTORING, COACHING, TRAINING

Three ways to grow people

Dimension	Mentoring	Coaching	Training
Focus	Career, long-term growth	Performance, specific goal	Defined skill or knowledge
Who leads	Experienced mentor shares	Coach asks, coachee finds	Instructor teaches
Duration	Months to years	Weeks to months	Hours to days
Style	Advice, role model	Powerful questions	Structured content
Project use	Stakeholder new to agile	Team adopting Scrum	New tool roll-out

ECO: “Organise and act on mentoring opportunities” sits in People Task 5 (align expectations).

Mnemonic: “Mentors Model, Coaches Question, Trainers Teach.”

PEOPLE TASK 6 · MANAGE STAKEHOLDER EXPECTATIONS

Keep satisfaction measured, not assumed

Identify internal and external customer expectations

Align and maintain outcomes to those expectations

Monitor satisfaction and respond as needed

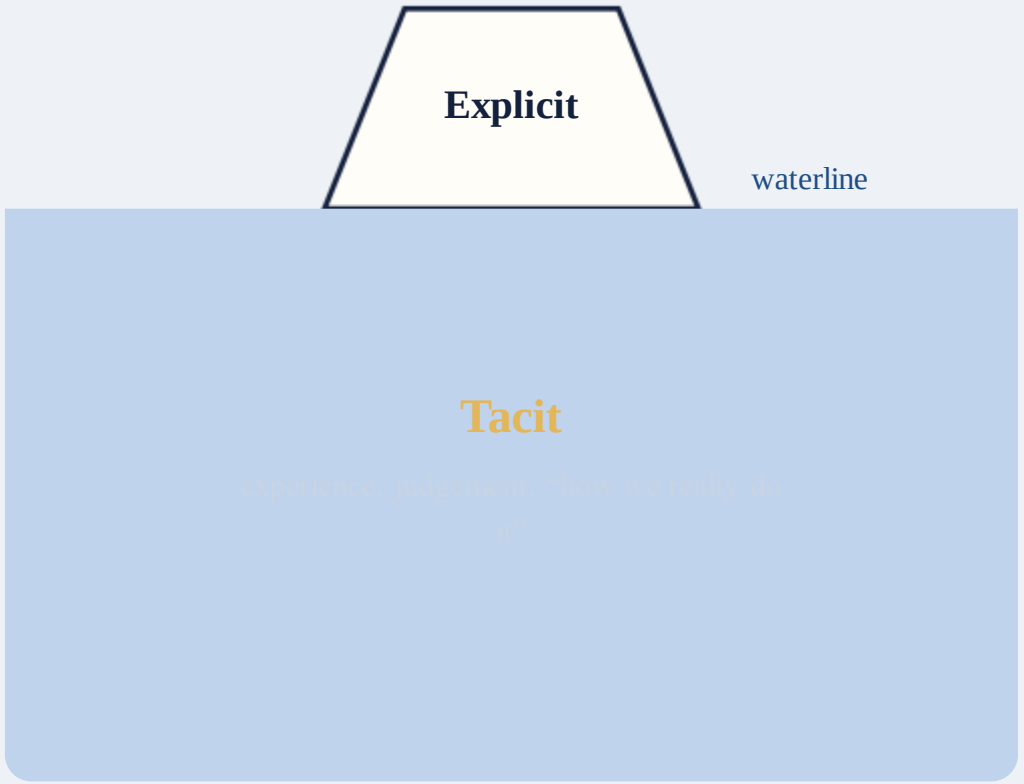
Measurement tools

Net Promoter Score · CSAT surveys · sprint review feedback · mood marbles / happiness index · adoption analytics · acceptance-criteria pass rates.

Exam tip: under-promise and over-communicate; surprises destroy trust faster than bad news.

Exam trap: pleasing a stakeholder with an unapproved extra feature is gold plating, not expectation management.

The knowledge iceberg



Explicit
Codified: documents, databases, manuals. Transfer by sharing artifacts.

Tacit
In people's heads. Transfer by interaction: pairing, shadowing, storytelling, communities of practice.

Mnemonic: "Explicit is Exported; Tacit needs Talking."

KNOWLEDGE MANAGEMENT TOOLKIT

Tools matched to knowledge type

For explicit knowledge

Lessons-learned register · wiki and knowledge base · decision log · runbooks · templates (OPAs) · searchable repositories.

For tacit knowledge

Pair / mob programming · job shadowing · communities of practice · storytelling · knowledge cafés · workshops · networking.

When to capture

Continuously, not only at closure: after each sprint, milestone, incident or decision.

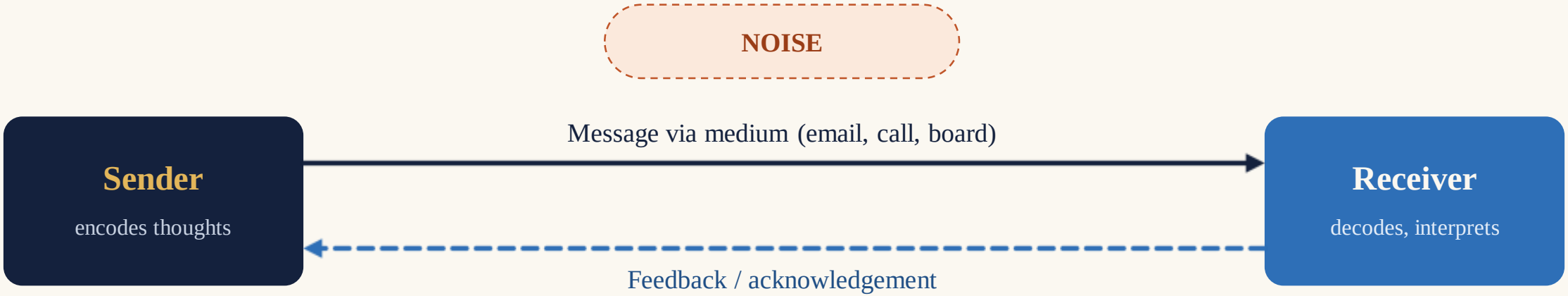
AI assist

AI can summarise meetings and search past lessons, but humans must verify accuracy and protect confidential data.

Exam tip: a key expert is about to leave? Arrange structured handover and pairing now; documentation alone will not capture tacit knowledge.

PEOPLE TASK 8 · ANNOTATED COMMUNICATION MODEL

Sender, message, noise, receiver



Noise = distance, language, culture, jargon, emotion, technology issues, information overload

Sender's job: make it clear and complete, then confirm understanding.

Receiver's job: receive fully, acknowledge, respond.

Note: acknowledgement ≠ agreement.

COMMUNICATION METHODS

Interactive, push, pull

Interactive

Real-time, multi-directional: meetings, calls, video, stand-ups. **Most effective** for understanding and conflict.

Push

Sent to specific recipients: email, memo, report, voicemail. Sent ≠ understood.

Pull

Recipients access at will: intranet, wiki, dashboard, knowledge base. Large audiences.

Formal written: contracts, plans, notices. **Informal verbal:** hallway chats. Choose by risk and audience.

Mnemonic "I-P-P": Interactive = talk; Push = send; Pull = fetch.

FORMULA · COMMUNICATION CHANNELS

Channels grow with the square of team size

FORMULA

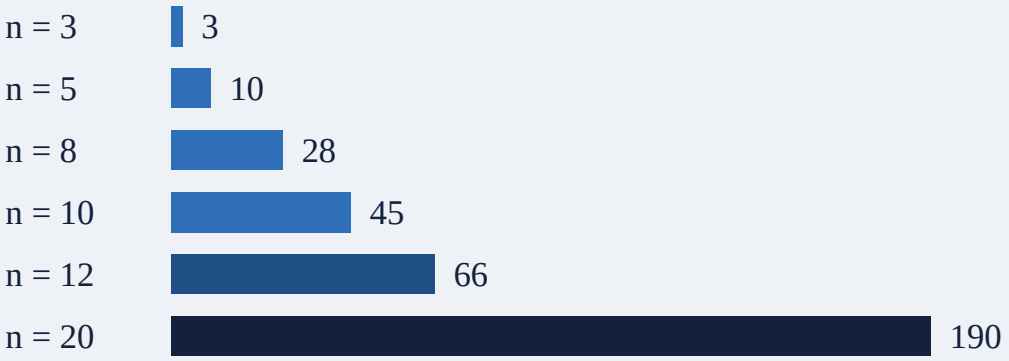
$$n\,(n-1)\,/\,2$$

n = number of people, **including you**

Worked example: team grows from 8 to 12.

$8 \times 7 / 2 = 28 \cdot 12 \times 11 / 2 = 66$

Added channels = $66 - 28 =$ **38**



Bars to scale (3 px per channel)

Exam trap: read whether the question asks for total channels or *additional* channels, and whether the PM is counted.

COMMUNICATIONS MANAGEMENT PLAN

Who gets what, when, how and from whom

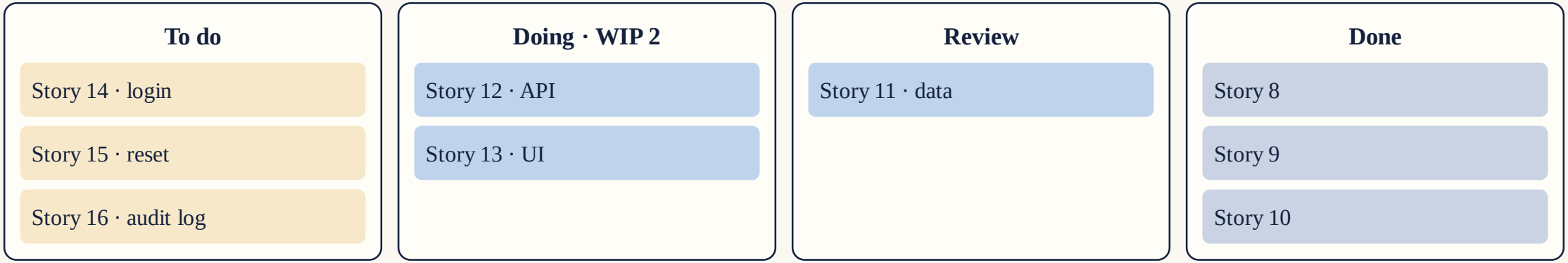
Communication	Audience	Frequency	Method	Owner	Format
Status report	Sponsor, steering	Fortnightly	Push	PM	RAG + EVM
Daily stand-up	Team	Daily	Interactive	Team	15 min
Sprint review	Stakeholders	Each sprint	Interactive	PO	Live demo
Project dashboard	All staff	Live	Pull	PMO	Web
Risk escalation	Sponsor	On threshold	Interactive	Risk owner	Call + memo
Change decision	Affected parties	Per CR	Push	PM	Formal notice

Link to RACI: the “C” and “I” columns tell you who must be consulted or informed: a ready-made audience list.

“5 W + H”: What, Who, When, Why, Which channel, How.

TRANSPARENCY · INFORMATION RADIATORS

Make work visible to everyone



Examples: task/Kanban boards, burndown and burnup charts, impediment boards, release roadmaps, risk boards.

Purpose: highly visible, low-ceremony, current. Replaces many status emails (big visible charts).

MODULE FOUR · DOMAIN II · 41%

Process: plan, deliver, measure

04

Integration, scope, value, resources, procurement, finance and EVM, quality, schedule, status, agile frameworks and closure.

MODULE MAP · PROCESS

Ten tasks, one integrated plan

TASK 1 · THE HUB

Integrated PM plan

Every other task feeds and is governed by it.

Task 2
Scope

Task 3
Value delivery

Task 4
Resources

Task 5
Procurement

Task 6
Finance · EVM

Task 7
Quality

Task 8
Schedule

Task 9
Status · metrics

Task 10
Closure

Route through this module: integration → scope → value → resources → procurement → finance & EVM → quality → schedule → status → agile frameworks → closure.

PRE-PROJECT BUSINESS DOCUMENTS

Business case and benefits plan

Business case

Answers: *why* invest?

- Business need and problem / opportunity
- Options analysed (incl. do nothing)
- Recommended option
- Costs, benefits, risks
- Financial measures (NPV, IRR, payback)

Benefits management plan

Answers: *how* will value be realised?

- Target benefits and metrics
- Timeframe (often post-project)
- Benefits owner
- Strategic alignment
- Assumptions and risks

Owner: the sponsor. Neither is a PM-owned project document, but the PM keeps both in view.

ECO: "Examine the business value throughout the project" and "verify a measurement system is in place to track benefits."

ANNOTATED ARTIFACT · PROJECT CHARTER

The birth certificate of the project

PROJECT CHARTER · [Project name]

Purpose and justification · Measurable objectives and success criteria · High-level requirements and scope · Key deliverables · Summary milestones · Summary budget · High-level risks · Key stakeholders · Assumptions and constraints · Exit criteria · **Assigned PM and authority level**

Approved by: **[Sponsor]** Date: [__]

1 · Authorises the project formally

2 · Names the PM and their authority

3 · Links the project to strategy

Trap: the sponsor (or initiator) signs it, not the PM. No charter? Do not start work.

PROCESS TASK 1 · INTEGRATED PROJECT MANAGEMENT PLAN

Anatomy of the PM plan

3 BASELINES

- **Scope baseline**
  scope statement + WBS + WBS dictionary
- **Schedule baseline**
- **Cost baseline**

Together = performance measurement baseline (PMB)

SUBSIDIARY PLANS

Scope	Requirements	Schedule
Cost / finance	Quality	Resource
Communications	Risk	Procurement
Stakeholder	Change mgmt	Configuration

Plus: life cycle description, development approach, performance reviews.

Trap: once baselined, the plan changes only through change control. Project *documents* (logs, registers) update freely.

PROCESS TASK 1 · EXECUTION STRATEGY

From approach to execution strategy

Assess
Needs, complexity, magnitude, critical info (e.g. sustainability)

→

Recommend approach
Predictive, adaptive or hybrid

→

Recommend strategy
Phasing, releases, in-house vs vendor, team structure

→

Integrate
One plan; dependencies and gaps checked

Complexity signals

Many interdependencies, novel technology, many stakeholders, regulatory scrutiny, distributed teams, ambiguity.

Delivery strategy options

Big bang vs phased roll-out · pilot first · MVP then scale · parallel run · vendor-led build.

Mnemonic "A-R-R-I": Assess, Recommend approach, Recommend strategy, Integrate.

PROCESS TASK 1 · DATA-INFORMED DECISIONS

Collect and analyse before you decide

Alternatives analysis

Compare options to choose an approach, resource or design.

Cost–benefit analysis

Is the benefit worth the cost? Basis for change and response decisions.

Root cause analysis

Five whys, fishbone: fix causes, not symptoms.

Trend analysis

Is performance improving or degrading over time?

Variance analysis

Planned vs actual: cause, size and action.

What-if / simulation

Scenario and Monte Carlo analysis of outcomes.

Mnemonic "ACRTVW": "All Clever Researchers Trust Verified Work."

AI angle: AI can surface patterns, but validate data quality and keep a human accountable.

PROCESS TASK 2 · COLLECT REQUIREMENTS

Elicitation techniques

Interviews
One-to-one, deep, confidential

Focus groups
Moderated, pre-qualified users

Workshops
JAD, QFD: cross-functional

Questionnaires
Large, dispersed audiences

Observation
Job shadowing; finds hidden needs

Prototypes
Storyboards, mock-ups, early feedback

Benchmarking
Compare to best practice

Context diagram
System and its actors

Requirement types: business · stakeholder · solution (functional, non-functional) · transition · project · quality.

Trap: requirements must be measurable, testable, traceable, complete, consistent and acceptable to key stakeholders.

ARTIFACT · REQUIREMENTS TRACEABILITY MATRIX

Trace every requirement end to end

ID	Requirement	Business objective	WBS ref	Test case	Owner	Status
R-01	Login with MFA	Reduce fraud	1.2.1	TC-14	[Lead]	Done
R-02	Export to PDF	Audit readiness	1.3.2	TC-22	[Lead]	Active
R-03	WCAG accessibility	Compliance	1.4.0	TC-31	[QA]	Active
R-04	Energy-efficient hosting	Sustainability	2.1.3	TC-40	[Ops]	Planned



Why it matters: prevents scope creep (no requirement without an objective) and gaps (no requirement without a test).

PROCESS TASK 2 · DEFINE SCOPE

The project scope statement

Product scope description
Features and functions of the product, service or result.

Deliverables
Unique, verifiable outputs, including project management reports.

Acceptance criteria
Conditions that must be met before deliverables are accepted.

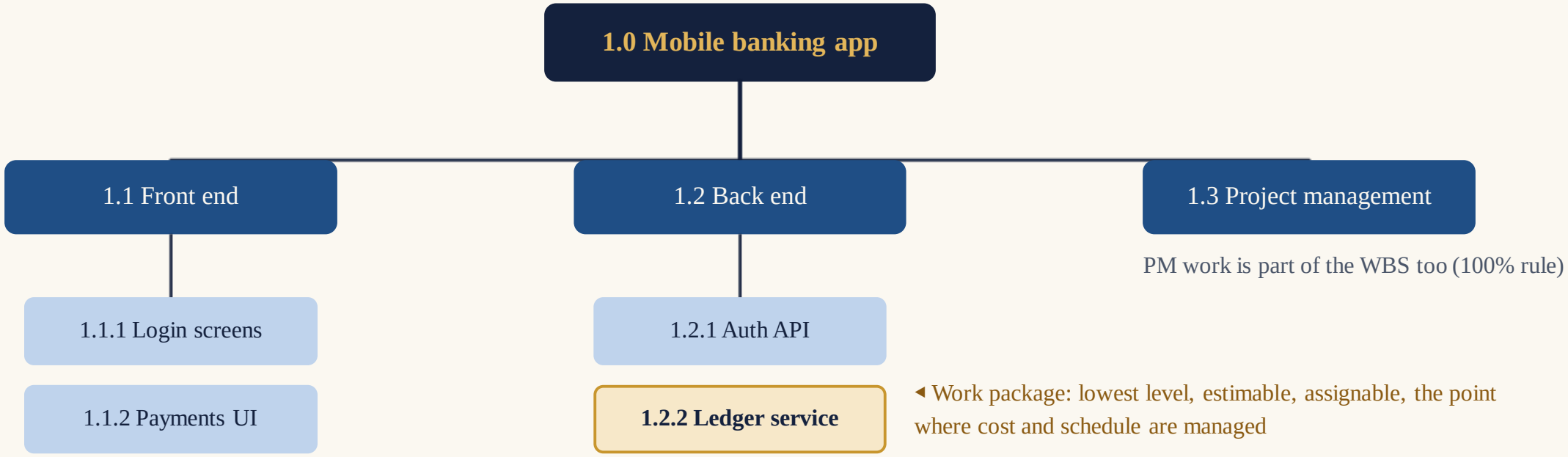
Exclusions
What is explicitly out of scope. The best scope-creep defence.

Mnemonic "PDAE": "Please Define All Exclusions." Plus assumptions and constraints.

ECO: "Obtain stakeholder agreement on project scope." Agreement comes before decomposition.

ANNOTATED FIGURE · WORK BREAKDOWN STRUCTURE

Decompose deliverables, not activities



Level 1 = project · Level 2 = major deliverables / phases · Level 3+ = components → work packages. Code of accounts (1.2.2) gives each element a unique ID.

Rules that make a WBS work

100% rule
The WBS captures all the work, and only the work. Children sum to exactly 100% of the parent.

8/80 heuristic
A work package is roughly 8 to 80 hours of effort, or fits one reporting period.

Built by the team
Created with the people who do the work; drives buy-in and completeness.

Rolling wave
Near-term work decomposed in detail; far-term kept as planning packages.

WBS dictionary entry: code of account · description of work · assumptions and constraints · responsible organisation (→ RAM!) · schedule milestones · associated activities · resources required · cost estimates · quality requirements · acceptance criteria · technical references · agreement information.

CRITICAL DISTINCTION

Control quality first, then validate scope



Aspect	Control Quality	Validate Scope
Concern	Correctness against quality requirements	Acceptance against scope / acceptance criteria
Who	Project team / QC staff	Customer or sponsor
Agile equivalent	Definition of Done, testing	Sprint review, PO acceptance

Mnemonic: “Q before V — the team Checks, the customer Accepts.”

SCOPE CONTROL

Scope creep versus gold plating

Scope creep

Uncontrolled expansion of scope **requested by stakeholders** without adjustments to time, cost and resources.

Cause: weak change control, vague requirements.

Gold plating

Extra features **added by the team** that the customer did not request, hoping to please.

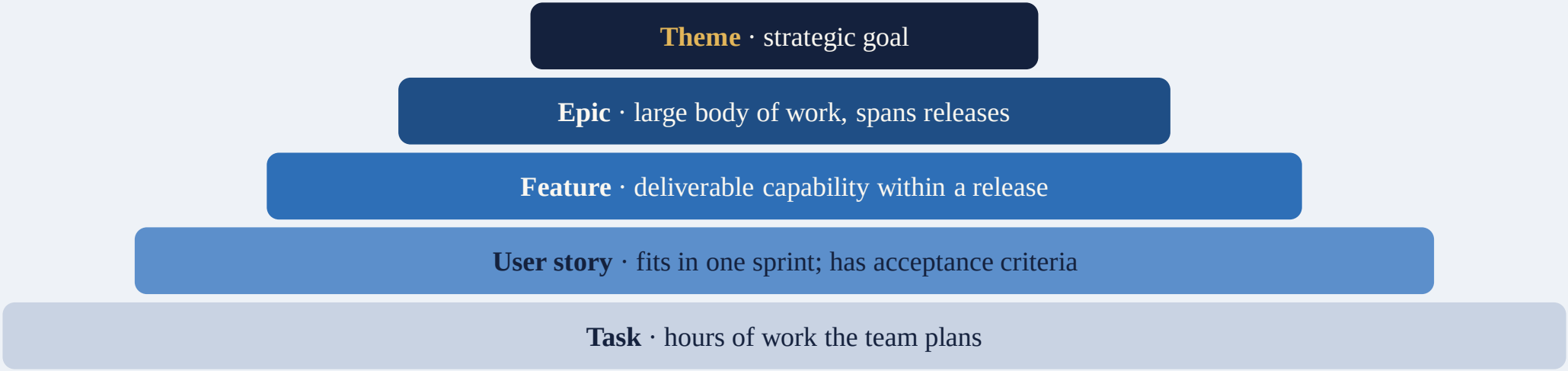
Cause: enthusiasm, poor discipline.

PMI view: both are undesirable. Deliver what was agreed; route new ideas through change control or the backlog.

Mnemonic: “Creep comes from Customers; Gold comes from the Group.”

AGILE SCOPE · THE BACKLOG HIERARCHY

Theme → epic → feature → story → task



Story format: As a [role], I want [capability] so that [benefit].
Acceptance: Given [context], when [action], then [result].

INVEST: Independent, Negotiable, Valuable, Estimable, Small, Testable. **3 Cs:** Card, Conversation, Confirmation.

PROCESS TASK 3 · HELP ENSURE VALUE-BASED DELIVERY

Six enablers of value delivery

1

Identify value components with key stakeholders

2

Prioritise work by value and stakeholder feedback

3

Assess opportunities to deliver value incrementally

4

Examine business value throughout the project

5

Verify a measurement system tracks benefits

6

Evaluate delivery options to demonstrate value

Mnemonic "I Prefer Incremental Examination, Measured & Evaluated" — Identify, Prioritise, Incremental, Examine, Measure, Evaluate.

PRIORITISATION · MOSCOW AND FRIENDS

Deciding what comes first

M

Must have

Non-negotiable; without it the release fails.

S

Should have

Important, but a workaround exists.

C

Could have

Nice to have, if time allows.

W

Won’t have (now)

Agreed out for this release.

100-point method
Each stakeholder distributes 100 points.

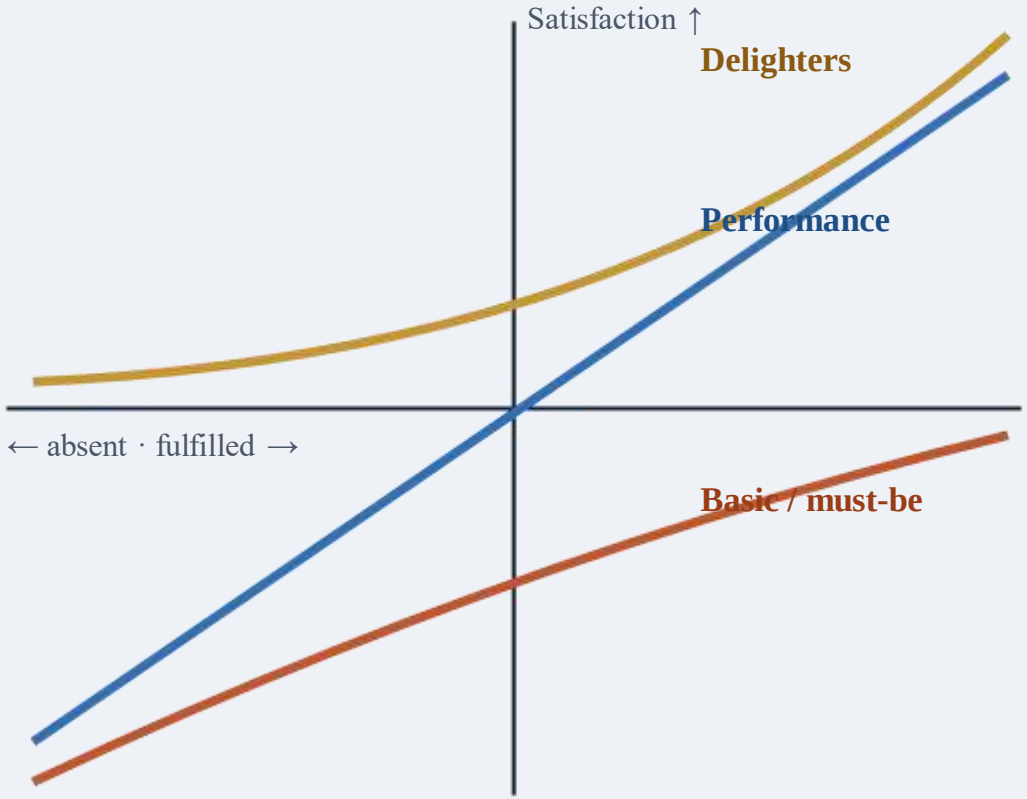
Dot voting
Limited dots placed on preferred items.

WSJF
Cost of delay ÷ job size (SAFe).

Exam tip: the product owner owns backlog order; the team estimates. Value, risk, dependencies and cost of delay all inform order.

ANNOTATED FIGURE · KANO MODEL

Not all features delight equally



Basic: expected; absence angers, presence goes unnoticed (brakes on a car).

Performance: more is better, linearly (battery life).

Delighters: unexpected; absence costs nothing, presence excites.

"B-P-D": Basics first, Performance next, Delighters last. Delighters decay into basics over time.

INCREMENTAL VALUE · MVP AND MMF

Build to learn, then build to earn



Iteration 1
usable, simple

Iteration 2
feedback applied

Iteration 3
market-ready

Iteration 4
full value

MVP · minimum viable product
Smallest thing that validates a hypothesis with real users. Purpose: **learn**.

MMF · minimum marketable feature
Smallest unit that delivers value customers will pay for or use. Purpose: **earn**.

MEASURING VALUE

Benefits measurement system

Benefit	Metric (KPI)	Baseline	Target	When	Owner
Faster onboarding	Days to activate	[__]	[__]	Q+2	[Ops lead]
Fewer support calls	Calls / 1,000 users	[__]	[__]	Q+3	[Support]
Lower emissions	kg CO_2e / transaction	[__]	[__]	Y+1	[ESG lead]

Leading indicators

Predict future value: adoption rate, trial sign-ups.

Lagging indicators

Confirm value after the fact: revenue, cost saved.

SMART targets

Specific, Measurable, Achievable, Relevant, Time-bound.

The resource management plan

- **Identification** of team and physical resources
- **Acquisition** approach (internal, external, virtual)
- **Roles & responsibilities**: role, authority, responsibility, competence — shown as a RAM / RACI
- **Project organisation charts** (OBS)
- **Team development, recognition, release**
- **Resource control**: physical resources, inventory, logistics

MNEMONIC · "ARC"

For each role define **A**uthority, **R**esponsibility, **C**ompetence.

ECO ENABLERS

Define and plan resources from requirements; manage and optimise needs versus availability.

Resource breakdown structure (RBS): resources by category and type: labour, material, equipment, supplies.

ORGANISATION CHARTS AND POSITION DESCRIPTIONS

Three formats for documenting roles

Hierarchical



OBS, RBS: reporting lines and units.

Matrix

R	A	C	I

RAM / RACI: work linked to people.

Text-oriented

Position descriptions: detail of each role.

Exam tip: if the question asks for a chart that shows *who does what work*, the answer is a matrix-based chart (RAM/RACI), not a hierarchical OBS.

ACQUIRING RESOURCES

How people join the project

Pre-assignment
Named in the charter or proposal (key expert promised to a client).

Negotiation
With functional managers, other PMs, vendors. Common in matrix structures.

Acquisition
Hiring or contracting externally when skills are not available in-house.

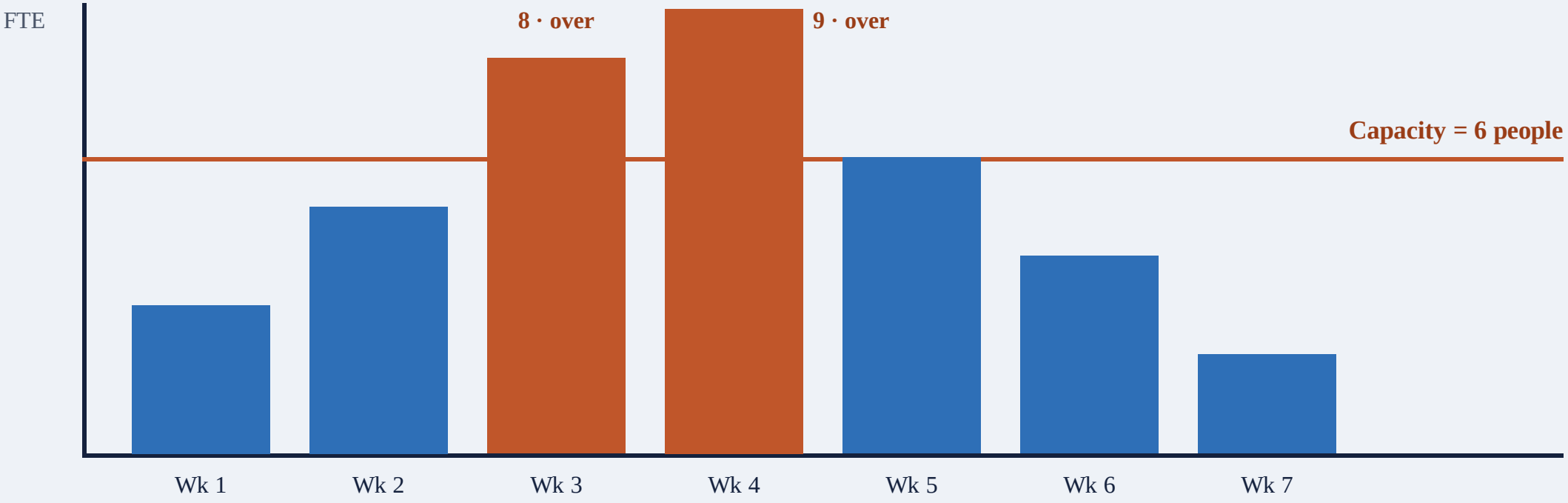
Virtual teams
Expands the talent pool across locations and time zones.

"PNAV": Pre-assign, Negotiate, Acquire, Virtual.

Trap: the assigned person lacks a skill? Train or pair first; replacing them is not step one.

ANNOTATED FIGURE · RESOURCE HISTOGRAM

Spot over-allocation at a glance



Read it: weeks 3–4 exceed capacity (orange, labelled “over”) — level or smooth (next slide), add resources, or re-sequence.

RESOURCE OPTIMISATION

Leveling versus smoothing

Resource leveling

Adjusts start and finish dates to respect resource limits.

Can change the critical path and extend the end date.

Use when resources are fixed or over-allocated.

Resource smoothing

Evens out demand peaks within available float.

Does not change the critical path or end date.

May not resolve every over-allocation.

Mnemonic: “Leveling can Lengthen; Smoothing Stays (inside float).”

PROCESS TASK 5 · PLAN PROCUREMENT

Make-or-buy analysis

Reasons to make

Idle capacity · protect IP · core competency · control quality · build skills.

Reasons to buy

Lack of skill · speed · transfer risk · lower cost · small volumes.

WORKED EXAMPLE · lease or buy

Lease: £200 per day. Buy: £5,000 purchase + £40 per day to run.

Break-even: 200d = 5,000 + 40d → 160d = 5,000 → **d = 31.25 days**

Need it more than ~31 days? Buy. Fewer? Lease.

Exam tip: set the two total-cost equations equal and solve for the break-even quantity.

PROCUREMENT DOCUMENTS

RFI, RFQ, RFP and the SOW

RFI

Request for information

Explore the market and capabilities before defining needs.

RFQ

Request for quotation

Price for a well-defined, standard item. Selection mainly on price.

RFP

Request for proposal

Complex needs; sellers propose solutions. Multi-criteria selection.

Procurement statement of work (SOW): describes the item in enough detail for sellers to decide whether they can provide it. Types: **performance** (what it must achieve), **functional** (purpose / characteristics), **design** (exact specification). A **terms of reference (TOR)** is used for services.

Mnemonic: "Inform → Quote → Propose" = rising complexity. SOW types "PFD": Performance, Functional, Design.

SOURCE SELECTION · WEIGHTED SCORING

Choosing the seller objectively

Criterion	Weight	Vendor A (score / wtd)	Vendor B (score / wtd)	Vendor C (score / wtd)
Technical approach	40%	8 / 3.2	6 / 2.4	9 / 3.6
Price	30%	7 / 2.1	9 / 2.7	5 / 1.5
Past performance	20%	9 / 1.8	7 / 1.4	8 / 1.6
Sustainability	10%	6 / 0.6	5 / 0.5	9 / 0.9
Total	100%	7.7 · winner	7.0	7.6

Other inputs: screening (pass/fail), independent estimates, seller ratings, presentations.

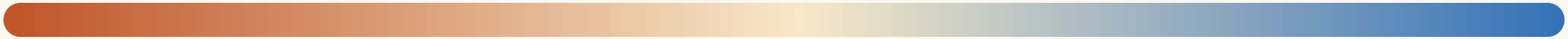
Trap: lowest price is not automatically best value; follow the published criteria.

ANNOTATED FIGURE · CONTRACT TYPES AND RISK

Who carries the cost risk?

← Seller bears most risk

Buyer bears most risk →



FFP	FPIF	FP-EPA	T&M	CPIF	CPAF	CPFF
Firm fixed price. Scope well defined.	Fixed price + incentive; ceiling price.	Economic price adjustment; long, multi-year.	Hybrid: rate × time + materials. Small, urgent.	Cost + incentive fee shared on performance.	Cost + award fee, buyer's subjective judgement.	Cost + fixed fee. Scope unclear.

Mnemonic: "Fixed — the Seller sweats; Cost — the Buyer bleeds; T&M sits in the middle."

Choose by scope clarity: clear → fixed price; unclear → cost-reimbursable; staff aug → T&M.

WORKED EXAMPLE · FIXED PRICE INCENTIVE FEE

How incentive sharing works

Target cost	Target fee	Share ratio	Ceiling price
£100k	**£10k**	**80 / 20** buyer/seller	**£120k**

Under-run · actual £90k

Saving £10k × 20% = £2k extra fee

Fee £12k · Price = 90 + 12 = **£102k**

Over-run · actual £110k

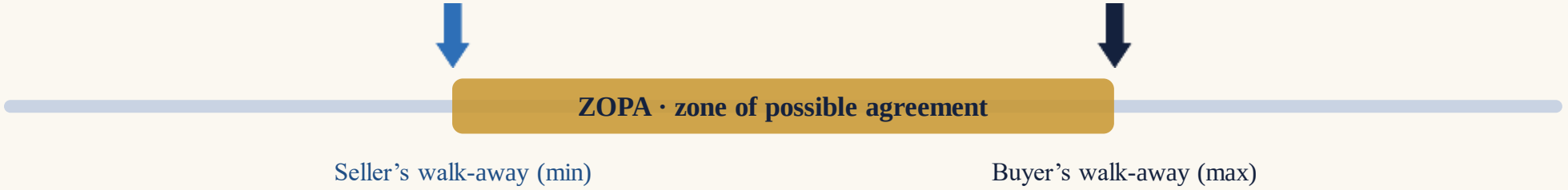
Overrun £10k × 20% = £2k fee loss

Fee £8k · Price = 110 + 8 = **£118k** (< ceiling)

Point of total assumption (PTA) = (Ceiling − Target price) ÷ Buyer share + Target cost = (120 − 110) ÷ 0.8 + 100 = **£112.5k**. Above this cost, the seller absorbs every extra pound.

NEGOTIATION STRATEGY

Negotiate interests, not positions



BATNA
Best alternative to a negotiated agreement: your leverage. Know yours, estimate theirs.

Principled
Separate people from problem; focus on interests; invent options; use objective criteria.

Goal
Fair agreement and a lasting relationship (win–win).

Control procurements

Evaluate performance
Inspections, audits, performance reviews against SLAs and the SOW; EVM on large contracts.

Claims administration
Contested changes. Prefer negotiation; then ADR (mediation, arbitration); litigation last.

Contract change control
Only authorised people (often procurement) change the contract, in writing.

Verify objectives met
Formal acceptance, final payment, closure documentation.

Trap: seller misses a deadline? Check the contract terms first, then act per its remedies; do not threaten informally.

Dispute ladder "N-M-A-L": Negotiate, Mediate, Arbitrate, Litigate.

AGILE AND ADAPTIVE CONTRACTING

Contracts that welcome change

Capped T&M
Pay for time up to a ceiling; buyer gains flexibility, risk bounded.

Graduated fixed price
Higher rate for early delivery, lower for late; shared risk and reward.

Fixed-price increments
Each user story or release is priced; small commitments.

Money for nothing, change for free
Swap equal-size backlog items freely; buyer may stop early with a fee.

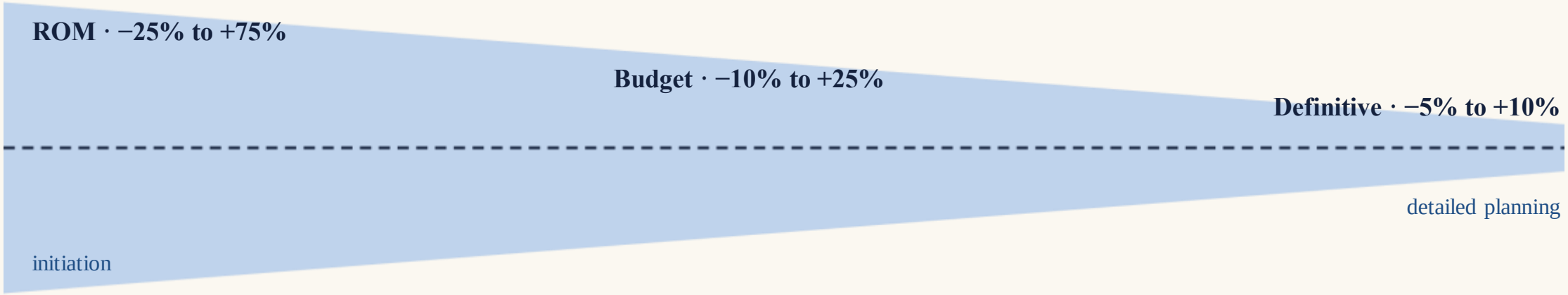
Multi-tiered
Stable terms in a master agreement; changing items in an SOW or schedule.

Team augmentation
Supplier staff join the buyer's team; most collaborative.

> **Agile value:** "Customer collaboration over contract negotiation" — contracts still matter, but they should enable collaboration.

PROCESS TASK 6 · ESTIMATE COSTS

Estimate ranges tighten over time



Analogous
Top-down from similar past project. Fast, least accurate.

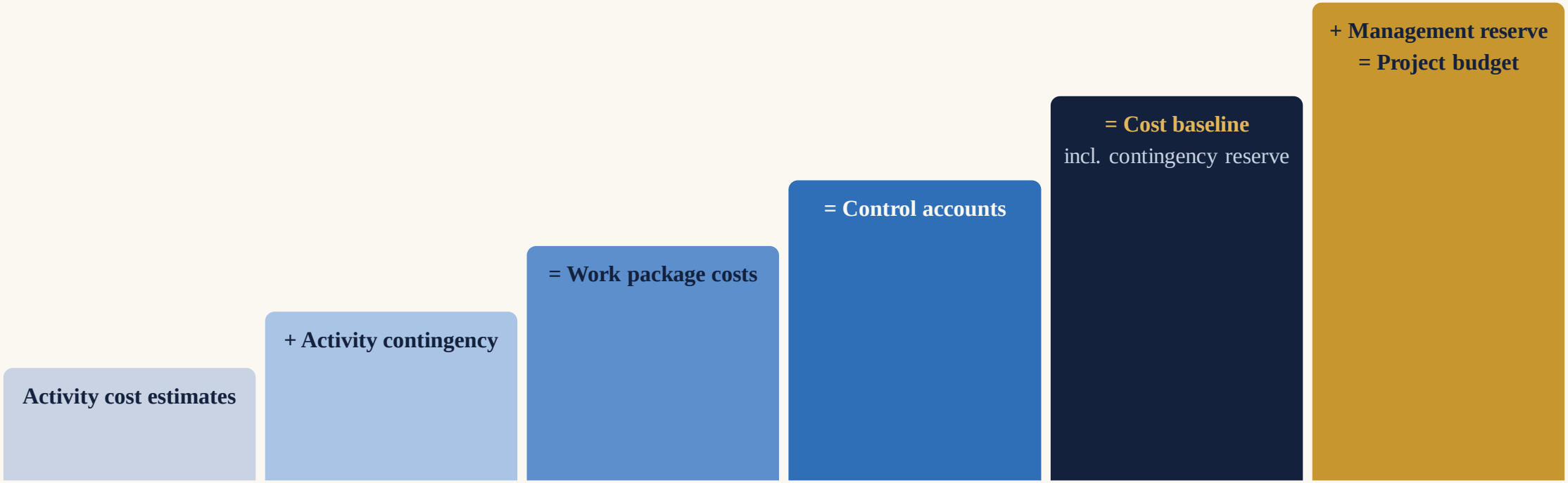
Parametric
Unit rate × quantity (£/m²). Accurate if model is sound.

Bottom-up
Sum work packages. Most accurate, slowest.

Three-point
Optimistic, most likely, pessimistic (PERT).

ANNOTATED FIGURE · BUDGET BUILD-UP

From activity estimates to project budget



Mnemonic "All Clever Workers Count Budgets Monthly": Activity → Contingency → Work package → Control account → Baseline → Management reserve.

RESERVES · CRITICAL DISTINCTION

Contingency versus management reserve

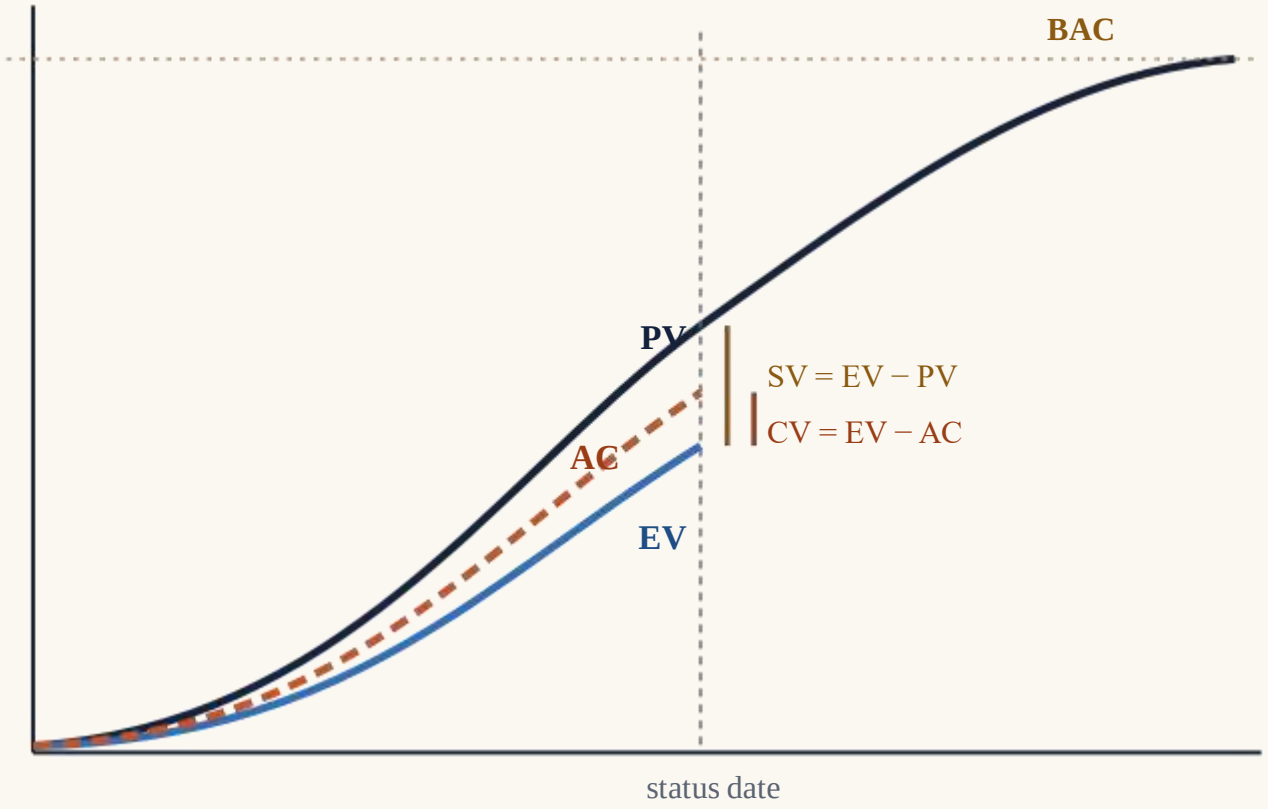
Aspect	Contingency reserve	Management reserve
Covers	Known-unknowns (identified risks)	Unknown-unknowns (unidentified)
In cost baseline?	Yes	No (in budget only)
Who controls	Project manager, per risk plan	Sponsor / management
How sized	Reserve analysis, EMV, Monte Carlo	Percentage per policy
Use requires	Risk occurs; draw down per plan	Change request + approval

Mnemonic: “Contingency is for the Catalogued; Management reserve for the Mystery.”

ECO: “Quantify risk and contingency financial allocations” and “manage financial reserves.”

ANNOTATED FIGURE · EARNED VALUE S-CURVE

PV, EV and AC on one chart



PV · planned value: budgeted cost of work scheduled

EV · earned value: budgeted cost of work performed

AC · actual cost of work performed

Reading: EV below PV → behind schedule; AC above EV → over budget.

EARNED VALUE MANAGEMENT · STATUS FORMULAS

Variances and indices

Measure	Formula	Good	Bad
Cost variance (CV)	EV − AC	Positive · under budget	Negative · over budget
Schedule variance (SV)	EV − PV	Positive · ahead	Negative · behind
Cost perf. index (CPI)	EV ÷ AC	> 1 · under budget	< 1 · over budget
Schedule perf. index (SPI)	EV ÷ PV	> 1 · ahead	< 1 · behind

MNEMONIC · "EV comes first"

Variance = EV **minus**; index = EV **divided by**. **C**ost pairs with **AC**; **S**chedule pairs with **PV** ("Plan Schedule").

RULE OF THUMB

Negative variance or index below 1 = bad news. Positive or above 1 = good news.

EARNED VALUE MANAGEMENT · FORECASTING

EAC, ETC, VAC and TCPI

Use when…	Formula	Meaning
Current CPI will continue	EAC = BAC ÷ CPI	Most common default
Variance was a one-off	EAC = AC + (BAC − EV)	Remaining work at plan rate
Both CPI and SPI matter	EAC = AC + (BAC − EV) ÷ (CPI × SPI)	Fixed deadline pressure
Original plan is invalid	EAC = AC + bottom-up ETC	Re-estimate remaining work
Cost still to go	ETC = EAC − AC	Estimate to complete
Variance at the end	VAC = BAC − EAC	Negative = expected overrun
Efficiency needed to hit BAC	TCPI = (BAC − EV) ÷ (BAC − AC)	> 1 = harder; use EAC in place of BAC once a new EAC is approved

Mnemonic "TCPI = Work left ÷ Money left": (BAC − EV) is work left; (BAC − AC) is money left.

WORKED EXAMPLE · EVM END TO END

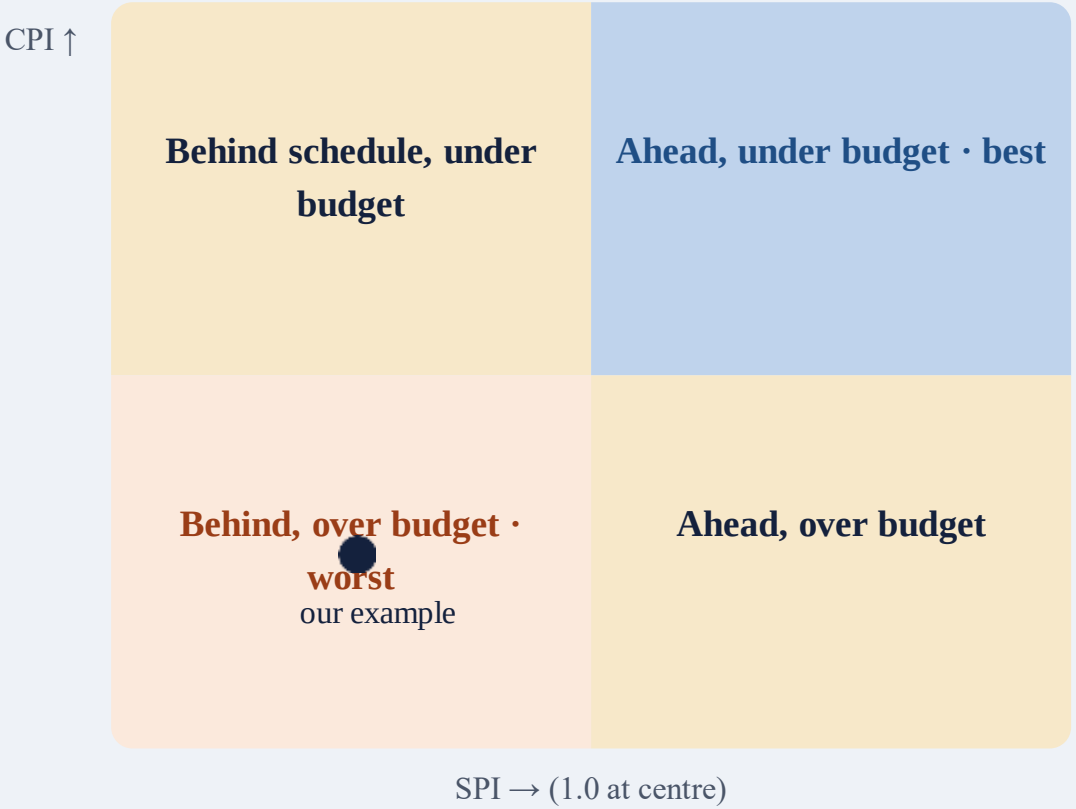
A ten-month, £100k project at month 5

BAC = **£100k**, spent evenly over 10 months. At month 5 the plan was 50% done; the team has completed **40%** and spent **£45k**.

PV = 50% × 100 **£50k**	EV = 40% × 100 **£40k**	AC (given) **£45k**	CV = 40 − 45 **−£5k**
SV = 40 − 50 **−£10k**	CPI = 40 ÷ 45 **0.89**	SPI = 40 ÷ 50 **0.80**	EAC = 100 ÷ 0.889 **£112.5k**
ETC = 112.5 − 45 **£67.5k**	VAC = 100 − 112.5 **−£12.5k**	TCPI = 60 ÷ 55 **1.09**	Verdict **Behind and over budget**

ANNOTATED FIGURE · CPI / SPI QUADRANT

Read project health in one glance



Our example: CPI 0.89, SPI 0.80 → bottom-left quadrant.

Ahead but over budget often means crashing or overtime is burning money.

Behind but under budget often means resources were not yet assigned.

Mnemonic: "Up-right is all right; down-left is all wrong."

Present value and NPV

PRESENT VALUE

$$PV = FV \div (1 + r)^n$$

NET PRESENT VALUE

$$NPV = \Sigma\ PV(\text{inflows}) - \text{investment}$$

Decision: NPV > 0 → accept; choose the **highest** NPV. Years are already built in.

Worked example · r = 10%

Invest £100k now; inflows £45k at end of years 1, 2, 3.

Year 1: 45 ÷ 1.10 = 40.91

Year 2: 45 ÷ 1.21 = 37.19

Year 3: 45 ÷ 1.331 = 33.81

Σ PV = 111.91 → **NPV = +£11.91k** → accept

BCR = 111.91 ÷ 100 = **1.12**

PROJECT SELECTION · OTHER MEASURES

IRR, payback, ROI and BCR

Measure	Definition	Choose	Our example
IRR	Discount rate where NPV = 0	Highest (above hurdle rate)	≈ 16.6%
Payback period	Time to recover the investment	Shortest	100 ÷ 45 ≈ 2.2 years
ROI	(Gain − cost) ÷ cost	Highest	(135 − 100) ÷ 100 = 35% undiscounted
BCR	PV benefits ÷ PV costs	Highest, > 1	1.12
NPV	Σ PV inflows − investment	Highest, > 0	+£11.9k

Mnemonic: “Everything High, except Payback — Payback is Pacey (shortest).”

Trap: simple payback ignores time value of money and cash after payback.

FINANCIAL CONCEPTS YOU MUST RECOGNISE

Sunk cost, opportunity cost, depreciation

Sunk cost

Money already spent and unrecoverable. **Ignore it** when deciding whether to continue.

Opportunity cost

Value of the best alternative given up by choosing another option.

Depreciation

Straight-line: equal each year. Accelerated (double declining, sum-of-years): faster early.

Fixed vs variable

Fixed: same regardless of volume (rent). Variable: scales with output (materials).

Direct vs indirect

Direct: attributable to the project. Indirect: shared overhead.

Life-cycle costing

Total cost of ownership, including operations and disposal.

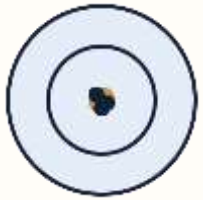
Trap: “We’ve spent £2m, so we must finish” is the sunk-cost fallacy. Decide on future value only.

Quality vs grade · precision vs accuracy

Quality: degree to which characteristics fulfil requirements.

Grade: category for products with the same use but different features.

Low grade can be fine. **Low quality is always a problem.** A basic phone that never fails = low grade, high quality.



Precise + accurate



Precise, not accurate



Accurate, not precise



Neither

COST OF QUALITY (COQ)

Pay now to prevent, or pay later to fix

Cost of conformance

Prevention
Training, documenting processes, equipment, time to do it right.

Appraisal
Testing, inspections, destructive test loss.

Cost of nonconformance

Internal failure
Found by the team: rework, scrap.

External failure
Found by customers: warranty, liability, lost business.

Mnemonic "PAIE": "Prevention And Inspection Earlier" — Prevention, Appraisal | Internal, External. Cost rises left to right.

2026 ECO: "Manage cost of quality (CoQ) and sustainability" — waste and rework have environmental cost too.

THE SEVEN BASIC QUALITY TOOLS

Know each tool’s single job

Cause-and-effect
Fishbone / Ishikawa: root causes of a problem

Flowchart
Process steps; where defects can occur (SIPOC)

Check sheet
Tally sheet to collect defect data

Pareto chart
Ranks causes by frequency: the vital few

Histogram
Distribution of data (shape, spread)

Control chart
Is the process stable over time?

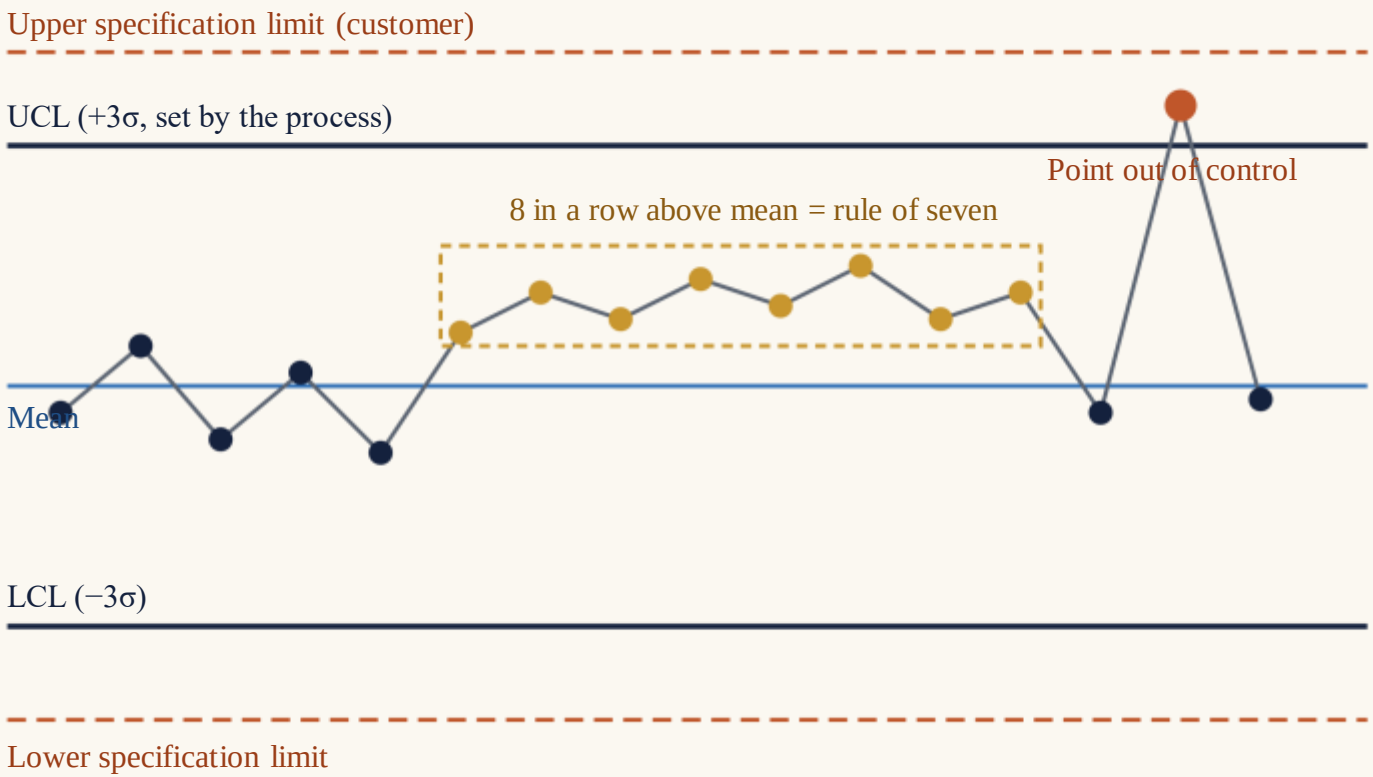
Scatter diagram
Correlation between two variables

Mnemonic
“Cats Find Cheese, Pigs Hate Cold Showers”

Exam tip: “Which tool shows the most frequent defect types?” → Pareto. “Is the process in control?” → control chart. “Do two variables relate?” → scatter.

ANNOTATED FIGURE · CONTROL CHART

Limits, specs and the rule of seven



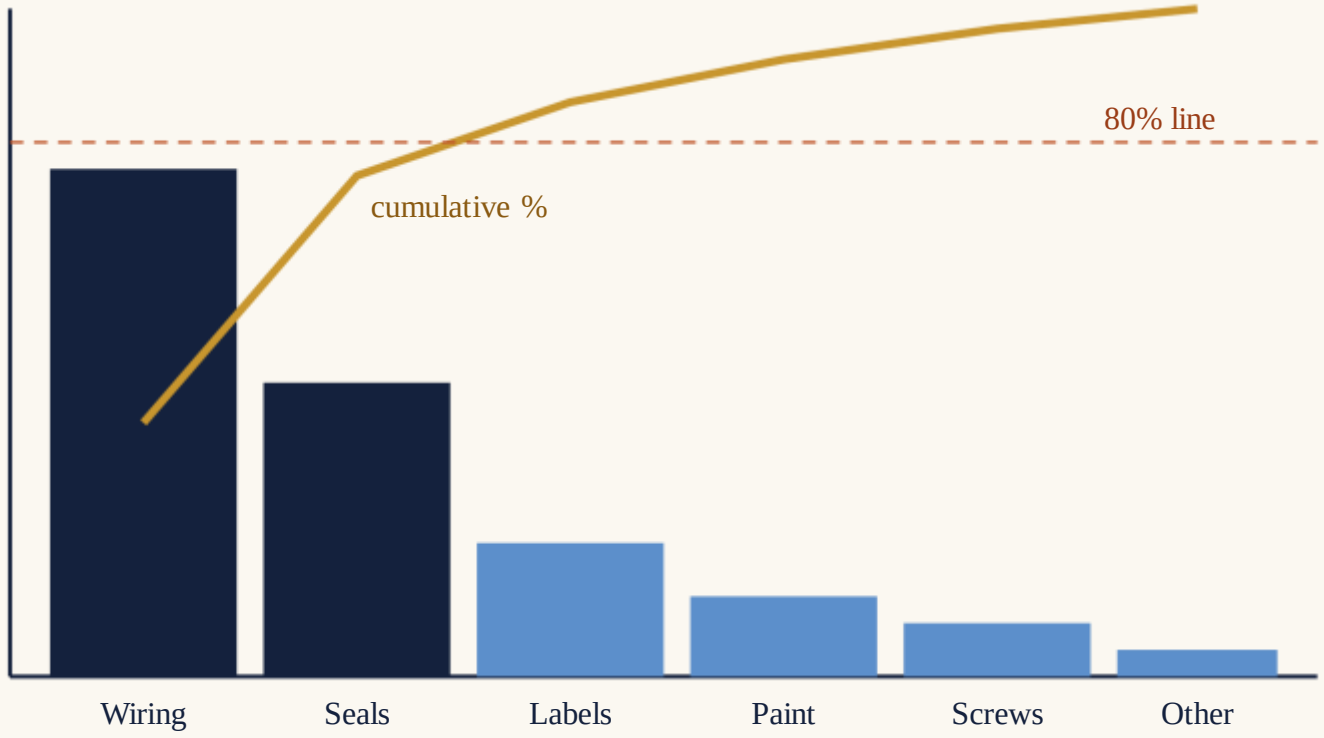
Out of control if: a point beyond a control limit, or 7+ consecutive points on one side of the mean.

Control limits = voice of the process. **Spec limits** = voice of the customer.

Action: investigate assignable (special) cause; don't tamper with common-cause variation.

ANNOTATED FIGURE · PARETO CHART

The vital few versus the useful many



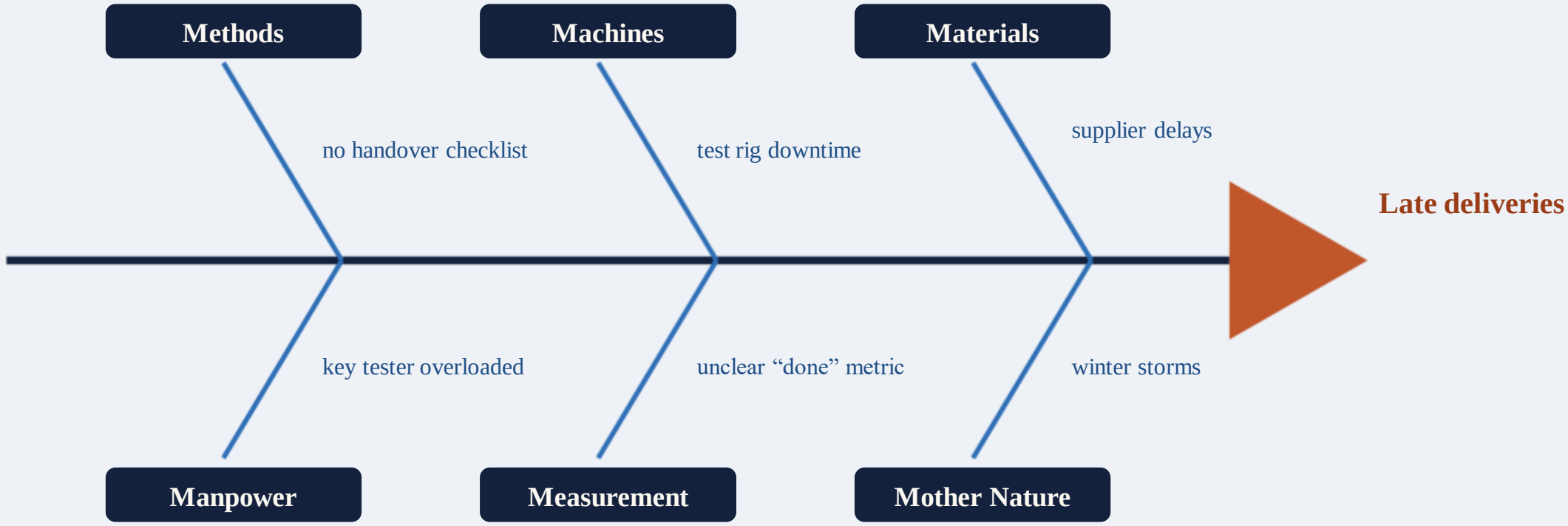
Bars sorted **descending** by frequency; line shows cumulative %.

Read: wiring + seals (navy) ≈ 80% of defects → fix these first.

80/20 rule: roughly 80% of problems come from 20% of causes (Juran).

ANNOTATED FIGURE · ISHIKAWA (FISHBONE) DIAGRAM

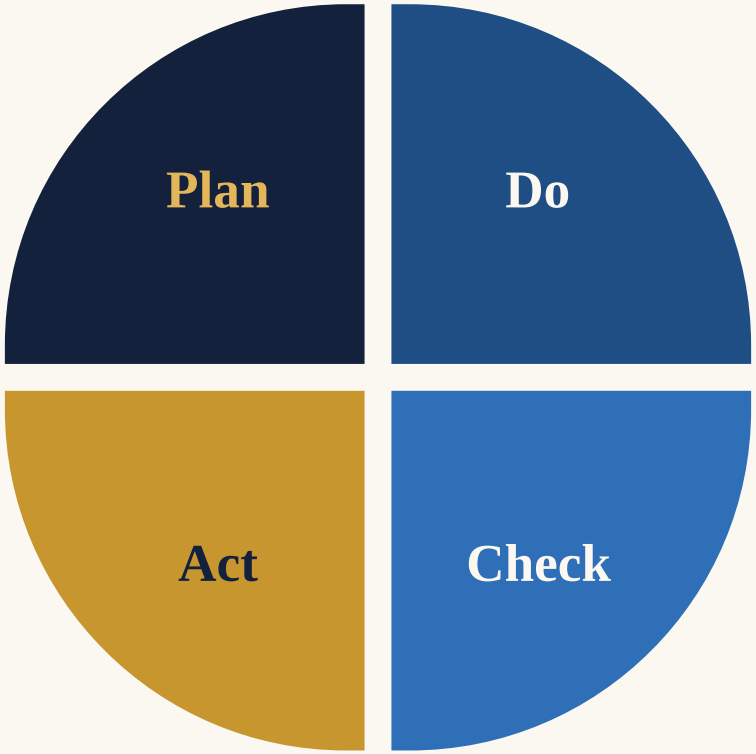
Tracing the root cause



Mnemonic “6 Ms”: Methods, Machines, Materials, Manpower, Measurement, Mother Nature. Combine with “5 Whys” to dig deeper on each bone.

CONTINUOUS IMPROVEMENT

PDCA, Kaizen, Six Sigma, Lean



PDCA (Deming)
Iterative cycle of small, tested improvements.

Kaizen
Continuous small improvements by everyone.

Six Sigma · DMAIC
Define, Measure, Analyse, Improve, Control.

Lean
Remove waste; value stream mapping.

Lean wastes "DOWNTIME": Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra processing.

MANAGE QUALITY VS CONTROL QUALITY

Assurance of the process, control of the product

Manage quality (assurance)

Are we following the right processes? **Process-focused**, preventive. Tools: quality audits, process analysis, design for X, problem solving.

Control quality

Does the output meet requirements? **Product-focused**, detective. Tools: inspection, testing, control charts, statistical sampling.

Quality audit: independent review that policies and processes are followed; finds good practices too.

Attribute sampling: pass/fail. **Variable sampling**: measured on a scale.

Regulatory compliance: ECO enabler; audits confirm it.

Mnemonic: “Manage the Method; Control the Creation.”

PROCESS TASK 8 · PLAN AND MANAGE SCHEDULE

The predictive scheduling chain

1 **Define activities** → 2 **Sequence** → 3 **Estimate durations** → 4 **Develop schedule** → 5 **Baseline & control**

Outputs of activity definition

Activity list, activity attributes, milestone list (milestones have zero duration).

ECO enablers

Schedule per approach; coordinate with other projects and operations; use benchmarks and history; baseline; analyse variation.

Mnemonic "Dogs Sit, Eat, Dream, Bark": Define, Sequence, Estimate, Develop, Baseline/control.

Four logical relationships

FS



Finish-to-start · most common. Pour before cure.

SS



Start-to-start. Level concrete starts once pouring starts.

FF



Finish-to-finish. Testing finishes when coding finishes.

SF



Start-to-finish · rarest. Old system runs until new one starts.

Dependency	Meaning	Example
Mandatory (hard logic)	Contractually or physically required	Foundation before walls
Discretionary (soft, preferred)	Best practice; can be changed to fast-track	Design review before coding
External / Internal	Outside vs inside the team’s control	Permit approval / own testing

ANNOTATED FIGURE · LEADS AND LAGS

Pulling forward, waiting deliberately

Lead · overlap (−)



Successor starts 2 days **before** predecessor finishes: FS − 2d. Landscaping starts before the punch list ends.

Lag · wait (+)



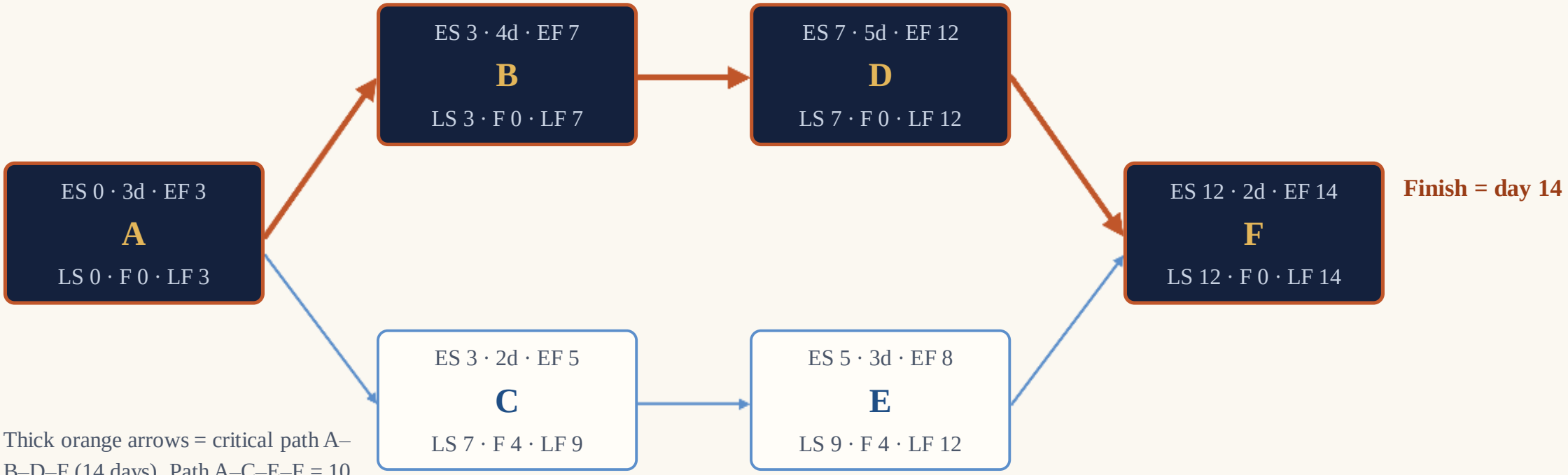
Successor waits 3 days **after** predecessor: FS + 3d. Concrete must cure before framing.

Mnemonic: “Lead = Leap ahead (−); Lag = Linger (+).”

Trap: a lag is not an activity; it consumes time but no resources or cost.

ANNOTATED FIGURE · CRITICAL PATH METHOD

Forward pass, backward pass, critical path



Thick orange arrows = critical path A–B–D–F (14 days). Path A–C–E–F = 10 days.

FLOAT (SLACK) FORMULAS

Total float, free float, project float

Total float

LS − ES or LF − EF

Delay possible without delaying the project end.

Free float

ES(next) − EF(this)

Delay possible without delaying any successor's early start.

Project float

Imposed date − EF

Delay possible before missing a client deadline.

From our network	Total float	Free float
Activity C	LS 7 − ES 3 = 4	ES(E) 5 − EF(C) 5 = 0
Activity E	LS 9 − ES 5 = 4	ES(F) 12 − EF(E) 8 = 4
Activities A, B, D, F	0 · critical	0

Mnemonic: "Total = Late minus Early; Free = Next's Early minus my Early Finish."

CRITICAL PATH FACTS

Six truths about the critical path

1. It is the **longest** path; it sets the **shortest** possible duration.
2. Activities on it usually have **zero** total float.
3. There can be **more than one**; more critical paths = more risk.
4. It can **change** as work progresses or durations shift.
5. Near-critical paths (low float) deserve monitoring too.
6. Compress **only** critical activities to shorten the project.

Mnemonic: “Longest path, Least float, Lots of risk.”

Choose the technique by what you know

Predictive

- **Analogous** · top-down from history
- **Parametric** · rate × quantity (10 m/day × 200 m)
- **Three-point** · triangular or PERT (next slide)
- **Bottom-up** · aggregate detailed estimates
- **Expert judgement** · Delphi, SMEs

Agile (relative)

- **Story points** · relative effort, complexity, risk
- **Planning poker** · Fibonacci cards 1, 2, 3, 5, 8, 13…
- **T-shirt sizing** · XS, S, M, L, XL
- **Affinity estimating** · group items by size quickly
- **Ideal days** · uninterrupted effort days

Trap: padding estimates hides risk. Estimate honestly; handle uncertainty with explicit reserves. Those who do the work estimate it.

FORMULA · THREE-POINT ESTIMATING

PERT (beta) and triangular

PERT / BETA

(O + 4M + P) ÷ 6

σ = (P − O) ÷ 6

TRIANGULAR

(O + M + P) ÷ 3

equal weights

Worked example: O = 4 days, M = 6 days, P = 14 days.

PERT = (4 + 24 + 14) ÷ 6 = **7.0 days** · σ = (14 − 4) ÷ 6 = **1.67 days** · Triangular = 24 ÷ 3 = **8.0 days**

Range at ±1σ ≈ 68%: 5.33 to 8.67 days; ±2σ ≈ 95%: 3.67 to 10.33 days.

Mnemonic: “Most likely Matters 4× more.” For a path, add the variances (σ²), then square-root — never add σ directly.

SCHEDULE COMPRESSION

Crashing versus fast tracking

Crashing



Add resources to critical activities (overtime, extra staff). **Raises cost.** Pick the lowest crash cost per day first.

Fast tracking

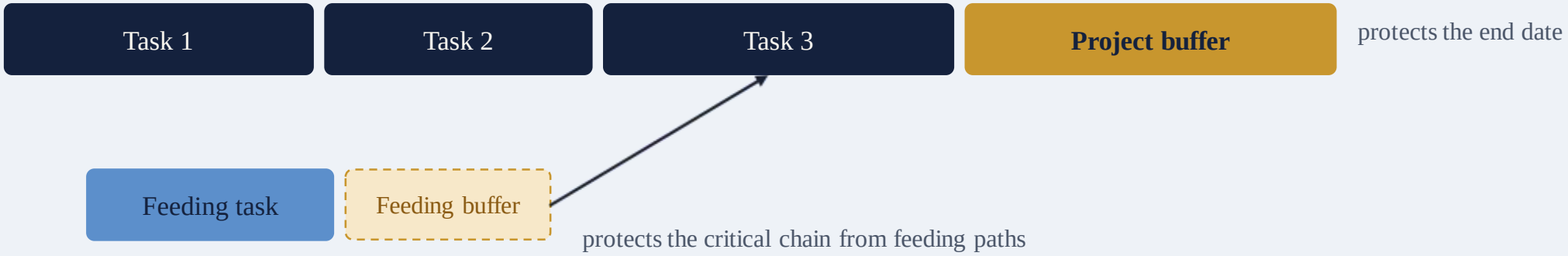


Overlap activities normally done in sequence. **Raises risk and rework.** Only on discretionary dependencies.

Crash option	Days saved	Extra cost	Cost per day
Activity B (critical)	2	£6,000	£3,000 · best
Activity D (critical)	2	£9,000	£4,500
Activity C (non-critical)	1	£1,000	No benefit · has float

ANNOTATED FIGURE · CRITICAL CHAIN METHOD

Pool the padding into buffers



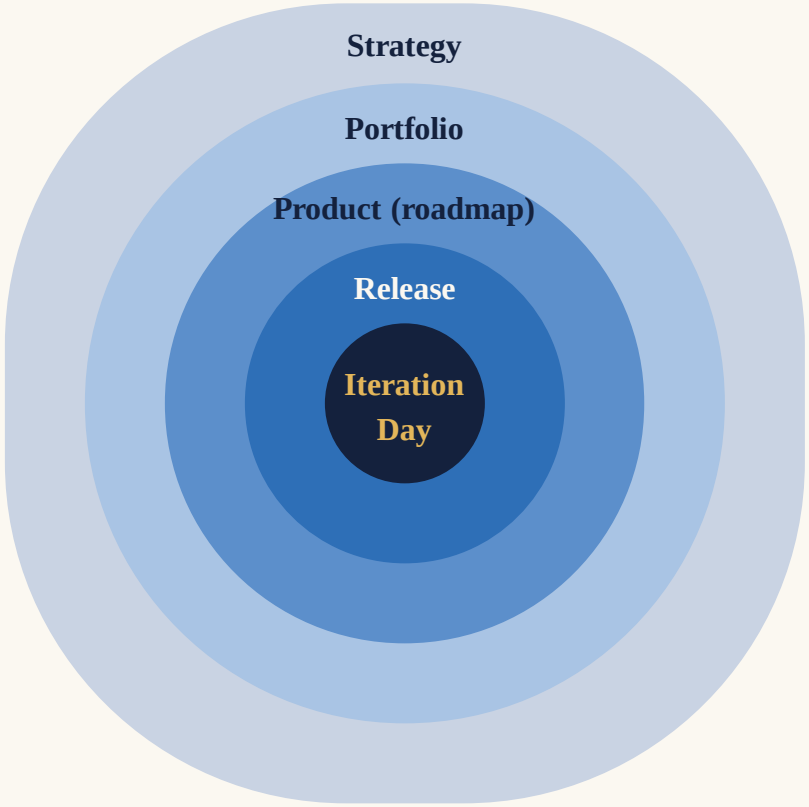
Considers resources: the critical chain is the longest resource-constrained path.

Aggressive estimates (≈50% confidence), safety moved to buffers.

Manage by buffer burn, not by individual task dates.

AGILE SCHEDULING · PLANNING ONION

Plan at the right horizon



Roadmap: themes and features over quarters.

Release plan: which features ship when, from backlog size ÷ velocity.

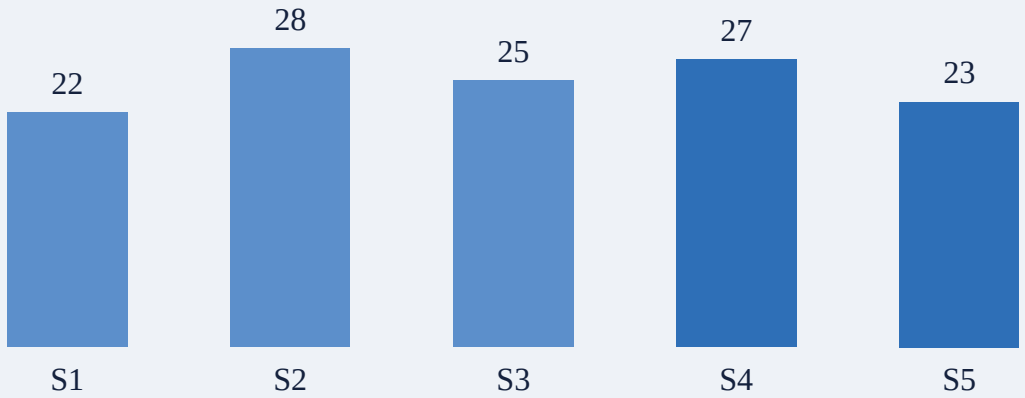
Iteration plan: sprint goal and committed stories.

Daily plan: stand-up coordination.

"SPPRID": Strategy, Portfolio, Product, Release, Iteration, Day — outer to inner.

VELOCITY AND FORECASTING

Forecast releases from real throughput



Completed story points per sprint (8 px per point). Average ≈ 25.

FORECAST

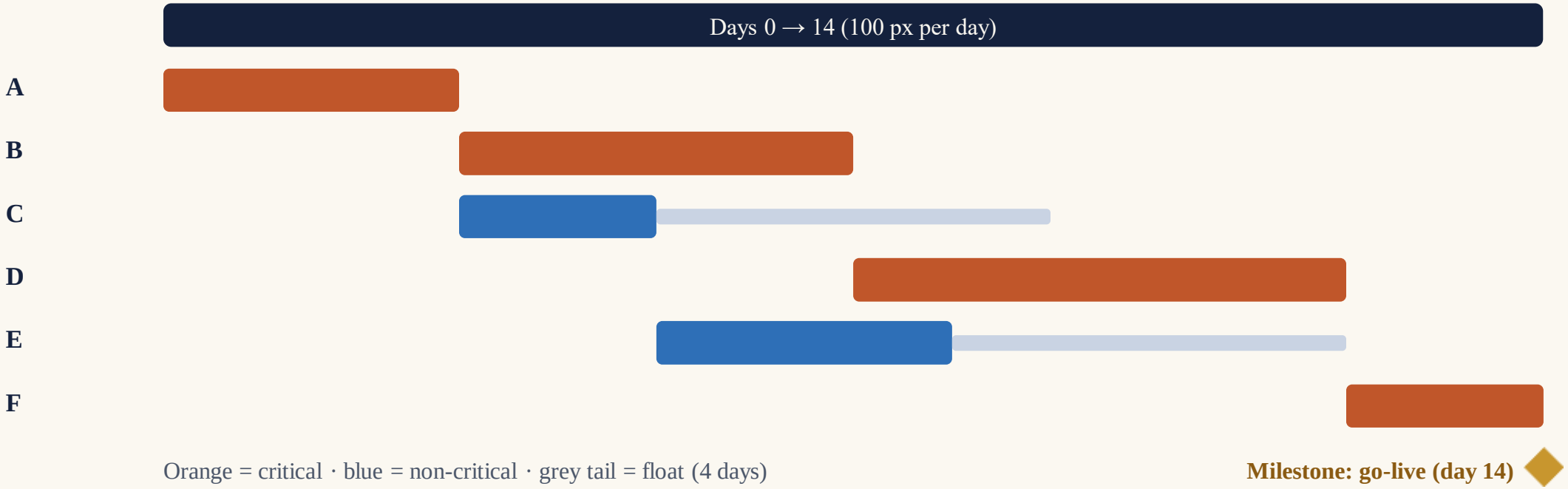
Remaining backlog 200 pts ÷ 25 pts/sprint = **8 sprints** (2-week sprints ≈ 16 weeks)

Use a range: best (28) to worst (22) gives ≈ 7.1 to 9.1 sprints.

Traps: velocity is not a performance target; never compare teams' velocities; only fully "done" stories count.

ANNOTATED FIGURE · GANTT CHART FROM OUR NETWORK

Bars, float and milestones



PROCESS TASK 9 · EVALUATE PROJECT STATUS

Metrics and artifact management

Delivery
Velocity, throughput, cycle time, lead time, WIP, defect rate.

Baseline
SV, CV, SPI, CPI, milestone hit rate, EAC.

Value & stakeholders
NPS, adoption, benefits realised, morale.

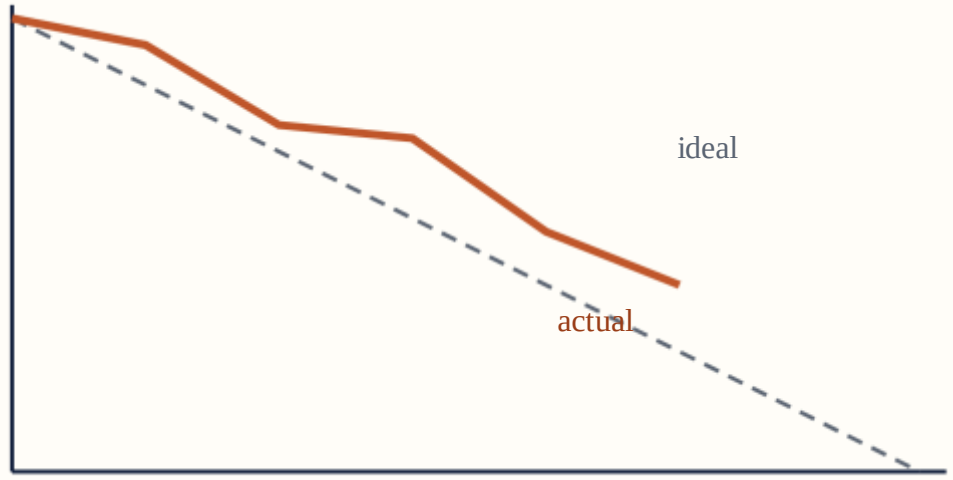
Artifact management (ECO enablers)
Identify and tailor needed artifacts · ensure they are created, reviewed, updated and documented · ensure accessibility · continually assess the effectiveness of artifact management.

Good metrics are "SMART-ER": meaningful, few, owned, trended, acted upon.

Trap: watermelon reporting — green outside, red inside. Report the truth.

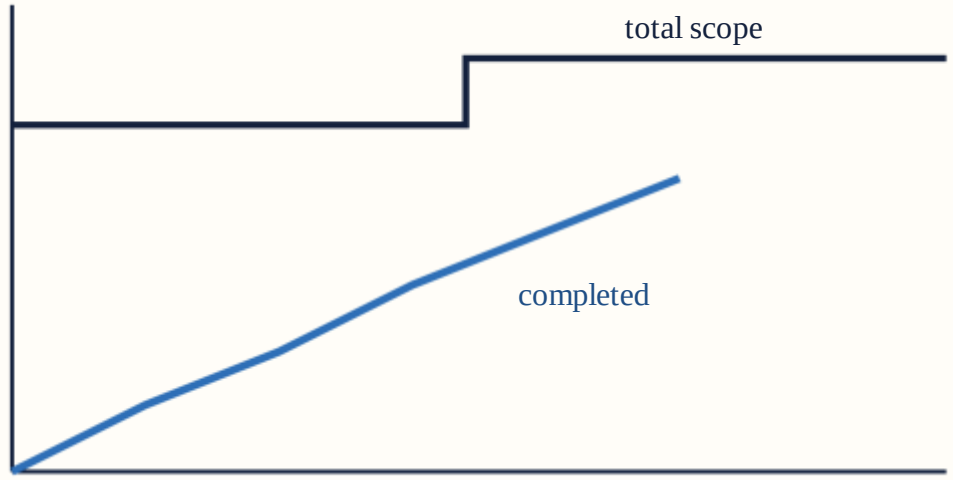
ANNOTATED FIGURES · BURNDOWN AND BURNUP

Two views of agile progress



Burndown · work remaining

Actual (orange) above ideal (dashed) → behind.

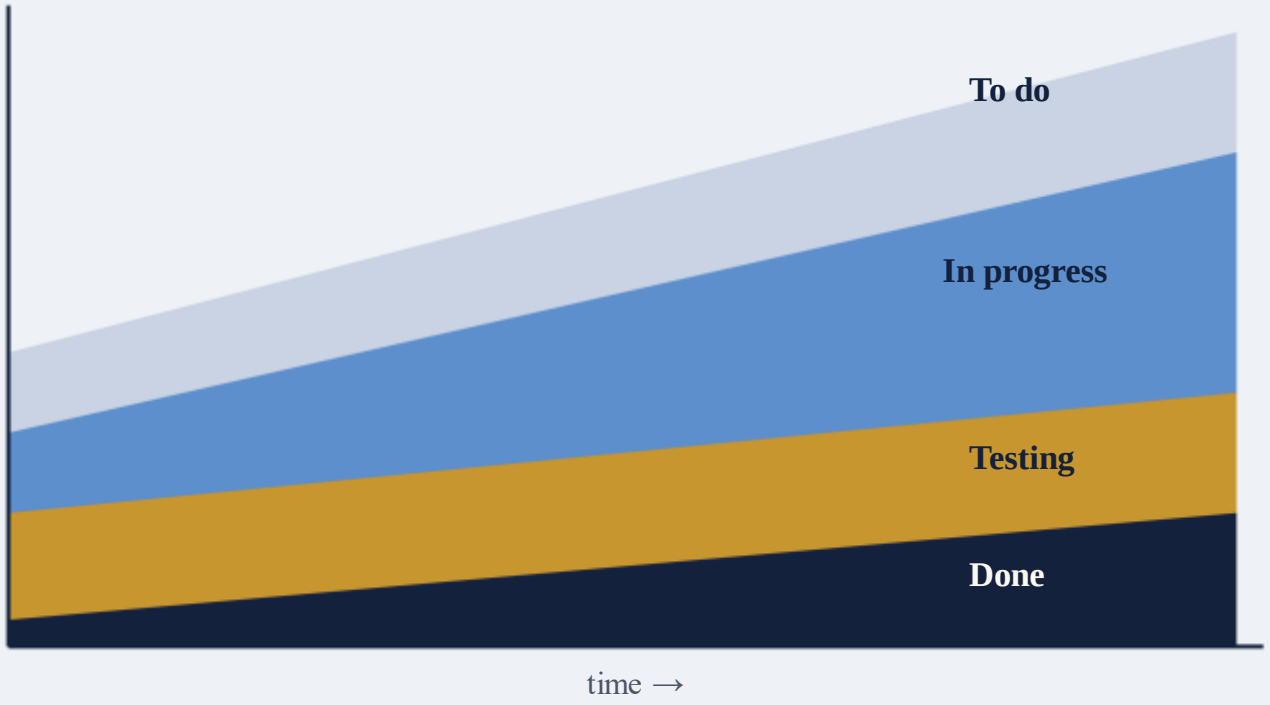


Burnup · work done vs scope

The step in the scope line shows added scope.

ANNOTATED FIGURE · CUMULATIVE FLOW DIAGRAM

Seeing bottlenecks in flow



Vertical width of a band = WIP in that state.

Horizontal width = approximate cycle / lead time.

Widening band = bottleneck (here: the in-progress band widens while Done grows slowly).

Fix: lower WIP limits; swarm on the constraint.

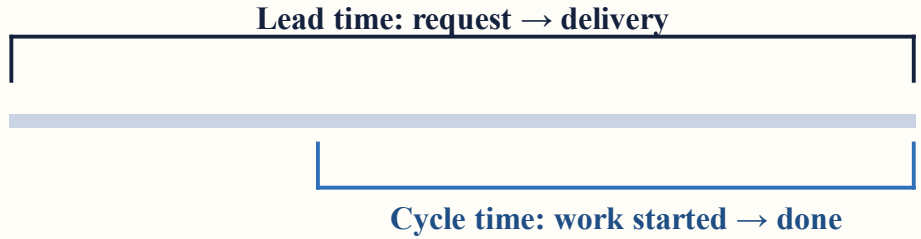
FORMULA · FLOW METRICS

Little’s Law, lead time and cycle time

LITTLE’S LAW

Cycle time = WIP ÷ Throughput

Worked example: 30 items in progress, team finishes 10 per week → cycle time = 30 ÷ 10 = **3 weeks**. Halve WIP to 15 → **1.5 weeks**.



Mnemonic: “Less WIP, Quicker Ship.” Lead time is what the customer feels; cycle time is what the team controls.

COMMUNICATING STATUS

A one-page status report that works

[Project] · status as of [date] — **Overall: AMBER**

Schedule	Cost	Scope	Risk
AMBER · SPI 0.94	GREEN · CPI 1.02	GREEN · 2 CRs approved	RED · supplier delay

Achieved: [milestones met] · **Next period:** [planned work] · **Decisions needed:** [from sponsor, by date] · **Top risks & issues:** [with owners and actions]

"SANDR": Status, Achievements, Next steps, Decisions, Risks.

Always pair RAG with words and numbers, never colour alone.

Four values: left over right

Individuals and interactions	over	processes and tools
Working software	over	comprehensive documentation
Customer collaboration	over	contract negotiation
Responding to change	over	following a plan

Mnemonic "I Want Coffee Right-now": Individuals, Working software, Customer collaboration, Responding. The right side still has value; the left has more.

THE TWELVE AGILE PRINCIPLES · PARAPHRASED

Twelve principles in four clusters

Customer & value

1 Satisfy through early, continuous delivery · 2 Welcome changing requirements · 3 Deliver frequently, shorter is better

People & collaboration

4 Business and developers work together daily · 5 Trust motivated individuals · 6 Face-to-face is most effective

Progress & pace

7 Working software measures progress · 8 Sustainable pace, indefinitely · 9 Technical excellence and good design

Simplicity & improvement

10 Simplicity: maximise work not done · 11 Self-organising teams · 12 Regularly reflect and adjust

Mnemonic "VPPS": Value, People, Pace, Simplicity — three principles each.

SCRUM · THREE ACCOUNTABILITIES

Who is accountable for what in Scrum

Product Owner

Accountable for maximising product value; orders the product backlog; one person, not a committee.

Scrum Master

Accountable for Scrum’s effectiveness; servant leader, coach, removes impediments.

Developers

Accountable for a usable increment each sprint; cross-functional; decide how to do the work.

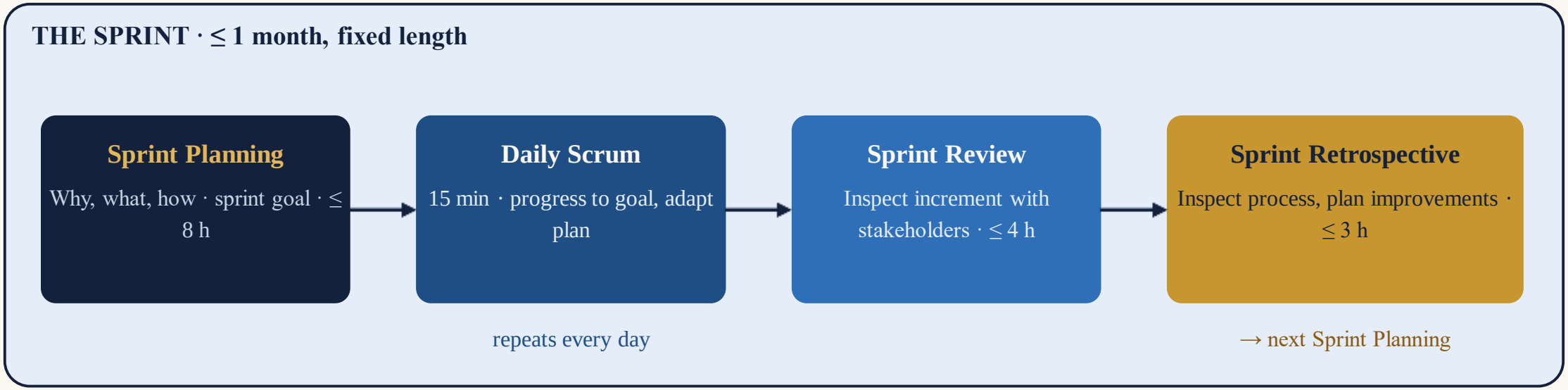
Scrum team size: typically 10 or fewer people. No sub-teams or hierarchies.

Trap: the Scrum Master does not assign tasks or own the backlog.

Link to Module 6: the Scrum Guide itself uses the word “accountabilities” — one clear owner per outcome, exactly the RACI “A” idea.

ANNOTATED FIGURE · THE SCRUM SPRINT CYCLE

Five events inside one timebox



Mnemonic "Sprints Pack Daily Reviews & Retros": Sprint · Planning · Daily · Review · Retrospective. Timeboxes shown are maximums for a one-month sprint.

SCRUM · ARTIFACTS AND THEIR COMMITMENTS

Each artifact has a commitment

Product Backlog
Ordered list of everything needed in the product.

Commitment: Product Goal

Sprint Backlog
Sprint goal + selected items + plan; owned by developers.

Commitment: Sprint Goal

Increment
Usable step toward the product goal; additive.

Commitment: Definition of Done

Definition of Done (DoD): shared quality standard for every increment (tested, reviewed, documented).

Definition of Ready (DoR): team-agreed criteria for a story to enter a sprint (not in the Scrum Guide, but common).

Mnemonic "PSI → PSD": Product backlog→Product goal, Sprint backlog→Sprint goal, Increment→Definition of Done.

KANBAN METHOD

Visualise, limit WIP, manage flow

1 · Visualise the workflow
Board with columns per state.

2 · Limit WIP
Pull new work only when capacity frees.

3 · Manage flow
Measure cycle time, remove blockers.

4 · Make policies explicit
Column entry/exit criteria.

5 · Feedback loops
Replenishment, delivery reviews.

6 · Improve collaboratively
Evolve experimentally.

"Very Little Mess, Please, For Improvement" — Visualise, Limit, Manage, Policies, Feedback, Improve.

Scrum vs Kanban: Kanban has no fixed iterations or prescribed roles; it is a pull system ideal for support and ops work.

Engineering practices that make agility safe

Pair programming
Two developers, one workstation.

Test-driven dev
Write the failing test first.

Continuous integration
Merge and test many times daily.

Refactoring
Improve design without changing behaviour.

Collective ownership
Anyone can improve any code.

Small releases
Frequent, low-risk deployment.

Simple design
Build only what is needed now.

Sustainable pace
No chronic overtime.

XP values "CSFCR": Communication, Simplicity, Feedback, Courage, Respect.

Technical debt: shortcuts that slow future work; pay it down with refactoring.

SCALING AGILE · RECOGNISE BY NAME

Frameworks for teams of teams

Framework	Key idea	Signature element
SAFe	Enterprise-scale alignment across agile release trains	PI (program increment) planning, WSJF
LeSS	One product backlog, one PO, many Scrum teams	Minimal extra roles
Scrum of Scrums	Team ambassadors coordinate dependencies	Daily or regular cross-team sync
Disciplined Agile (DA)	Toolkit of choices; context counts (PMI-owned)	Choose your WoW (way of working)
Spotify model	Squads, tribes, chapters, guilds	Autonomy with alignment

Exam tip: with several agile teams sharing dependencies, a Scrum of Scrums or joint planning is the expected coordination answer.

PROCESS TASK 10 · MANAGE PROJECT CLOSURE

The closure checklist

- Confirm closure criteria are met
- Obtain formal stakeholder acceptance
- Validate readiness for transition
- Close procurements and contracts
- Close financials and accounts
- Final lessons learned / retrospective
- Archive records; update OPAs
- Release resources; celebrate

Mnemonic "CAT-P-FLAR": Criteria, Acceptance, Transition, Procurement, Financials, Lessons, Archive, Release.

TRANSITION READINESS

Handing value to operations

Plan early
Transition requirements captured with scope.

→

Prepare
Training, runbooks, support model, warranty.

→

Validate
Readiness review, go/no-go, pilot.

→

Hand over
Sign-off; benefits owner takes over.

Readiness criteria: operations staffed and trained, documentation accepted, support SLAs agreed, data migrated, rollback plan tested.

Next-phase transition: the same logic applies when handing one phase's outputs to the next team.

Trap: "Throw it over the wall" at the end. Involve operations as stakeholders from the start.

LESSONS LEARNED AND RETROSPECTIVES

Learn continuously, not just at the end

Start / Stop / Continue
Simple, quick retrospective format.

Sailboat
Wind (helps), anchors (slows), rocks (risks), island (goal).

4Ls
Liked, Learned, Lacked, Longed for.

Retrospective flow (Derby & Larsen): Set the stage → Gather data → Generate insights → Decide what to do → Close the retrospective.

"Some Good Gifts Deserve Celebrating" — Set, Gather, Generate, Decide, Close.

Output: a lessons-learned register, archived as an OPA; and concrete improvement actions.

EARLY TERMINATION

When the project stops early

Common causes

Business case no longer valid · strategy change · funding cut · unresolvable risk · regulation change · value already achieved (agile).

What the PM still does

Run closure anyway: document status and reasons, close contracts (termination clauses), capture lessons, archive, release resources.

Agile view: stopping once enough value is delivered is a success, not a failure.

Trap: a cancelled project still needs formal closure; never just walk away.

MODULE FIVE · DOMAIN III · 26%

Business Environment: govern, comply, adapt

05

Governance, compliance, change control, issues and impediments, risk, continuous improvement, organisational change, external forces, AI and sustainability.

MODULE MAP · BUSINESS ENVIRONMENT

Inside-out and outside-in forces

INSIDE THE ORGANISATION

1 · Governance
2 · Compliance
3 · Change control
4 · Impediments & issues
5 · Risk
6 · Continuous improvement
7 · Organisational change



OUTSIDE THE ORGANISATION

8 · External business environment

Regulation · technology · geopolitics · markets

AI developments · sustainability expectations

Thread to follow: external change → assess impact on scope/backlog → risk or change request → governance decision within thresholds → organisational change support.

BE TASK 1 · DEFINE AND ESTABLISH PROJECT GOVERNANCE

Governance: the rules of the game

Structure
Sponsor, steering committee, PMO, CCB, design authority.

Rules & procedures
Decision rights, approval levels, stage gates.

Reporting
What is reported, to whom, how often.

Ethics
Code of conduct, conflicts of interest, AI use policy.

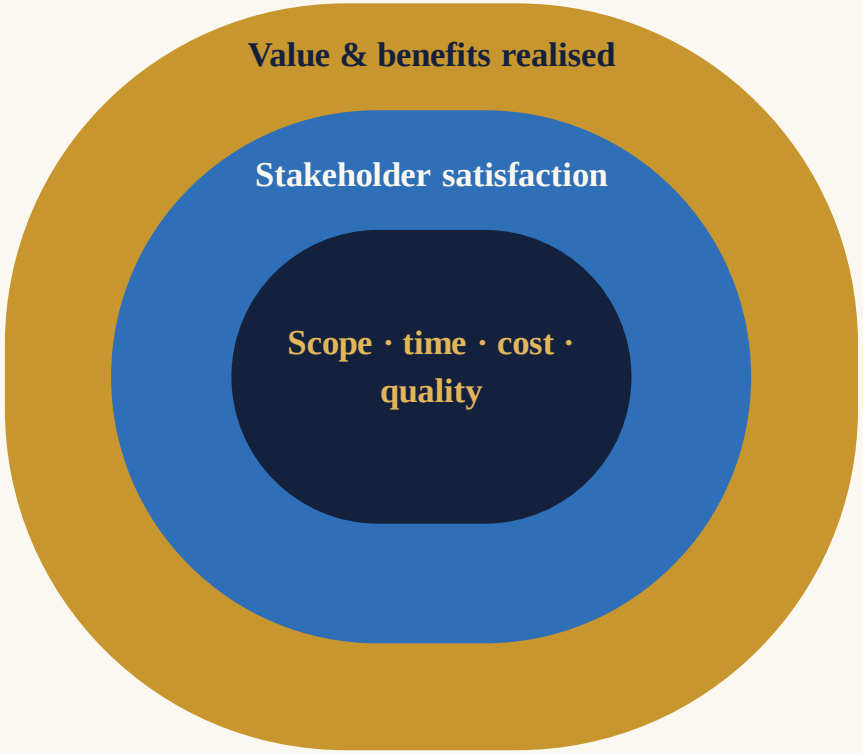
Policies via OPAs
Use templates and procedures that exist; don't reinvent.

Escalation
Paths and thresholds (next slides).

Mnemonic "SRREPE": "Smart Rules Rarely Ever Produce Errors" — Structure, Rules, Reporting, Ethics, Policies, Escalation.

DEFINING SUCCESS METRICS

Success is broader than the triple constraint



Project management success: delivered as planned (inner ring).

Project success: delivered value worth the effort and expense (outer ring).

Agree success metrics with the sponsor **at the start** and record them in the charter.

Mnemonic: “On time is nice; worth it is the price.”

ANNOTATED FIGURE · ESCALATION PATHS AND THRESHOLDS

Decide at the lowest competent level

Portfolio board	Variance > 20% or strategic impact
Sponsor / steering	Variance 10–20%, management reserve, scope change
Project manager	Variance 5–10%, contingency draw, re-plan
Team / lead	Variance < 5%, task-level adjustment

Threshold values are illustrative; each organisation sets its own.

Why thresholds: empower teams, keep leaders focused, avoid surprise.

Exam rule: within your threshold, act; above it, escalate with analysis and options.

Trap: escalating without facts, impact and a recommendation.

KEY GOVERNANCE ROLES

Sponsor, steering committee, PM

Role	Core responsibilities	Typical RACI letter
Sponsor	Funds, champions, signs charter, owns business case, removes organisational blocks	A for project outcome and benefits
Steering committee	Strategic direction, gate decisions, resolves escalations above PM authority	A for gate decisions; C on major change
Project manager	Leads planning and delivery, integrates, reports, manages within thresholds	A for project management deliverables
PMO	Standards, support, assurance, portfolio data	C / I
Change control board	Approves or rejects change requests	A for change decisions

Exam tip: the sponsor is the PM's first escalation point for issues beyond the PM's authority, especially resources and priorities.

Compliance categories

Security & privacy
Data protection, cyber, GDPR-type laws

Health & safety
Worker and public safety rules

Sustainability
Environmental, ESG reporting

Regulatory
Industry rules, licences, standards

Confirm requirements → Classify categories → Find threats → Apply methods → Measure compliance

Mnemonic "SHSR": "Safe Hands Save Reputations" — Security, Health & safety, Sustainability, Regulatory.

COMPLIANCE IN PRACTICE

Methods, consequences and measurement

Methods
Compliance requirements in the RTM · checklists · audits · training · design reviews · privacy impact assessments · supplier clauses.

Consequences of non-compliance
Fines · legal action · shutdown · reputational damage · safety incidents · loss of licence.

Measuring
Audit findings closed · % controls tested · incidents · certification status.

Exam rule: compliance is non-negotiable. If the sponsor asks to skip a regulatory step to save time, the answer is to explain the consequences and keep compliance; tailoring cannot remove it.

Mnemonic "MCM": Methods prevent, Consequences motivate, Measurement proves.

BE TASK 3 · ANNOTATED FIGURE · INTEGRATED CHANGE CONTROL

The predictive change control flow

1 **Change identified** → 2 **Evaluate impact** → 3 **Identify options** → 4 **Submit CR** → 5 **CCB decides** → 6 **Update & inform**

Approved: update plan, baselines, documents; implement; communicate.

Rejected: record in change log; inform requester with reasons.

Deferred: logged with a review date.

"I Eat Oranges So Cats Understand" — Identify, Evaluate, Options, Submit, CCB, Update.

Trap: never implement first and document later — even for the sponsor.

CHANGE SCENARIOS · WHAT WOULD PMI DO?

Change requests under pressure

Scenario	Best response
Sponsor verbally asks for a new feature	Evaluate impact, then submit a formal CR
Team member made a change without approval	Assess impact; take it through change control; coach on process
CCB rejected a change the customer wants	Communicate decision and reasons; log it
Emergency fix needed to stop safety risk	Act per emergency procedure; document and ratify afterwards
Change within the PM's authority threshold	PM approves per plan; log and communicate

CCB: formally chartered group that reviews, approves, defers or rejects changes. Its authority is in the change management plan.

Order of moves: Analyse → Options → Formal CR → Decision → Update → Tell.

CONFIGURATION MANAGEMENT

Change control versus configuration control

Change control

Manages changes to the **project**: baselines, plans, scope, schedule, cost.

Configuration control

Manages the **product's** specifications and versions: which design, which build, which document version.

Identification of configuration items

Status accounting: what changed, when, by whom

Verification & audit: correct versions in use

Mnemonic "ISV": Identify, Status, Verify. "Change = the Plan; Configuration = the Product."

Change flows through the backlog



Mid-sprint: the sprint goal is protected. New work normally waits for the next sprint; if the goal becomes obsolete, only the PO may cancel the sprint.

Hybrid: backlog changes inside agreed release scope need no CCB; changes to budget, contract or milestone go through governance.

Mnemonic: “Predictive → CCB. Agile → PO. Hybrid → it depends on the threshold.”

BE TASK 4 · CRITICAL DISTINCTION

Risk, issue, impediment

Risk

Uncertain future event; may or may not happen. Threat or opportunity.

Tool: risk register

Issue

Current condition **already happening** that affects objectives.

Tool: issue log

Impediment

Obstacle **blocking the team** from progressing (agile term).

Tool: impediment board

Risk → it occurs → **Issue** → blocks team → **Impediment**

Mnemonic: “Risk = Maybe; Issue = Now; Impediment = Stuck.” ECO: “recognise when a risk becomes an issue.”

ARTIFACT · ISSUE LOG

Every issue gets an owner and a date

ID	Description	Priority	Owner	Action	Due / status
I-07	Vendor API down for 2 days	High	[Tech lead]	Escalate to vendor; mock API	[date] · Open
I-08	Tester on sick leave	Medium	[PM]	Borrow tester from team B	[date] · Open
I-09	Legal review delayed	High	[Sponsor]	Sponsor to expedite	[date] · Closed

Prioritise by impact on objectives and urgency; **collaborate** with relevant stakeholders on the approach (ECO).

Owner rule: one named owner per issue — an issue owned by everyone is owned by no one (see Module 6).

REMOVING IMPEDIMENTS

Evaluate, prioritise, intervene, reassess

1

Evaluate impact

How much does it slow flow or value?

2

Prioritise & highlight

Make it visible on the board.

3

Intervene

Remove or minimise; escalate if outside reach.

4

Reassess

Continually, until cleared.

Typical impediments: waiting for approvals, missing environments, unclear requirements, dependencies on other teams, tool access.

Surfaced at: daily scrum, retrospectives, board blockers, one-to-ones.

Mnemonic "EPIR": "Every Problem Is Removable."

The risk management cycle

1 **Plan risk mgmt** → 2 **Identify** → 3 **Qualitative** → 4 **Quantitative** → 5 **Plan responses** → 6 **Implement** → 7 **Monitor**

Risk
An uncertain event that, if it occurs, has a positive or negative effect on objectives.

Individual vs overall
Single events vs the effect of all uncertainty on the project.

Risk owner
One person accountable for monitoring and responding.

Mnemonic "Please Identify Quality Questions, Plan, Implement, Monitor" — the seven risk processes in order.

IDENTIFYING RISKS

Techniques and the cause–risk–effect format

Brainstorming
Team generates risks freely

Checklists
From past projects (OPAs)

SWOT
Strengths, weaknesses, opportunities, threats

Assumption analysis
What if an assumption is false?

Prompt lists
PESTLE, TECOP, VUCA

RBS
Risk breakdown structure by category

Interviews / Delphi
Expert, anonymous input

Root cause analysis
Find underlying drivers

Metalanguage: "Because of [cause], [uncertain event] may occur, which would lead to [effect on objectives]."

Because the vendor is new, the API may be delivered late, delaying integration by 3 weeks.

ARTIFACT · RISK REGISTER

The living record of uncertainty

ID	Risk statement	P	I	Score	Owner	Response	Trigger
R-01	Poor IT security → data breach	0.3	0.8	0.24	[CISO]	Mitigate: pen test	Audit fail
R-02	New vendor late with API	0.5	0.4	0.20	[Lead]	Mitigate: mock API	Missed demo
R-03	Grant funding may be awarded early	0.3	0.6	0.18	[Sponsor]	Exploit: pre-plan	Letter
R-04	Carbon rules tighten for data centres	0.2	0.6	0.12	[ESG lead]	Accept · monitor	Consultation

Also record: category, proximity, residual and secondary risks, contingency plan, fallback plan, status.

Score: probability × impact. Watch-list low scores; respond to high ones.

ANNOTATED FIGURE · PROBABILITY AND IMPACT MATRIX

Qualitative risk analysis

P \ I	0.05	0.10	0.20	0.40	0.80
0.9	.045 L	.09 M	.18 H	.36 H	.72 H
0.7	.035 L	.07 M	.14 M	.28 H	.56 H
0.5	.025 L	.05 M	.10 M	.20 H	.40 H
0.3	.015 L	.03 L	.06 M	.12 M	.24 H
0.1	.005 L	.01 L	.02 L	.04 L	.08 M

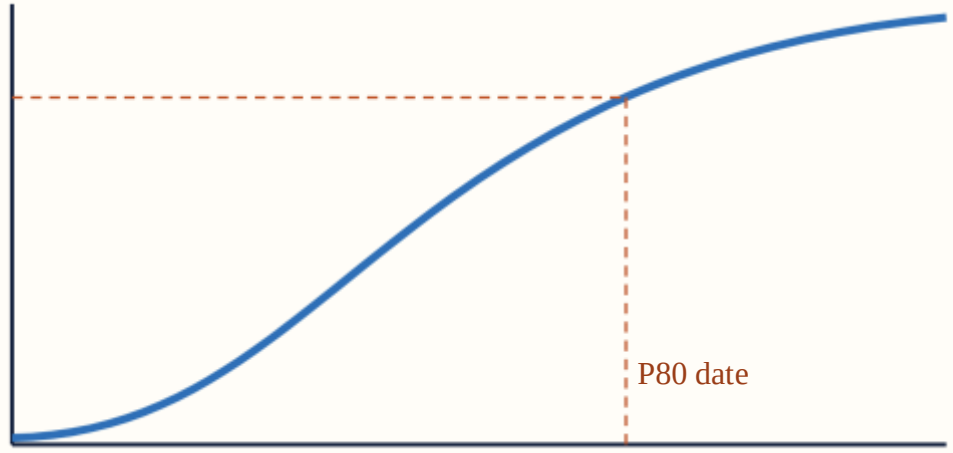
Score = P × I. Thresholds (illustrative): H ≥ 0.18 · M 0.05–0.17 · L < 0.05.

Also assess: urgency, proximity, manageability, connectivity, data quality.

Mnemonic: “Qualitative = Quick ranking; Quantitative = Quantified numbers.”

QUANTITATIVE RISK ANALYSIS

Monte Carlo and the tornado diagram



Monte Carlo S-curve

Thousands of simulated runs → 80% chance to finish by the marked date (P80).



Tornado (sensitivity) diagram

Longest bar on top = variable with the greatest effect on the outcome.

FORMULA · EXPECTED MONETARY VALUE

EMV = probability × impact

$$EMV = \Sigma (P \times I)$$

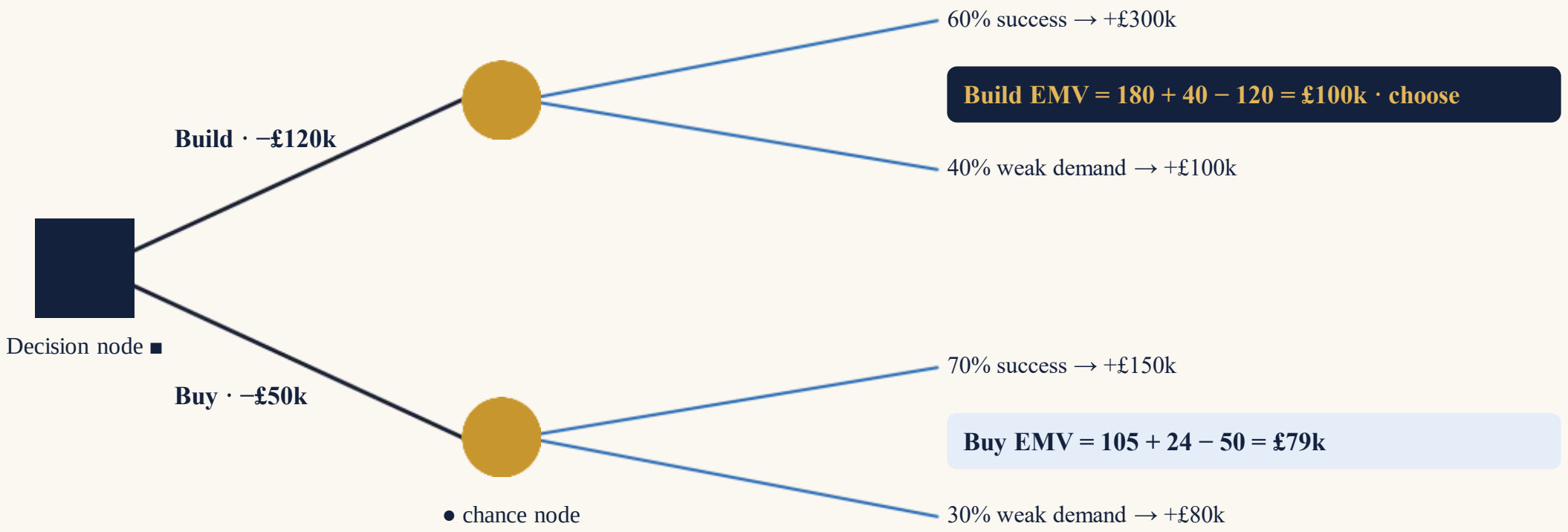
threats negative (−), opportunities positive (+)

Risk	Probability	Impact	EMV
Threat: supplier failure	30%	−£50,000	−£15,000
Threat: design rework	10%	−£40,000	−£4,000
Opportunity: bulk discount	20%	+£20,000	+£4,000
Net EMV → contingency reserve			−£15,000

Exam tip: the net EMV of identified risks is a common way to size the contingency reserve (here £15,000).

ANNOTATED FIGURE · DECISION TREE ANALYSIS

Build or buy? Let EMV decide



RISK RESPONSES · THREATS

Five ways to respond to a threat

Escalate

Outside PM's authority or scope → to program/portfolio owner.

Avoid

Eliminate the threat: change plan, scope or approach.

Transfer

Shift ownership: insurance, fixed-price contract, warranty.

Mitigate

Reduce probability or impact: testing, redundancy, prototypes.

Accept

Active (contingency reserve) or passive (deal with it if it happens).

Mnemonic "Every Angry Teacher Must Assess": Escalate, Avoid, Transfer, Mitigate, Accept.

Trap: transfer moves ownership, not the risk itself; and it usually costs a premium.

RISK RESPONSES · OPPORTUNITIES

Five ways to respond to an opportunity

Escalate
Value lies beyond the project → to a higher owner.

Exploit
Make sure it happens (probability → 100%).

Share
Partner with a third party better able to capture it.

Enhance
Increase probability or impact.

Accept
Take it if it comes; no proactive action.

Threat strategy	Mirror opportunity strategy
Avoid (P → 0)	Exploit (P → 100%)
Transfer (give to others)	Share (partner with others)
Mitigate (reduce P or I)	Enhance (increase P or I)

Mnemonic "Every Excited Student Eagerly Accepts": Escalate, Exploit, Share, Enhance, Accept.

RISK ATTITUDES

Appetite, tolerance, threshold

Appetite

How much uncertainty the organisation is **willing** to take for reward. Qualitative.

Tolerance

The acceptable **range** of variation (e.g. ±5% budget).

Threshold

The **specific point** beyond which action or escalation is required.

Risk-averse: pays to reduce uncertainty (regulated, safety).

Risk-neutral: decides on expected value (EMV).

Risk-seeking: accepts more uncertainty for upside (start-ups).

Mnemonic "ATT = Attitude": Appetite (how hungry), Tolerance (how much wiggle), Threshold (where the line is).

MONITORING RISKS

Keep the register alive

Risk reassessment
Regularly identify new risks, close old ones, re-score.

Risk audit
Are responses effective? Is the process followed?

Reserve analysis
Is remaining contingency enough for remaining risk?

Trend & variance
EVM and burn charts signal emerging risk.

Workarounds
Unplanned responses to unidentified risks that occur.

Communicate
ECO: status of risk impact to stakeholders.

Exam tip: a risk owner reports that a trigger has fired → implement the planned response. An unforeseen problem occurs → workaround, then add lessons and update the register.

BE TASK 6 · CONTINUOUS IMPROVEMENT

From project lessons to organisational assets

Utilise lessons
Check past lessons before planning.

→

Update CI processes
Improve how the organisation improves.

→

Update OPAs
Templates, checklists, historical data.

Project level: retrospectives, PDCA, Kaizen events, quality improvements.

Organisation level: PMO standards, maturity models, communities of practice.

Mnemonic "LUCO": Lessons Used, CI updated, OPAs refreshed — the learning organisation loop.

BE TASK 7 · SUPPORT ORGANISATIONAL CHANGE

Read the culture before you change it

Assess culture
Values, decision style, tolerance for risk, hierarchy, appetite for change, past change history.

Evaluate impact
How does a reorganisation, merger or new strategy affect this project, and what actions are needed?

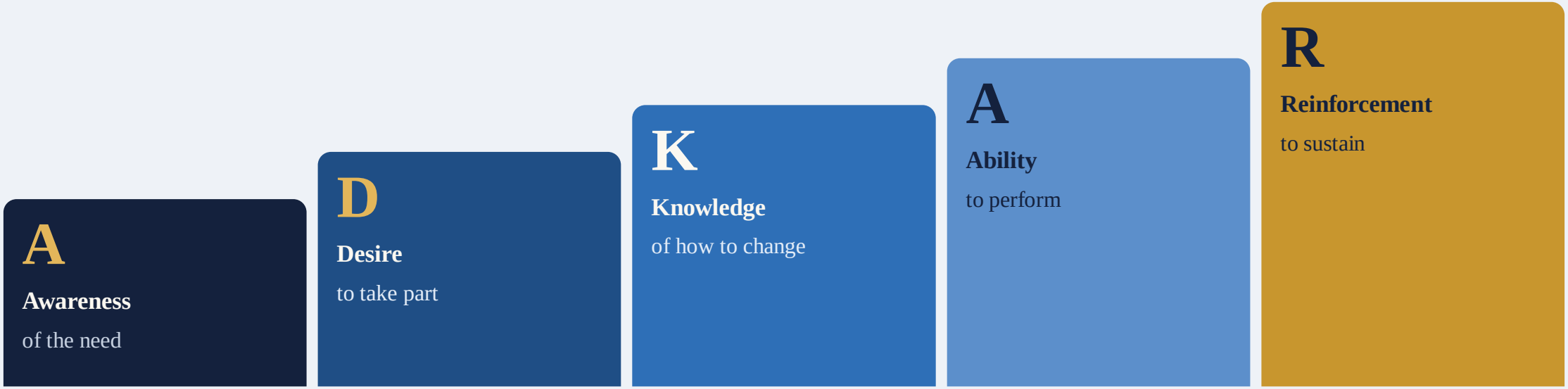
Change curve
Shock → denial → frustration → experimentation → decision → integration. Support people through each stage.

Resistance
Is information; uncover its cause (fear, loss, workload) and respond with involvement and support.

Exam tip: the project's outputs are only valuable if people adopt them. Plan adoption as deliberately as delivery.

INDIVIDUAL CHANGE · PROSCI ADKAR®

Five building blocks of personal change



Diagnostic use: find the first missing block. Training (K) fails if people lack desire (D).

ORGANISATIONAL CHANGE · KOTTER'S EIGHT STEPS

Leading change across the organisation

1 Create urgency

2 Build a guiding coalition

3 Form a strategic vision

4 Enlist a volunteer army

5 Remove barriers

6 Generate short-term wins

7 Sustain acceleration

8 Institute change

Navy = create a climate · blue = engage and enable · gold = implement and sustain

"Under Grey Skies, Volunteers Remove Snow, Securing Infrastructure" — the 8 steps in order.

Other models: Lewin (unfreeze–change–refreeze), Satir change curve, Bridges transition.

BE TASK 8 · EVALUATE EXTERNAL BUSINESS ENVIRONMENT CHANGES

Scan the horizon with PESTLE

P
Political
elections, sanctions, trade

E
Economic
inflation, FX, rates

S
Social
demographics, values

T
Technological
AI, platforms, obsolescence

L
Legal
new laws, export controls

E
Environmental
climate, emissions rules

Survey external changes → **Assess & prioritise** impact on scope / backlog → **Continually review** for new impacts

Mnemonic: “PESTLE = Pests Eat Small Tomatoes, Lettuce, Eggplants.”

AI IN PROJECT MANAGEMENT · NEW IN 2026

AI as assistant, PM as accountable

Planning
Draft WBS, estimates from history, schedule scenarios, risk prompts.

Monitoring
Anomaly detection in EVM, forecasting, sentiment in stakeholder feedback.

Communication
Meeting summaries, report drafts, translation, knowledge search.

HUMAN-IN-THE-LOOP RULE

AI may **recommend**; a named human is **accountable** for the decision. In RACI terms, AI is never the "A".

EXAM TRAP

"Accept the AI forecast as final" or "paste confidential data into a public tool" — both wrong.

RESPONSIBLE AI USE

Six checks before you rely on AI

Privacy & security
Approved tools only; no confidential data leaks.

Accuracy
Verify outputs against sources and data.

Fairness & bias
Check for biased recommendations.

Transparency
Disclose AI use to stakeholders.

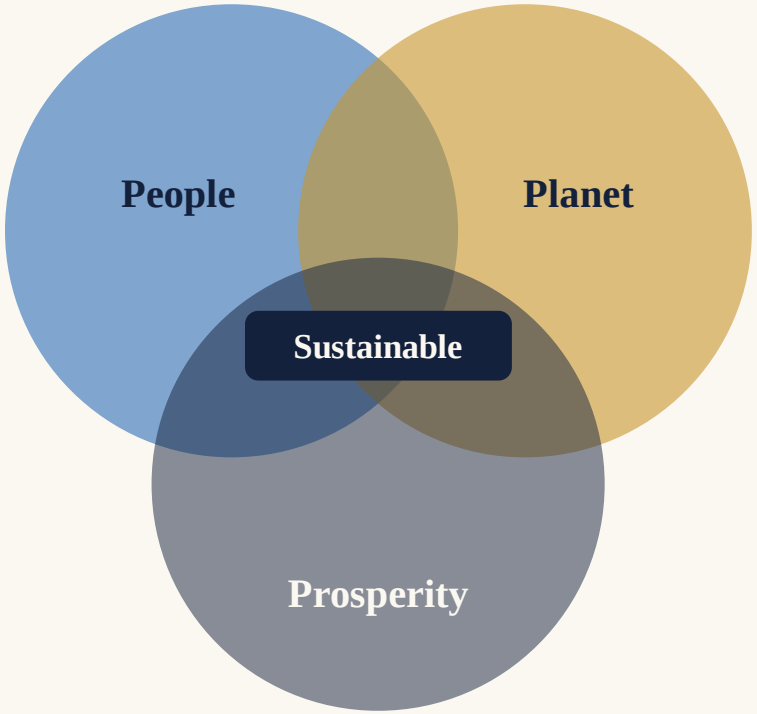
Accountability
A named human owns each decision.

IP & compliance
Licences, copyright, AI regulation.

Mnemonic "PAF-TAI": "Please Always Fact-check; Then Assign Individually" — Privacy, Accuracy, Fairness, Transparency, Accountability, IP.

SUSTAINABILITY ACROSS THE LIFE CYCLE

The triple bottom line in practice



Plan: sustainability as a critical information requirement (Process 1).

Procure: supplier criteria with environmental scores.

Quality: CoQ and sustainability; reduce waste and rework (Process 7).

Comply & risk: sustainability categories and risks (BE 2, BE 5).

"3 Ps, 4 places": People-Planet-Prosperity in Plan, Procure, Quality, Comply.

MODULE SIX · DEEP DIVE

06

RAM and RACI: who is actually accountable?

Two project management terms. One common misunderstanding. And a critical distinction that determines how effectively teams work together.

IMAGINE THIS PROJECT

Everyone is busy. Meetings are happening. Tasks are being completed.

Then a **critical deliverable is delayed** — and nobody can answer one simple question:

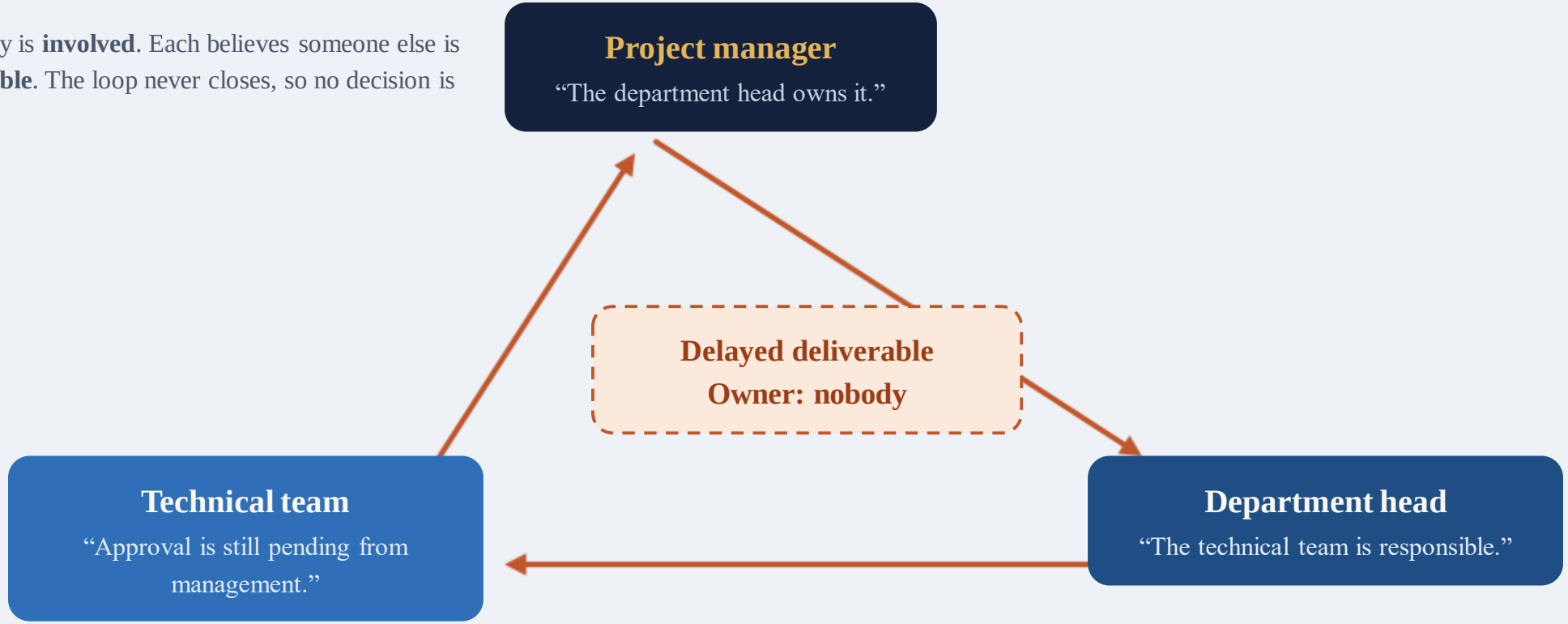
> ***"Who is actually accountable for this?"***

Everyone is involved. Ownership is unclear.

ANNOTATED FIGURE · THE CIRCLE OF ASSUMED OWNERSHIP

Everyone points somewhere else

Each party is **involved**. Each believes someone else is **accountable**. The loop never closes, so no decision is made.



WHY IT MATTERS

The hidden cost of unclear ownership

Decision latency
Questions wait for an owner who doesn't exist; schedules slip silently.

Duplicate or dropped work
Two people do the same task, or nobody does it.

Conflict and blame
Energy shifts from solving to defending.

Communication noise
Everyone copied on everything; nobody acts.

Weak escalation
Without an owner, risks are never raised to the right level.

The remedy
Make roles explicit per deliverable: a Responsibility Assignment Matrix.

ECO link: People Task 3 enabler "establish clear roles and responsibilities within the team"; Process Task 4 "define and plan resources".

TERM 1 · RESPONSIBILITY ASSIGNMENT MATRIX

What is a RAM?

A grid that connects **project work** with the **individuals, roles or organisational units** responsible for delivering it.

Its one question

"Who is assigned to which part of the project?"

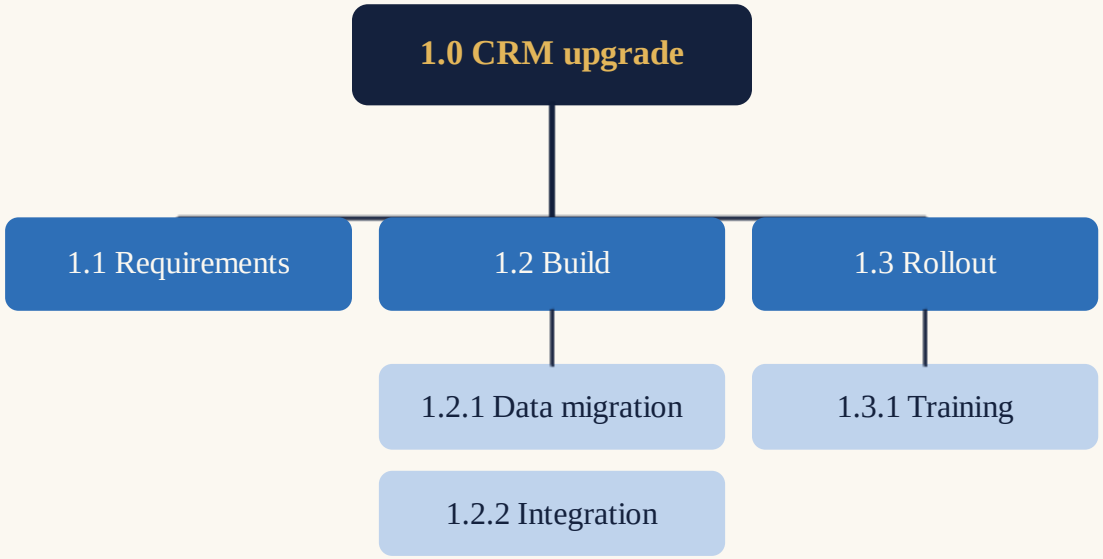
Its shape

Rows = work (deliverables, work packages, activities). Columns = people, roles or units. Cells = the assignment.

Key idea: RAM is a **category** of tool. What goes in the cells — an X, hours, or letters like R-A-C-I — defines which *type* of RAM you have.

BUILDING BLOCK 1 · WBS

The WBS answers “what work?”



Work Breakdown Structure: defines project deliverables and work packages.

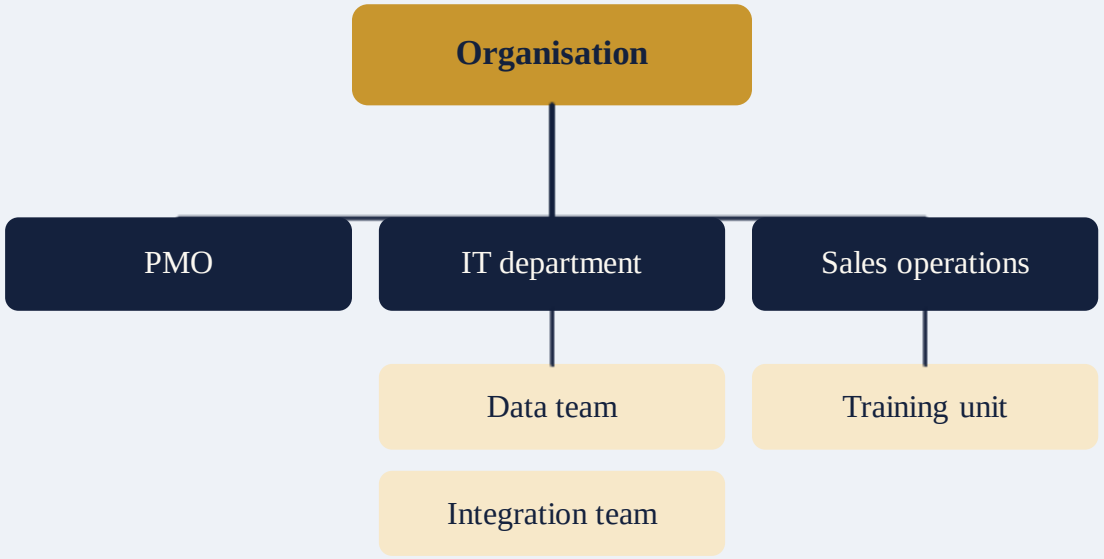
Deliverable-oriented, hierarchical, 100% of scope.

Says nothing about **who** does the work.

This WBS becomes the **rows** of our RAM on the coming slides.

BUILDING BLOCK 2 · OBS

The OBS answers “which unit?”



Organisational Breakdown Structure: identifies the organisational units responsible for performing the work.

Hierarchical, like the WBS, but of people and departments.

Says nothing about **which work** each unit owns.

This OBS becomes the **columns** of our RAM.

ANNOTATED FIGURE · WBS × OBS = RAM

Where work meets the organisation

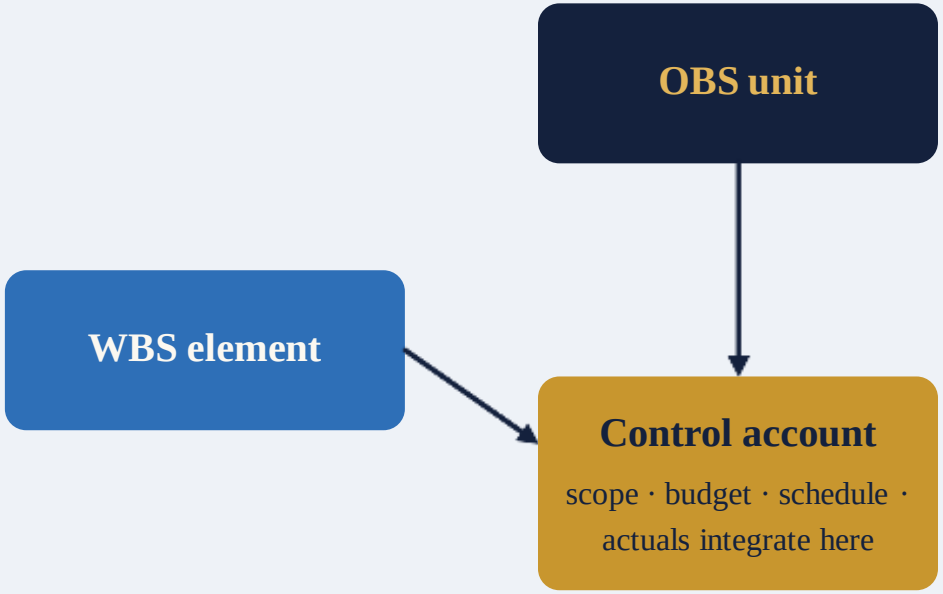
WBS ↓ · OBS →	PMO	Data team	Integration	Training
1.1 Requirements	X			
1.2.1 Data migration		X		
1.2.2 Integration			X	
1.3.1 Training				X

Each X = this unit is assigned to this work package.

Highlighted cell: Data team × Data migration — the intersection is a **control account** (next slide).

This simple “X” grid is already a valid RAM: an **assignment matrix**.

The RAM intersection is where control happens



Control account: a management control point where scope, budget, actual cost and schedule are integrated and compared to earned value.

Control account manager (CAM): the single person accountable for that intersection.

Takeaway: even at the cost-control level, PMI insists on **one owner** per intersection — the same idea as RACI's single "A".

Same grid, different cell contents

Assignment (X)

X		
	X	X

Who is involved. Simplest.

Effort / allocation

40h	8h	
	50%	20%

How much effort each contributes.

Role-based (RACI)

A	R	I
C	**A/R**	I

In what capacity each is involved.

Levels: high-level RAM (deliverables × departments) for governance; detailed RAM (work packages or activities × named people) for the team.

Mnemonic: “X says Who, Hours say How much, Letters say How.”

THE COMMON MISUNDERSTANDING

MYTH

"RAM and RACI are two competing tools. Choose one."

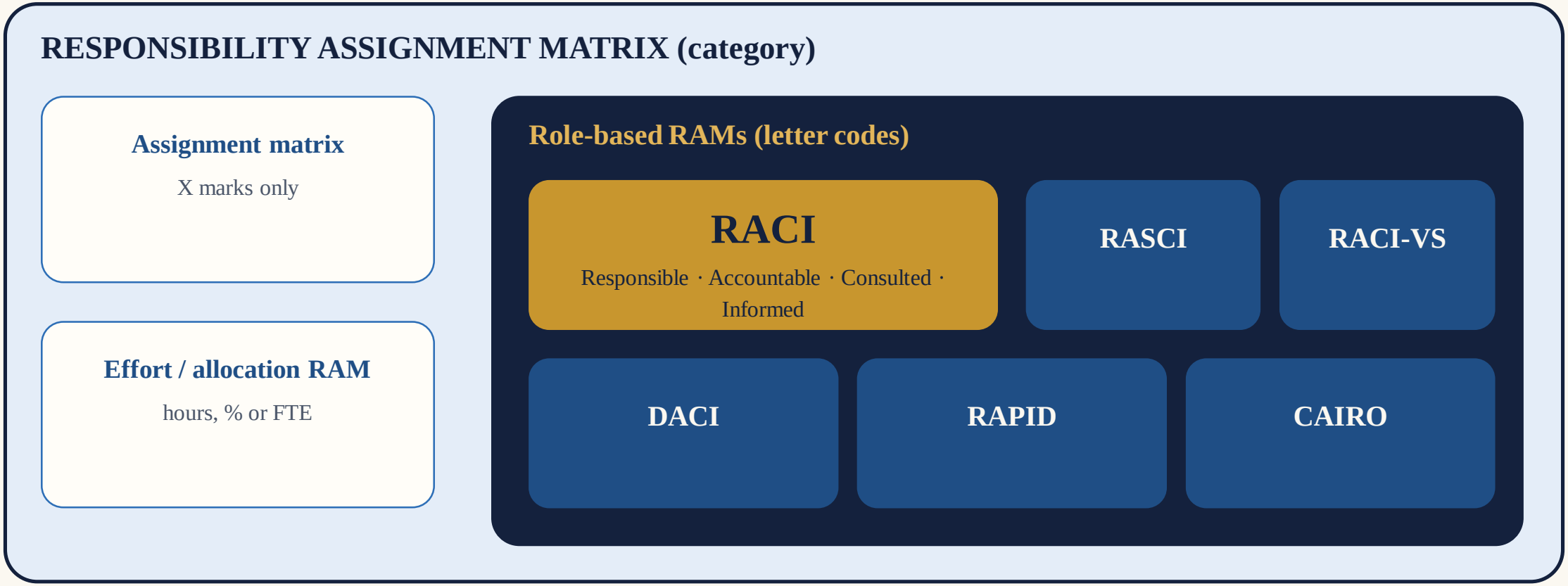
FACT

RACI is one specific *type* of Responsibility Assignment Matrix.

Every RACI chart is a RAM. Not every RAM is a RACI chart.

ANNOTATED FIGURE · THE RAM FAMILY TREE

RACI lives inside the RAM category



TERM 2 · THE RACI MODEL

Four letters, four distinct roles

R

Responsible

Does the work. Can be one or several people.

A

Accountable

Owns the outcome; approves; answers for it. **Exactly one.**

C

Consulted

Gives input before or during work. **Two-way** communication.

I

Informed

Kept up to date on progress or decisions. **One-way** communication.

Mnemonic: “R does it · A owns it · C advises on it · I hears about it.”

DEEP DIVE · R

R

Responsible

Definition

The person or people who perform the work to complete the task or deliverable.

Properties

- At least one R per activity
- Can be shared among several people
- Can be delegated or reassigned
- Reports progress to the A

Story check: the technical team was clearly R for the data migration. Doing the work was never the problem.

DEEP DIVE · A

A

Accountable

Definition

The one person who ultimately answers for the correct and complete delivery, approves the work, and owns the decision.

Properties

- **Exactly one** A per activity or deliverable
- Cannot be shared or split
- Cannot be delegated away (the work can)
- Must have the authority to decide

Story check: nobody was the A for the migration. That single empty cell caused the whole delay.

DEEP DIVE · C AND I

Consulted versus Informed

C Consulted

Team ⟷ Expert

Two-way dialogue. Opinions sought **before** decisions or work. Subject-matter experts, legal, security, affected managers.

Too many Cs slow decisions to a crawl.

I Informed

Team → Audience

One-way update. Told **after** decisions or at milestones. Downstream teams, executives, support desk.

Too many Is create inbox noise and ignored updates.

THE GOLDEN RULES OF RACI

Six rules that keep a RACI honest

1. **Exactly one A** per row — never zero, never two.
2. **At least one R** per row; someone does the work.
3. The **A must hold authority** to approve and decide.
4. Use **names or specific roles**, not vague groups.
5. **Keep C and I lean**; involve only who adds value.
6. **Agree it with the people named**, then publish it.

MNEMONIC · "ONE A TO RULE EACH ROW"

If you can only remember one RACI rule for the exam, remember this one.

HOW TO BUILD A RACI · STEP BY STEP

Seven steps from blank grid to agreement

1
List deliverables / activities from the WBS (rows)

2
List roles or people from the OBS (columns)

3
Assign the single A for each row first

4
Assign the R(s) who will do the work

5
Add C and I sparingly

6
Run horizontal and vertical checks

7
Validate with everyone named; baseline in the resource plan

Then: review at gates, on team changes, and after escalations.

Mnemonic "Rows, Columns, A, R, CI, Check, Confirm": "Really Clever Analysts Rarely Create Confusing Charts."

CASE STUDY · BEFORE · THE BROKEN MATRIX

What the delayed project actually had

Deliverable	Project manager	Dept head	Technical team	Management
1.1 Requirements	A, R	A	C	I
1.2.1 Data migration	C	C	R	C
1.2.2 Integration	I	R	R	R
1.3.1 Training	I	I	I	I

1.1: two As → conflicting approvals

1.2.1: no A at all → the delay story

1.2.2: three Rs, no A → diffusion

1.3.1: all I, no R or A → orphaned

CASE STUDY · AFTER · THE REPAIRED MATRIX

One A per row, clear R, lean C and I

Deliverable	Project manager	Dept head	Technical team	Management
1.1 Requirements	A, R	C	C	I
1.2.1 Data migration	C	A	R	I
1.2.2 Integration	I	C	A, R	I
1.3.1 Training	A	C	C	I
Go-live approval	R	C	C	A

Now the question has an answer: the department head is accountable for the migration and owns the decision to unblock it.

Management moves from vague C to clear I — except go-live, where it holds the single A.

RACI ANALYSIS 1 · READ ACROSS EACH ROW

Horizontal analysis: is each task covered?

Pattern in a row	What it signals	Fix
No A	Accountability gap; nobody decides	Assign one owner with authority
Two or more As	Conflicting approvals; turf wars	Choose one; others become C
No R	Nobody does the work	Assign a doer (A may also be R)
Many Rs	Diffusion; possible over-staffing	Split the task or name a lead R
Many Cs	Slow decisions, meeting overload	Keep only essential experts
Every cell filled	Over-involvement	Convert most to I or blank

Mnemonic “No A, no way; two As, no peace.”

RACI ANALYSIS 2 · READ DOWN EACH COLUMN

Vertical analysis: is each person set up to succeed?

Pattern in a column	What it signals	Fix
Lots of As	Decision bottleneck; overload	Delegate accountability for some rows
Lots of Rs	Person over-allocated	Rebalance; check resource histogram
No R or A at all	Role may be unnecessary	Confirm need; release or move to I
No empty cells	Too involved in everything	Remove low-value Cs and Is
A without authority	Accountable but powerless	Grant authority or reassign A

Mnemonic: “Rows check the work; columns check the worker.”

Link: vertical analysis complements the resource histogram from Module 4.

THE CRITICAL DISTINCTION

Responsible is not Accountable

R

Does the work
Can be many people
Can be delegated
Answers to the A
Focus: **execution**

≠

A

Owns the outcome
Exactly one person
Cannot be delegated away
Answers to the sponsor / governance
Focus: **ownership and decisions**

Memory hook: “You can share the work. You cannot share the blame.” Responsibility is *delegable*; accountability is *not*.

GOING DEEPER · FOUR RELATED CONCEPTS

Responsibility, accountability, authority, competence

Responsibility

The duty to perform assigned work.

Accountability

Answerability for the result, including consequences.

Authority

The right to make decisions, apply resources and sign approvals.

Competence

The skill and capacity to do the role well.

Balance rule: accountability without authority is a trap; authority without accountability is a risk. Pair them in the same person, and give that person the competence (or support) to succeed.

OTHER ROLE-BASED RAMS

RACI’s siblings and when to use them

Model	Letters	Best for
RACI	Responsible, Accountable, Consulted, Informed	General default
RASCI	RACI + Support (helps the R)	Large teams with helpers
RACI-VS	RACI + Verifier, Signatory	Regulated, quality-critical work
CAIRO	RACI + Out of the loop (explicitly excluded)	Reducing noise
DACI	Driver, Approver, Contributors, Informed	Product and decision making
RAPID	Recommend, Agree, Perform, Input, Decide	Strategic decisions (Bain)

Common thread: every variant keeps a single decider — the A, the Approver, or the D. That is the non-negotiable core of any role-based RAM.

SIDE BY SIDE

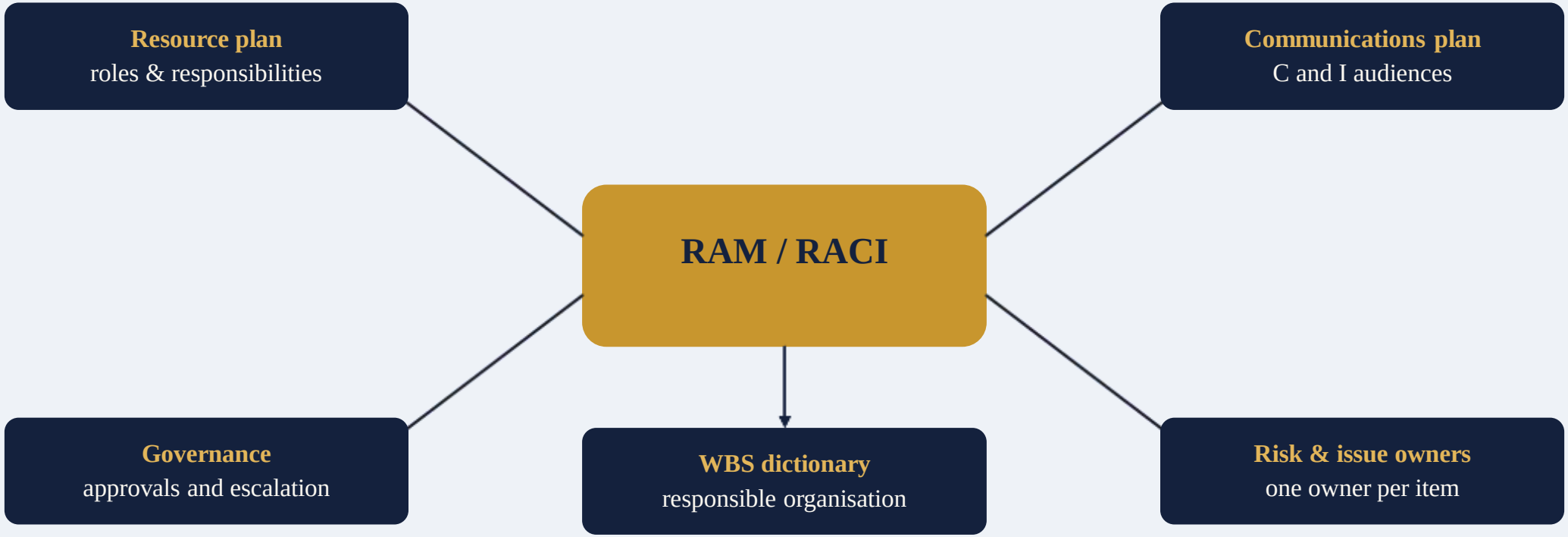
RAM (category) versus RACI (type)

Dimension	Responsibility Assignment Matrix	RACI chart
What it is	The general family of matrices	One specific member of that family
Core question	Who is assigned to which work?	In what role is each person involved?
Cell content	X, hours, %, or role codes	R, A, C, I only
Typical axes	WBS × OBS (or activities × people)	Deliverables/activities × named roles
Strength	Flexible; links scope to organisation; supports control accounts	Makes ownership and communication explicit
Rule set	Depends on the type chosen	One A per row; at least one R

Mnemonic "Every RACI is a RAM; not every RAM is a RACI." Like "every square is a rectangle."

WHERE RAM/RACI CONNECTS IN THE PLAN

One matrix, many uses



RACI IN AGILE AND HYBRID TEAMS

Clear boundaries, self-organised inside

Use RACI for

- Interfaces with other teams and vendors
- Governance decisions (release go-live, budget)
- Compliance sign-offs
- Product decisions (PO is the A for backlog order)

Avoid RACI for

- Micro-assigning sprint tasks to individuals
- Overriding the team's choice of how to work
- Replacing the team charter and conversation

Scrum mapping: PO = A for product value and backlog order; Developers = A for the increment and R for its tasks; Scrum Master = A for Scrum effectiveness. Stakeholders = C at review, I via radiators.

PITFALLS AND BEST PRACTICES

How RACI charts fail — and how to prevent it

- **Created once, never updated** → review at gates and team changes.
- **Too granular** (hundreds of rows) → keep to deliverables and key decisions.
- **Groups instead of people** ("IT") → name a role or person.
- **Imposed without agreement** → workshop it with those named.
- **Used to assign blame** → use it to prevent gaps, not to punish.
- **Hidden in a file** → publish it where the team works.

Mnemonic "LIVE": a good RACI is Lean, Individual-named, Validated by all, and Evergreen.

MODULE 6 CHECK · EXAM-STYLE SCENARIO

A deliverable is late. The RACI shows two people marked “A” and three marked “R”. The sponsor asks who will fix it. What should the PM do first?

A. Ask both accountable people to report to the sponsor jointly

B. Work with stakeholders to agree one accountable owner, update the RACI and communicate it

C. Take over as accountable owner without consultation

D. Add a fourth responsible person to speed up the work

Answer: B. Two As break the golden rule; the root cause is unclear ownership. A keeps the conflict; C bypasses collaboration and may lack authority; D adds Rs without fixing accountability (and may slow things, per Brooks’ law).

MODULE SEVEN

Exam readiness

07

Formula sheet, master mnemonic sheet, ten worked scenario questions, an eight-week study plan and an exam-day playbook.

FORMULA SHEET · PART 1 · EARNED VALUE

Every EVM formula on one page

Cost variance
CV = EV − AC

Schedule variance
SV = EV − PV

Cost performance index
CPI = EV ÷ AC

Schedule performance index
SPI = EV ÷ PV

EAC · typical
BAC ÷ CPI

EAC · atypical
AC + (BAC − EV)

EAC · CPI and SPI
AC + (BAC − EV) ÷ (CPI×SPI)

EAC · new estimate
AC + bottom-up ETC

Estimate to complete
ETC = EAC − AC

Variance at completion
VAC = BAC − EAC

TCPI (to BAC)
(BAC − EV) ÷ (BAC − AC)

TCPI (to EAC)
(BAC − EV) ÷ (EAC − AC)

FORMULA SHEET · PART 2 · EVERYTHING ELSE

Schedule, risk, communication, finance

PERT (beta)
(O + 4M + P) ÷ 6

Triangular
(O + M + P) ÷ 3

Standard deviation
σ = (P − O) ÷ 6

Total float
LS − ES = LF − EF

Free float
ES(next) − EF

Communication channels
n(n − 1) ÷ 2

Expected monetary value
EMV = P × I

Present value
FV ÷ (1 + r)ⁿ

Benefit–cost ratio
PV benefits ÷ PV costs

Little’s Law
Cycle = WIP ÷ throughput

Sprints remaining
Backlog ÷ velocity

FPIF point of total assumption
(Ceiling − target price) ÷ buyer share + target cost

MASTER MNEMONIC SHEET · PART 1

Structures and people

6 principles: Happy Villagers Quietly Accept Sunny Evenings

7 domains: Good Scouts Sail Far, Steering Rough Rivers

5 focus areas: I Pet Every Mild Cat

Org structures: Fix With Balance, Stay Productive

Conflict: Five Wise Sheep Can Cooperate

Power: Really Excellent Leaders Can Reward

Tuckman: Fresh Students Need Practice Always

Situational: Don’t Cry, Support Delegates

Maslow (up): Please Stop Sending Emails, Sam

Herzberg: Hygiene Holds, Motivators Move

Engagement: Under Rain, Nobody Stays Long

Power/interest: Satisfy · Manage · Inform · Monitor

MASTER MNEMONIC SHEET · PART 2

Process, environment and RACI

EVM: EV first; C↔AC, S↔PV; minus = variance, divide = index

Budget: All Clever Workers Count Budgets Monthly

Contracts: Fixed — seller sweats; Cost — buyer bleeds

CoQ: PAIE — Prevention, Appraisal | Internal, External

Quality tools: Cats Find Cheese, Pigs Hate Cold Showers

Compression: Crash costs Cash; Fast-track takes Risk

Threats: Every Angry Teacher Must Assess

Opportunities: Every Excited Student Eagerly Accepts

Change control: I Eat Oranges So Cats Understand

Risk / issue: Maybe · Now · Stuck

RACI: R does it · A owns it · C advises · I hears

Golden rule: One A to rule each row

PRACTICE QUESTION 1 OF 10 · PEOPLE · SINGLE RESPONSE

A team member has missed four daily stand-ups this sprint, and other developers are starting to complain in the team channel. What should the project leader do first?

A. Report the behaviour to the person’s functional manager

B. Meet the team member privately to understand the reasons

C. Remind everyone of the ground rules in the next stand-up

D. Move the stand-up to a time that suits that person

Answer: B. Assess before acting and talk directly and privately first (Module 1 ladder; ground rules, Module 3). A escalates too early; C addresses the group publicly before understanding the cause; D accommodates without diagnosis.

PRACTICE QUESTION 2 OF 10 · PROCESS · EVM INTERPRETATION

At the status date, CPI = 1.10 and SPI = 0.85. The end date is contractually fixed. Which action is most appropriate?

A. Report that the project is healthy because it is under budget

B. Request management reserve to cover a cost overrun

C. Analyse the critical path and consider crashing using the cost under-run

D. Re-baseline the schedule to match current progress

Answer: C. CPI > 1 = under budget; SPI < 1 = behind. With a fixed date, recover schedule on critical activities; the under-run can fund crashing. A ignores the delay; B misreads the cost signal; D hides the variance without approval.

PRACTICE QUESTION 3 OF 10 · PROCESS · AGILE

Mid-sprint, a senior stakeholder asks a developer directly to add a “small” reporting feature before the sprint review. What should happen?

A. The developer adds it, since it is small

B. The project leader submits a change request to the CCB

C. Redirect the request to the product owner to evaluate and order in the backlog

D. Cancel the sprint to include the new feature

Answer: C. The PO is accountable for backlog order; the sprint goal is protected. A is uncontrolled change; B applies predictive governance to an agile team’s backlog; D is drastic and only the PO may cancel, when the goal is obsolete.

PRACTICE QUESTION 4 OF 10 · BUSINESS ENVIRONMENT · CHANGE

On a predictive construction project, the sponsor verbally asks the PM to add a rooftop solar array. What should the PM do first?

A. Add it, because the sponsor funds the project

B. Evaluate the impact on scope, schedule, cost, risk and quality

C. Refuse because the scope baseline is approved

D. Ask the CCB to approve it immediately

Answer: B. Impact analysis comes before the formal CR. A bypasses control; C ignores a legitimate request; D skips the analysis the CCB needs to decide.

PRACTICE QUESTION 5 OF 10 · BUSINESS ENVIRONMENT · RISK

A supplier insolvency, recorded in the risk register with a planned response, has just occurred. What should the PM do?

A. Implement the planned risk response and log the issue

B. Create a workaround

C. Perform a new qualitative risk analysis

D. Escalate to the sponsor for a decision

Answer: A. The risk was identified and planned, so execute the plan; it is now an issue. B is for *unplanned* events; C is analysis already done; D is only needed if the response exceeds authority or fails.

PRACTICE QUESTION 6 OF 10 · PROCESS · SCHEDULE (GRAPHIC-BASED)

Using the Module 4 network (critical path A–B–D–F = 14 days; C has 4 days total float), activity C slips by 3 days. What is the effect on the project finish?

A. Finish moves to day 17

B. Finish moves to day 15

C. No change; finish stays at day 14

D. C becomes the only critical activity

Answer: C. A 3-day slip is within 4 days of total float, so the end date holds. E's remaining float drops to 1 day, because C and E share the path's float; monitor closely.

PRACTICE QUESTION 7 OF 10 · PEOPLE · CONFLICT

Two senior architects strongly disagree on the platform design. The decision is not due for two weeks, and both have valid points. Which approach is best?

A. Force a decision based on the PM’s preference

B. Facilitate a joint problem-solving session against agreed criteria

C. Split the difference: half of each design

D. Postpone the discussion until it resolves itself

Answer: B. Time is available and both views have merit: collaborate for a win–win. A forces without need; C compromise may produce an incoherent design; D withdraws from a real issue.

PRACTICE QUESTION 8 OF 10 · BUSINESS ENVIRONMENT · EXTERNAL CHANGE

A new data-residency regulation is announced that will apply before your product launches. Your hybrid project stores customer data overseas. What should you do first?

A. Assess its impact on scope and backlog with compliance experts

B. Continue; the rule is not yet in force

C. Move all data immediately to a local data centre

D. Ask the sponsor to lobby for an exemption

Answer: A. BE Task 8 then Task 2: survey, assess and prioritise impact on scope/backlog, then plan compliance actions, risks and change requests. B ignores a known future obligation; C acts before analysis; D is not the PM’s first move.

PRACTICE QUESTION 9 OF 10 · PROCESS / BE · AI

An AI tool produced activity estimates in minutes. The sponsor wants to baseline them today to save time. What should the PM do?

A. Baseline them; AI estimates are unbiased

B. Discard them and estimate from scratch

C. Validate them with the team and historical data before baselining

D. Add 20% padding to every AI estimate

Answer: C. AI can assist, but people who do the work estimate it and a human remains accountable. A over-trusts; B wastes a useful input; D hides uncertainty with padding.

PRACTICE QUESTION 10 OF 10 · PROCESS · PROCUREMENT

A buyer needs a vendor to start a large research-heavy build whose scope cannot yet be defined, and wants to motivate cost efficiency. Which contract type fits best?

A. Firm fixed price (FFP)

B. Fixed price with economic price adjustment

C. Cost plus incentive fee (CPIF)

D. Purchase order

Answer: C. Undefined scope points to cost-reimbursable; the incentive fee shares savings and overruns to motivate efficiency. A and B need clear scope; D suits simple commodity buys.

EIGHT-WEEK STUDY PLAN

Weight your time like the exam does

Week	Focus	Practice
1	Modules 1–2: ECO, mindset, PMBOK 8	Flashcards for principles, domains, structures
2	Module 3: People (33%)	40 People scenario questions
3	Module 4a: integration, scope, value, resources, procurement	40 questions; build a WBS and RACI by hand
4	Module 4b: finance, EVM, quality, schedule	Work every formula twice; CPM by hand
5	Module 4c: agile, status, closure	40 agile/hybrid questions
6	Module 5: Business Environment (26%)	40 governance/risk/change questions
7	Module 6 + full mock exam 1 (180 Q, timed)	Review every wrong and lucky answer
8	Module 7 + full mock exam 2; rest the day before	Target ≈ 75–80%+ before booking

EXAM-DAY PLAYBOOK

Answering technique for every question

1 Read the last line: first, best, next, most likely?

→

2 Spot context: predictive, agile or hybrid?

→

3 Find the real problem behind the symptom

→

4 Eliminate unethical, passive and escalate-first options

→

5 Pick the most proactive, collaborative, value-focused

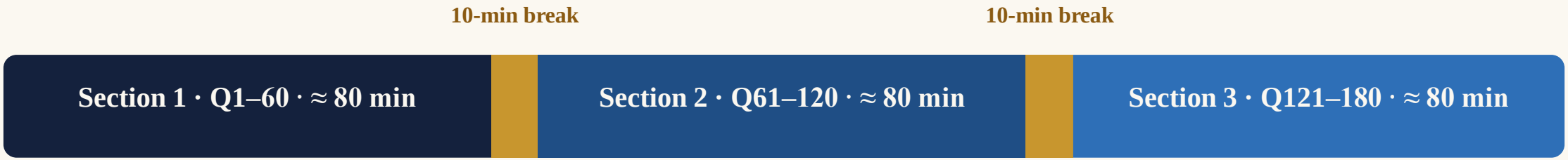
Before: sleep, ID documents, arrive early (or complete OnVUE system check and room scan), light meal, water.

During: flag and move on; never leave a blank; use both breaks to reset.

Mnemonic "Last, Context, Problem, Eliminate, Pick" — "Lucky Cats Purr Every Time."

ANNOTATED FIGURE · TIME BUDGET

240 minutes, three sections, two breaks



Breaks are not counted in the 240 minutes. Section split shown is the common 60/60/60 pattern; follow on-screen instructions on the day.

80 s
average per question (240 × 60 ÷ 180)

10 min
reserve at the end of each section for flagged items

2 min
cap per question before you flag and move on

FINAL WORD

Lead with value. Decide with data. Own the outcome.

People · serve, align, communicate

Process · plan, deliver, measure

Business Environment · govern, comply, adapt

And on every project: one accountable owner for every deliverable.